

Displacement and Infrastructure Provision: Evidence from the Interstate Highway System *

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September 13, 2026

Abstract

This paper studies the long-run effects of the Interstate Highway System on displaced individuals and their neighbors. I identify individuals in the 1950 census living in homes later destroyed by construction and those living near the constructed highway, and link them to administrative mortality records. Displaced individuals die younger and move to worse neighborhoods within the city, relative to comparable neighbors living farther from the highway (100 to 200 meters away). Longevity also declines among those living adjacent to the highway (within 100 meters). The cost of relocation and the erosion of housing wealth seem to explain these results.

JEL Codes: N92, R23, R42, J15, I14.

Keywords: Displacement, Infrastructure, Mortality, Relocation Assistance.

*First version: July 2022. I want to thank Victor Couture, Nathan Nunn, Réka Juhász, and Kevin Milligan for their invaluable mentorship and advice during this project, in particular, and the Ph.D., in general. I am also grateful to Lukas Althoff, Nate Baum-Snow, Pierre-Loup Beauregard, Leah Boustan, Leah Brooks, Justin Cook, Tom Davidoff, Jose-Antonio Espín-Sánchez, Óscar Fentanes, Andy Ferrara, Nicolas Gendron-Carrier, Ed Glaeser, Joshua Hausman, Stephan Heblich, Torsten Jaccard, Giulia Lo Forte, Florian Mayneris, Pascuel Plotkin, Fernanda Rojas-Ampuero, Valentina Rutigliano, Hanna Schwank, Fernando Secco, Raffaele Saggio, Cristine von Dessauer, Katherine Wagner, and Ron Yang; and the audience of several seminars and conferences for their detailed comments and suggestions. I acknowledge generous financial support from the Economic History Association's Sokoloff Fellowship and University Press Early Stage Dissertation Fellowship. All errors are my own.

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1. INTRODUCTION

Infrastructure provision is a key driver of economic growth and regional development, yet its construction often requires the displacement of large numbers of local residents.¹ The World Bank estimates that infrastructure construction displaced about 20 million people worldwide each year between 2011 and 2020 (Cernea and Maldonado, 2018). These costs fall disproportionately on vulnerable communities, which have the least economic and political power to oppose these projects (Penz et al., 2011). While there is abundant evidence on the benefits of infrastructure for cities and regions, we know remarkably little about what happens to the individuals displaced by these projects and to those living in these communities, particularly in the long run.

This paper studies the non-pecuniary costs of infrastructure provision for displaced individuals and for residents of the communities where these projects are built. To do so, I examine the construction of the Interstate Highway System, the largest public infrastructure project in U.S. history, which is estimated to have displaced more than one million people between 1956 and 1970 (Schwartz, 1975). Interstate construction proceeded rapidly and without any relocation assistance, leaving those whose homes were destroyed to bear the full costs of displacement without support. Despite its scale, the identities of displaced households were never systematically recorded, making it hard to empirically study the effects of displacement. Unlike the benefits of highway construction, the costs of displacement are borne by individuals, who are often no longer living in the same location. Furthermore, construction may have affected individuals in these neighborhoods who were not directly displaced, as a broad literature outside economics has explored how highway construction reshaped neighborhoods and communities (Rothstein, 2017; Lewis, 2013; Rose and Mohl, 2012).

To address these challenges, I develop a new method to identify individuals affected by Interstate construction in the 1950 full-count census. I geocode the universe of urban residents in metropolitan areas and classify displaced individuals as those living in homes that were later destroyed by highway construction. Beyond those displaced, the resulting dataset allows me to observe all those living in affected neighborhoods prior to highway construction. The method suggests that by 1970, more than 456,000 people were displaced by Interstate construction in metropolitan areas, with millions more now living in close proximity to the highway. With this information in hand, I follow individuals after highway construction by linking the 1950 census to administrative mortality records from 1988 to 2007.

I start by documenting that construction disproportionately affected vulnerable indi-

¹The Interstate Highway System increased trade and growth in the U.S., raising welfare (Duranton et al., 2014; Allen and Arkolakis, 2014). Similar patterns arise for port development (Ducruet et al., 2024; Brooks et al., 2021) and for airport hubs (Redding et al., 2011).

viduals. By comparing individuals who lived in the same city and at a similar distance from the city center, I find that those displaced and those living within two kilometers of the highway were more likely to be Black, lived in lower-value housing, and were more likely to live in redlined areas. The finding that Black individuals lived closer to future highways is not explained by the sorting of Black individuals into city centers (Boustan, 2010), which were the same areas planners were trying to connect. I find no such differences when looking at the placement of the Yellow Book routes, the federal plans that followed similar routing guidelines. Instead, deviations from the federal plans resulted in construction disproportionately affecting Black individuals. These findings are consistent with historical accounts documenting that state officials deliberately used highway construction as a way of displacing Black individuals from their neighborhoods (Rose and Mohl, 2012).

To identify causal long-run effects of Interstate construction, I compare displaced individuals and individuals living near highway routes with observationally similar neighbors living farther away. This exercise faces a tradeoff: selecting a comparison group living far from the highway ensures that comparison households are not affected by construction, but may result in a comparison group that is very differently selected. I conduct an empirical exercise that provides guidance for the research design. By looking at how long-run outcomes vary with distance to the highway, I show that negative spillovers from highway construction are localized among those living within 100 meters. In contrast, I find no effect for those living beyond that distance. As a result, I compare those displaced and those adjacent to the highway (within 100 meters) to unaffected neighbors living farther away (between 100 and 200 meters). After accounting for factors that shaped routing decisions, namely race and proxies for household income, I show that those displaced and those adjacent to highways were well-balanced relative to those living 100 to 200 meters away, not only on a broad set of pre-construction characteristics, but also on changes in characteristics from 1940 to 1950.

Beyond the main comparison to the neighbors living farther away, I implement two complementary research designs, as well as a battery of robustness checks, to address the idea that the results could be driven by other potentially correlated factors. First, I match those displaced and those adjacent to individuals living more than 2,000 meters from any built highway, using propensity score matching estimated from individual, household, and neighborhood characteristics. If spillovers affected neighbors in ways not captured by the distance analysis, these effects would have to extend beyond 2,000 meters, well beyond what the literature has found (Currie and Walker, 2011). Second, I use individuals who would have been displaced by, or lived adjacent to, federal routes that were planned but not built. Similar to those affected by construction, these individuals lived along roads and in communities deemed suitable for routing. Thus, the unobserved characteristics that led them to live in these areas should be balanced across the two groups. Finally, as a

robustness exercise, I use those living along major roads and those living between 500 and 600 meters from the highway as comparison groups. These exercises yield very similar results to the close neighbors comparison, ruling out concerns about omitted variable bias and spillovers and lending credibility to the causal interpretation of the results.

The main findings show that highway construction imposed high non-pecuniary costs on displaced individuals and on those living adjacent to the highway. Interstate construction reduced longevity for both treated groups, with displaced individuals experiencing a one-month reduction in life expectancy and those adjacent to the highway dying about twenty days earlier than their unaffected neighbors. When analyzing ex-ante predicted longevity constructed from detailed demographic and socioeconomic characteristics, I find no such effect, suggesting that these results are indeed due to highway construction rather than different underlying mortality risk. Because treatment is assigned based on the 1950 address, individuals could have moved out before construction began, which attenuates the estimates toward zero. I develop a correction that uses the predicted probability of remaining in the same home to scale the treatment, finding effects 2 to 2.7 times larger.

Highway construction also shaped where individuals lived. I study spatial mobility using individuals' last recorded residence, and characterize their neighborhood by constructing a quality index from characteristics measured in the 2000 census. By the time of death, nearly all displaced individuals had moved out of their 1950 neighborhood and relocated to lower-quality neighborhoods within the same city. Individuals living adjacent to the highway were also more likely to leave their original neighborhood, but I find no effect on either their probability of remaining in the city or the quality of their neighborhood at death. This contrast suggests that construction forced displaced individuals to move when they would otherwise have stayed in their homes, while those adjacent to the highway *chose* to move out. In this way, the findings are consistent with the Rosen-Roback model, in which inframarginal residents would attain lower utility if they were forced to move (Rosen, 1974; Roback, 1982).

These costs were not distributed equally across racial lines. Although construction disproportionately affected Black individuals, I find that the reduction in longevity for both treatment groups comes entirely from white individuals. I show that the racial difference cannot be explained by pre-existing differences between Black and white individuals. Reweighting white individuals in the sample to match Black individuals on gender, age, occupational score, and homeownership does little to shrink the effects. Instead, I find that most of the racial gap comes from differences in redlined status before construction. While displacement helped Black individuals relocate out of redlined areas, it hindered white individuals living in non-redlined areas. Therefore, displacement affected Black and white residents in opposite directions.

I test several explanations for why displaced and adjacent individuals died younger, and find that differences in labor market conditions and in neighborhood quality are unlikely

to account for the estimated effects. I find no differences in income or unemployment in the neighborhood where each individual died, measured in the decades they turned 40, 50, and 60. Displaced individuals and those adjacent did not relocate farther from the city center, where jobs were concentrated at the time. Moreover, the estimates are unchanged when I control for the quality of the origin neighborhood in 1950. Destination neighborhoods were not worse to begin with, as the decline in the quality index arises only decades after individuals moved. Lastly, differences in air pollution across destinations are too small to explain the effects.

Instead, I find that the mortality effect can be explained by the erosion of housing wealth for adjacent individuals but not for those displaced. Using linked 1940 and 1950 censuses to identify household tenure, I show that renters account for the entire longevity effect among those displaced. For those adjacent, the pattern reverses, with the effect coming from homeowners. For these individuals, the negative externalities arising from highways likely depressed the value of their properties. Consistent with this explanation, I find that the effect is concentrated in non-redlined neighborhoods, with no effect in redlined areas where there was little property value left to lose.

For displaced individuals the mortality effect was driven by the cost of relocation. Historians have argued that this process imposed not only an economic cost on those displaced, who had to pay for their move, but also an emotional cost, as established communities were destroyed (Rose and Mohl, 2012; Lewis, 2013). A change in policy occurred during the early 1960s when states gradually adopted laws providing relocation payments to displaced tenants. I assemble a dataset on the year in which each state enacted its first relocation law and find that, after enactment, those displaced experienced a gain in longevity that offsets the cost of displacement. In contrast, adjacent individuals were not entitled to compensation and show no such reversal, ruling out the concern that highways were later built through different communities. Finally, and although I cannot pin down the exact mechanism by which relocation assistance worked, I find that those displaced were three times as likely to relocate to public housing. This result suggests that the longevity reduction for those displaced may be explained by their need for housing security.

This paper contributes to the literature on the long-run effects of displacement on individuals by studying displacement resulting from policy decisions. While the literature has documented positive effects of displacement on education, earnings, and health when it encourages individuals to move to higher-opportunity locations or to invest in mobile assets (Becker et al., 2020; Nakamura et al., 2021; Deryugina et al., 2018; Deryugina and Molitor, 2020; Sarvimäki et al., 2022), I find that displacement generated by highway construction had large negative effects on the longevity of those displaced.² An important

²These papers study unanticipated events such as natural disasters or wars, which involve large changes to local economies with substantial general equilibrium effects, and often feature monetary assistance or

objective of ongoing research in this literature is to understand the sources of these heterogeneous effects of displacement and, in particular, how to mitigate them and protect affected individuals. My results suggest that the negative effects of displacement are not driven by changes in local economic conditions, since displaced individuals overwhelmingly remain in the same city, but rather by the disruption of established living arrangements in the absence of relocation assistance. Understanding the non-pecuniary costs of displacement caused by eminent domain is important, as the practice remains highly contested, particularly since *Kelo v. City of New London* (U.S. Supreme Court, 2005), which upheld its use for private projects deemed to benefit the public. By separating the effect of forced relocation from relocation assistance, I provide evidence on both the consequences of displacement and the role of policy in mitigating them.

I also contribute to a growing literature on the rising costs of infrastructure provision in the United States (Liscow, 2025; Mehrotra et al., 2024), which has focused on the budgetary costs of construction. I show that the early years of the Interstate program carried large non-pecuniary costs that these accounts leave out. A back-of-the-envelope calculation puts the mortality cost alone at roughly \$11,400 per displaced individual in 2010 dollars, or more than \$5.2 billion added to the cost of early Interstate construction, a lower bound that excludes the costs borne by nearby residents and by disrupted communities. That relocation assistance offsets much of this cost speaks to the design of compensation policies, which Brooks and Liscow (2023) identify as a significant driver of rising construction costs.

Furthermore, this paper contributes to the extensive literature on the consequences of Interstate construction (Baum-Snow, 2007; Duranton et al., 2014; Herzog, 2021; Brinkman and Lin, 2024; Bagagli, 2023; Valenzuela-Casasempere, 2024; Williams, 2024). While this literature focuses on the places that received highways, I study the consequences for individuals affected by construction. The distributional costs I document, for displaced individuals and for their neighbors alike, complement the heterogeneous aggregate welfare gains documented in previous work (Weiwu, 2025).

Finally, this paper contributes to the literature on the racial legacy of U.S. government policies, which has documented how federal programs narrowed income gaps (Derenoncourt and Montialoux, 2020; Aizer et al., 2020) or reinforced existing inequalities through their implementation (Althoff and Szerman, 2025). To the best of my knowledge, this is the first systematic quantitative evidence on the role of race in the placement of U.S. highways. Carter (2023) shows that Interstates in Detroit were routed through low-value properties, while I use micro-level data covering all metropolitan areas.

other support for affected individuals. By contrast, I study a policy-induced displacement that affected a small share of the population in each city and initially offered no relocation assistance. Rojas-Ampuero (2025) also finds negative effects of displacement, showing that slum-dwellers relocated to the city's outskirts die younger than those resettled within the same community.

2. HISTORICAL CONTEXT

This section provides historical context on the Interstate Highway System, the factors that shaped routing decisions, and the consequences for displaced households.

2.1 The Interstate Highway System

The Interstate Highway System was established by the Federal-Aid Highway Act of 1956. The bill authorized more than 70,000 kilometers of highways, funded 90% by the Federal Government, while leaving routing decisions to state and local officials. The legislation included no provisions for the relocation of individuals displaced by construction beyond condemnation payments, largely because policymakers feared these provisions would be too costly (Lewis, 2013).³ To facilitate congressional approval, the Federal Government released a set of proposed urban routes for 100 cities in the “Yellow Book.”⁴ These routes reflected federal engineers’ efforts to create a connected national network at minimum cost and were usually routed through primary roads. Consequently, they were not influenced by local political considerations beyond geography and land values (Weiwu, 2025). The Yellow Book routes, however, were only advisory. State highway agencies had full authority over the final placement of urban interstates, which allowed state and local officials to use highway placement to push for their own political priorities (Rose and Mohl, 2012, pg. 97).⁵

State officials frequently envisioned highways as instruments for clearing areas labeled “blighted,” often targeting predominantly Black neighborhoods (Guerra, 2025). In Miami, for example, Interstate 95 was routed directly through Overtown, the center of the city’s Black economic and cultural life. Planners rejected an alternative route along an abandoned railway right-of-way that would have minimized displacement (Rothstein, 2017).⁶ Contemporaneous assessments predicted that interstate construction would displace more than one million people, disproportionately affecting Black households (Rose and Mohl, 2012, pg. 96).

Because the 1956 Act made no provision for relocation, what a household received

³In 1956, Arthur Burns, chairman of the Council of Economic Advisers, warned that compensating displaced residents would be prohibitively expensive, as the highway program was expected to evict nearly one hundred thousand people per year (Archer, 2020).

⁴The report’s official title is “General Location of National System of Interstate Highways, Including All Additional Routes at Urban Areas.” Appendix Figure A.11 illustrates the maps for Atlanta, Detroit, Miami, and New Orleans.

⁵Appendix Figure A.12 compares planned and built highway networks for Atlanta, Detroit, Miami, and New Orleans. While constructed routes broadly matched the intended origins and destinations, there were notable deviations in their precise alignments relative to the initial plans.

⁶Similar patterns emerged in cities with relatively small Black populations. In St. Paul, Minnesota, Interstate 94 cut through the city’s Black neighborhood, displacing one-seventh of its African American residents. As one critic observed, “very few Black individuals lived in Minnesota, but the road builders found them” (Rose and Mohl, 2012, pg. 108).

when its home was destroyed depended entirely on tenure and the value of the property. Homeowners were paid “just compensation” at the fair market value of the real estate, an amount families frequently argued was below what their homes were worth (Rose and Mohl, 2012).⁷ Tenants, who made up the majority of residents in seized homes, received nothing (U.S. Department of Transportation, 1967). No payment covered the practical costs of relocation, such as the search for replacement housing, security deposits, or moving expenses. No agency was required to help families find a new home. Displaced households therefore financed and organized their own relocation, whatever their means.

The costs of construction also extended beyond the destroyed housing. Neighborhoods at the time concentrated family ties, churches, corner stores, and mutual aid, generating a strong sense of community that highway construction tore apart. Residents of East Tremont, split by the Cross-Bronx Expressway, recalled that the construction “left them with feelings of being isolated, left alone, that no one cared” (Caro, 1974). Ken Welch, the first Black mayor of St. Petersburg, described how I-275 destroyed the close-knit community where he grew up: “Folks were looking out for one another. When people were dispersed and sent all around the city, you lost that sense of community” (Briggs, 2023c). Some families avoided dispersal by moving in with relatives, but this strategy often came at the cost of overcrowding and strained household resources (Archer, 2020).

Households that were never displaced also bore costs. For those who remained beside the new road, the highway brought noise and pollution and erected walls between residents and the churches, schools, and stores on the other side (Schindler, 2015). In St. Petersburg, a former resident recalled that “the interstate highway was just a few blocks from our church. The highway prevented some people from going there” (Briggs, 2023c). Sometimes the institution itself did not survive. Moses Freeman, a resident of Chattanooga, Tennessee, recalled the fate of his community’s churches: “Churches were the basis of institutional life. [...] The freeway knocked all that down, dispersing everyone in different directions. [...] My church had to flee the area and go to the outskirts of a previously all-white neighborhood” (Briggs, 2023a).

Opinions diverged over the consequences of displacement. Supporters argued that relocation would move families out of deteriorating neighborhoods and into better housing, claiming that highway construction “launched otherwise contained Black folk into the suburbs” (Connolly, 2014, pg. 282–283). Critics countered that relocation rarely produced such benefits. Highway construction occurred amid peak levels of racial segregation (Cutler et al., 1999), sharply limiting the housing options available to displaced Black households. At the same time, interstate construction destroyed a large stock of affordable housing, worsening the constraints facing low-income families (Archer, 2020). Combined with

⁷In St. Paul, one displaced homeowner fought “for as long as he was financially able” to overturn the valuation placed on his property, “only to be defeated at every stage.” Part of the valuation rested on the state’s assertion that the homes were “inferiorly constructed,” yet ten years after acquisition the homes were still standing (Briggs, 2023b).

the absence of financial and advisory support, these constraints often pushed displaced families to the urban periphery or to emerging “second ghettos,” areas with lower-quality housing and limited access to employment (Archer, 2020).

These hardships did not go unnoticed. Local opposition to urban highway construction mounted, and Congress responded with relocation assistance requirements, though implementation was gradual and uneven. The first requirements arrived with the Federal-Aid Highway Act of 1962, seven years after construction began.⁸ The Act required states to offer relocation payments and advisory services as a condition for receiving federal highway funding. However, the legislation did not take effect until July 1965, when relocation assistance became available with a maximum payment of \$200 for displaced families. Even then, adoption varied substantially across states. By mid-1965, only fifteen states had established programs, as shown in Appendix Table F.1. States faced significant implementation barriers, including constitutional prohibitions on certain types of payments and limited administrative capacity for managing relocation services, as well as a lack of urgency from administrative agencies (Highway Research Board, 1967). These obstacles generated staggered adoption of relocation assistance programs up until 1971, when the Uniform Relocation Assistance and Real Property Acquisition Act mandated standardized procedures across all federal projects.⁹

Evidence from the Highway Research Board (1967) evaluation shows that the early program had limited reach. Of the nearly fifty thousand people displaced between April 1965 and October 1966, only 37% sought advisory assistance, and only 15% secured housing through this aid. Average relocation payments were roughly \$100, an amount insufficient to cover the costs of moving (Highway Research Board, 1967). In addition, states differed substantially in when they began implementing relocation payments and in the generosity of the support offered.¹⁰ For example, New York and West Virginia authorized relocation payments shortly after the 1962 Act, while states such as Texas and Indiana delayed implementation until the end of the decade, creating substantial cross-state differences in the support available to displaced households. By the time relocation programs became operational, most urban interstate segments had already been constructed, and the majority of displacement had already taken place.

Taken together, the qualitative evidence suggests that early highway construction imposed substantial and uneven costs on displaced households and on the neighbors who remained beside the road, with limited institutional support to mitigate these effects.

⁸Throughout early construction, homes were acquired by eminent domain. State highway departments negotiated purchases or condemned properties to secure right-of-way.

⁹Congress expanded relocation assistance in 1968 before the comprehensive 1971 reform.

¹⁰As of January 1, 1967, thirty-three states and the District of Columbia had authorized the payment of moving expenses for displaced households. Among these, twenty-one states aligned their payments with the reimbursement provisions of the Federal-Aid Highway Act of 1962, while thirteen adopted either more generous or less generous payment levels (Highway Research Board, 1967). Appendix Table F.1 reports the dates at which states introduced relocation payments.

3. DATA

The major empirical challenge is identifying individuals displaced by Interstate construction and observing their long-run outcomes. To do so, I geocode address information in the 1950 census for all urban residents in the 168 metropolitan areas and overlay the highway network to determine which individuals were living in homes under the right-of-way of a highway. I then link these individuals to administrative mortality records from 1988 to 2007.

3.1 Measuring highway-induced displacement

No systematic records of households displaced by highway construction exist. Therefore, I develop a novel approach to identify displaced individuals using historical address information from the 1950 U.S. census. Throughout this paper, I define displacement as those individuals whose homes were destroyed by highway construction, as they were living under what became the highway right-of-way.

The 1950 census recorded the address of each household living in urban areas. Following the guidelines developed by Logan and Zhang (2018), I standardize and geocode these addresses to obtain latitude and longitude coordinates for each urban household in the counties containing the 168 Standard Metropolitan Areas (SMAs) in 1950.¹¹ Within this set, I successfully geocode 58.53% of total households.¹² Panel (a) of Appendix Figure A.13 presents an example of the geocoded data, showing the location of homes in a neighborhood in Cleveland, Ohio, in 1950. Appendix Section A provides additional details on the sample and the geocoding process, as well as the performance by state.

Identifying displaced households requires the location and opening date of each highway segment. This information comes from the Federal Highway Administration's PR-511 database linked to the built highway network from OpenStreetMap (Baum-Snow, 2007; OpenStreetMap, 2017). To estimate the right-of-way acquisition for highway construction, I assume a two-lane highway had a width of 150 feet (45.7 meters), the minimum width used in the 1950s (Weingroff, 2017).¹³ For segments with more than two lanes, I add 12 feet (3.66 meters) per additional lane, equal to the average lane width (Federal Highway Administration, 2007).¹⁴ Using these width assumptions, I construct a buffer around

¹¹The geocoding is done using ArcGIS's Street Map Premium 2024 locator.

¹²Appendix Table A.3 shows that of the 66.7 million individuals, only 49.1 million have a valid street address. Therefore, the geocoding rate jumps to 73.6% of the individuals with a valid address. The sample of geocoded individuals is representative of the urban population in 1950, with similar distributions of race, gender, age, and employment.

¹³The modal and median number of lanes in the sample is two.

¹⁴If actual right-of-way widths exceeded my assumptions, some displaced individuals would be misclassified as adjacent, and the estimated spillover effects on adjacent individuals could partly reflect this misclassification. However, the estimated spillovers are robust to dropping individuals living within 25

each highway segment representing the right-of-way area and calculate each household's distance to the nearest segment. Displaced households are those whose 1950 residence fell within this right-of-way area. I apply the same procedure to planned highway routes from the 1955 Federal engineering maps, also known as the Yellow Book, which covered 100 cities (Bureau of Public Roads, US, 1955). Thus, for each household I observe whether it was displaced by highway construction, the distance to the nearest highway segment, and whether it lay within the planned route. Panel (b) of Appendix Figure A.13 shows the highway network in Cleveland, Ohio.

Identifying displaced individuals depends on accurate coordinates. Because highway construction destroyed street segments, and street names or numbering may have changed over time, using a modern geocoder to locate historical addresses can misclassify individuals' locations. I address this concern in three ways. First, I restrict the sample to addresses with high-quality matches, i.e., geocoding score above 85, which reduces classification error.¹⁵ Second, I construct a historical geocoder for nine cities using 1940 street grids that predate highway construction, providing benchmark coordinates from pre-highway street grids.¹⁶ The two geocoders yield highly correlated locations, with more than 65% of matches falling within 40 meters (Appendix Figure A.3) and with no directional bias in discrepancies between the two geocoders (Appendix Figure A.4). At the level of treatment classification, Appendix Figure A.5 shows that the two geocoders agree closely across distance bands, with more than 70 percent of observations classified in the same 100-meter bin. Third, in Appendix Figure A.1, I re-estimate the main results using the historical geocoder, yielding quantitatively similar estimates.

Together, these checks suggest that geocoding quality is high and that the remaining differences reflect classical measurement error, which attenuates the estimates. More details on these analyses can be found in Appendix Section A.4.

3.1.1 Outmigration before highway construction

Because treatment status is assigned based on residence in 1950 while construction occurred between 1956 and 1970, some individuals may have left their homes before construction started. If those displaced by construction had a higher underlying probability of moving out of their home, pre-construction moves would pose a threat to the analysis. Two considerations suggest this is unlikely. First, the sequence of events leaves little room for anticipation before 1950. I observe individuals' residences in April 1950, nearly three years before President Eisenhower took office and six years before the Federal-Aid Highway

meters of the highway (Appendix Figure A.1).

¹⁵ArcGIS's Street Map Premium provides a geocoding score from 0 to 100 that indicates the quality of the match. A score above 85 indicates a high-quality match, such as an exact match to a street segment or parcel. Appendix Figure A.1 shows that the results are robust to increasing the quality threshold to 98 out of 100.

¹⁶The historical street grids were created by the *Urban Transition Historical GIS Project* and are available at <https://s4.ad.brown.edu/Projects/UTP2/citymaps.htm>. Accessed February 10, 2026.

Act authorized the Interstate system. Moving in anticipation of construction would have required individuals to foresee the outcome of the 1952 election, the Eisenhower administration's prioritization of the Interstate Highway System, and the eventual location of Interstate routes. Passage of the Act remained uncertain even after Eisenhower took office, as an earlier highway proposal failed in the Senate in 1955.

Second, the quantitative evidence suggests that those displaced by construction did not have a different ex ante probability of moving out of their residence. Using the 5% sample of the 1960 census, I estimate the probability of residential mobility at 5-, 10-, and 20-year horizons.¹⁷ I then use the fitted model to predict mobility for each individual in the 1950 sample based on their observed characteristics. If those displaced had a different underlying probability of moving out before construction, this should be captured by this analysis. I find no such evidence. Appendix Figure A.6 shows that neither those displaced by the highway nor those living next to it had a different probability of moving out over 5-, 10-, and 20-year horizons.

I bound the bias that pre-construction moves could introduce into the estimated effects. Appendix Section A.6.1 presents the derivation. Because misclassification of treated individuals attenuates the estimates toward zero, the true effect is top bounded by the estimate scaled by the inverse of the probability that those affected remained in their residence. Using the linked 1940-to-1950 censuses to calculate these probabilities, I find that the true parameters lie between the estimates and 3.05 times that value. Section 6.1 corrects the estimates by the probability of staying, showing that the effects are between 2 and 2.7 times the uncorrected values, well within the bound. The estimates reported are, therefore, lower bounds on the true effects.

Finally, the results are larger for the subpopulations most likely to have remained in their homes. Appendix Figure A.9 shows that the coefficients grow in magnitude with the age of the cohort, with an estimated effect roughly twice as large for those born before 1925. Older cohorts are less mobile and more likely to have stayed in the same home through construction, so this pattern is what one-sided misclassification predicts. That these coefficients fall within the bound derived above is reassuring for the magnitude of the estimates. The estimated effects are also larger for displacement occurring before 1965 (Appendix Figure A.10), when less time had elapsed for individuals to leave their homes. Together, these results suggest that although outmigration before construction played a role, the scope for attenuation does not substantially alter the magnitude of the estimated displacement costs.

¹⁷The 1960 census is the first to report the year in which the family moved into their home. Because this variable is reported in bins, I can only estimate the probability of moving out of the same home over 5-, 10-, and 20-year horizons.

3.2 Linkage to administrative mortality records

To observe individuals after highway construction, I link the 1950 census to the Numerical Identification (NUMIDENT) database. These administrative mortality records contain detailed information on age at death, reliance on Social Security, and nine-digit ZIP codes of last residence. The database covers nearly all deaths among American citizens aged 65 and older between 1988 and 2007, though coverage is incomplete before 1988 (Breen and Goldstein, 2022). Accordingly, the analysis focuses on individuals 65 or older who died between 1988 and 2007.¹⁸

One advantage of NUMIDENT records is the inclusion of deceased parents' names, which allows the linkage of women who changed their last name after marriage. Additionally, nine-digit ZIP codes provide granular location indicators that refer to "a segment or one side of a street" (U.S. Postal Service, 2024). These ZIP codes allow me to observe the socioeconomic characteristics of neighborhoods at death by linking to the National Historical Geographic Information System (NHGIS) for the year 2000, near the midpoint of the mortality window. More details on the linkage process are provided in Appendix Section A.9.

3.3 Attrition and sample representativeness

The final sample corresponds to the set of individuals observed in 1950 whom I can both geocode and link to a death record. Of the 66.7 million individuals in the metropolitan 1950 census, only 49.1 million have a valid address. Of them, I am able to geocode 40.2 million individuals, of whom 3.52 million are linked to a NUMIDENT death record, roughly 8.9% of those geocoded (Appendix Table A.3). There are two distinct sources of attrition in play that can affect the representativeness of the final sample. The first is a failure to geocode the 1950 address or to match the record to the NUMIDENT. The second is the coverage of the NUMIDENT itself, which records deaths only between 1988 and 2007 and only at ages 65 and older. An individual therefore enters the matched sample only if they were young enough in 1950 to reach age 65 or older within this window, which mechanically makes the linked sample younger, with a mean age of 26.6 in the linked sample versus a mean of 33.1 for the full sample of urban residents. Since many women change their last names when marrying, the female share of the linked sample is lower than in the full sample. This is a known limitation of the linking algorithms (Abramitzky et al., 2024). These differences reflect limitations of the linking algorithm and the coverage window rather than selective migration. The matched sample is otherwise similar on race, employment, and household socioeconomic status.

¹⁸The last row of results in Appendix Figure A.16 shows that including individuals who passed away prior to 1988 does not alter the results.

Attrition would pose a threat to the validity of the results if it differed by treatment status. I do not find evidence of different linkage rates in the sample used. Appendix Table A.2 regresses an indicator for being linked to the NUMIDENT on the displaced and adjacent indicators, using the full sample of geocoded individuals and the controls of the main specification. Without controls, displaced individuals are 0.17 percentage points less likely to be linked, a gap that falls to 0.087 points once I condition on metropolitan area, race, birth year, gender, and the household head's occupational score. Against a base linkage rate of 8.9%, this gap is less than 1% of the linkage rate and is at most marginally significant (p -value = 0.098). Adjacent individuals are statistically indistinguishable from their unaffected neighbors in every specification. These differences are far too small to account for the displacement effects I estimate, and because the comparison is within neighborhood among individuals of the same race, gender, and age, the controls absorb any residual gap.

I perform two additional robustness checks to ensure that the results are not driven by the sample representativeness. First, I show that the imbalance on the linked sample does not drive the estimates. Following Bailey et al. (2020), I reweight each observation by the inverse of its estimated probability of being geocoded and linked, which restores the matched sample to the observable distribution of the 1950 population.¹⁹ The estimates are virtually unchanged, suggesting that the results are not driven by the sample of individuals who were successfully linked (Appendix Figure A.16). Second, I show that the results are not driven by the coverage of the mortality records. The NUMIDENT undercounts deaths among younger cohorts, which could bias the estimates if the undercount differs by treatment status. Reweighting the results using the mortality-adjusted weights from Breen et al. (2023), which correct for this differential coverage, shows that the results are robust (Appendix Figure A.16). Together, these two re-weighting exercises suggest that the results are not driven by the sample representativeness or the coverage of the mortality records.

4. WHO WAS AFFECTED BY HIGHWAY CONSTRUCTION?

The historical literature has argued that Interstate construction disproportionately affected low-income and Black communities (Rose and Mohl, 2012; Lewis, 2013; Rothstein, 2017; Archer, 2020). Yet, no systematic records of displaced individuals exist, leaving both their number and socioeconomic characteristics largely unknown.²⁰ The dataset I build allows

¹⁹The probability is calculated as a probit on race, age, gender, marital status, immigrant status, household size, number of children, region, and the household head's occupational score and age, including interactions between race, gender, and age. Weights are winsorized at the 1st and 99th percentiles to avoid extreme weights.

²⁰Federal and state agencies did not collect data on who was displaced, primarily because relocation payments were not required and because accurate counts would have imposed legal liability on planners

me to address both questions directly. First, I estimate that Interstate construction displaced 456,405 individuals between 1956 and 1970, or an average of 30,427 per year. These figures represent a lower bound, as the sample covers only urban residents in metropolitan areas and geocoding of the full household sample is incomplete. The estimates, however, align closely with official figures reported in 1967, when the federal government estimated that highway construction relocated an average of 33,070 individuals per year between April 1965 and October 1966 (U.S. Department of Transportation, 1967, pg. 2). The correspondence between my estimates and the official figures validates the accuracy of the geocoding procedure and, by extension, the methodology used to identify those displaced. Second, I use these records to describe the socioeconomic profile of displaced individuals and their communities, something the historical literature has been unable to do systematically.

To characterize how pre-construction socioeconomic characteristics vary with distance from the highway, I estimate the following equation:

$$y_i = \beta_0 \text{Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i \quad (1)$$

where y_i is an individual outcome measured in the 1950 census, Displaced_i indicates individuals whose homes were under the right-of-way path, and Bin_i^k are indicators for 100-meter distance bins measured from the future highway. For instance, Bin_i^{500} equals one if the individual lived between 400 and 500 meters from a future Interstate in 1950. This approach treats individuals symmetrically regardless of which side of the highway they lived on, and avoids the misclassification that can arise from using census tracts, which vary substantially in shape and area. I set the cap at 7,500 meters (4.7 miles) and use individuals living between 7,400 and 7,500 meters as the comparison group. In my baseline specification, $\mathbf{X}_i$ includes metropolitan area fixed effects and the distance from the individual's home to the city center, since highways were built to connect downtown areas and Black residents disproportionately lived near city centers during this period (Boustan, 2010). Appendix Figure B.2 presents the results without conditioning on distance to the city center. Standard errors are clustered at the metropolitan area level.

Figure 1 shows that Interstates were routed closer to where Black families lived. In each panel, the blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. Panel (a) shows that displaced residents were 7 percentage points more likely to be Black than the comparison group, and that this gap remains above 5 percentage points for residents living within 2,000 meters (1.25 miles) of a future Interstate. Panel (b) shows that the same gradient does not hold for foreign-born residents, indicating that the racial pattern is not driven by highways passing through minority communities generally.

and authorities (Schwartz, 1975, pg. 235).

The results also suggest that Interstates were built through lower-income communities. Panel (c) shows no meaningful differences when looking at employment rates. Panel (d) shows that displaced residents earned approximately 0.05 log points less in annual earnings than the comparison group, with the gap decaying quickly with distance. In Appendix Figure B.1, I show that the income gap is not driven by differences in hours worked. Most of the income gap appears to be driven by differences in the type of jobs held, as residents near future Interstates were employed in occupations with lower occupational income scores.

State highway planners sought to minimize right-of-way acquisition costs, a motive that predicts routing highways through neighborhoods with lower land values. While the 1950 census did not collect housing information, I draw housing characteristics from the 1940 census. I use census-to-census linkages provided by Ruggles et al. (2025) and keep individuals living in the same house in 1950 as in 1940.²¹ Figure 2 presents the results on home values, rents, homeownership, and redlining status. Panels (a) and (b) show that Interstates were built in communities with a lower cost of land, consistent with planners minimizing acquisition costs. Displaced properties were roughly 0.25 log points cheaper than those of the comparison group, whether measured by home values or rents, and the gap declines steadily with distance from the future Interstate. Panel (c) shows that the homeownership rate did not vary with distance to the highway. Finally, Panel (d) shows that highways were also more likely to be routed through redlined neighborhoods, with displaced individuals and those living within 1,000 meters of a future Interstate being 10 percentage points more likely to live in a redlined area. There are two considerations to keep in mind when interpreting these results. First, the 1940 census reports property values and rents but does not record housing characteristics such as unit size, so I cannot construct a quality-adjusted price measure. Second, home values in affected areas may have diverged from the rest of the city between 1940 and 1950 if housing markets anticipated the construction of highways.

How avoidable was routing highways closer to low-income and Black communities? The sorting of Black families into low-rent downtown areas, combined with the stated policy goal of connecting city centers to the broader highway network, suggests that any counterfactual Interstate network might also have been routed closer to Black communities. To examine this possibility, I analyze the characteristics of communities along the Yellow Book planned routes in Appendix Figure B.3. Like the network that was built, these plans likely followed major roads and sought to minimize construction costs. I find no evidence that planned highways were disproportionately routed through Black or immigrant neighborhoods, with estimates within 2,000 meters of a planned route that are small in magnitude and statistically insignificant. For the subsample matched to 1940,

²¹Individuals are considered to live in the same house across censuses if their residences are within 30 meters of each other. The results do not change when I use more restrictive or more lenient thresholds.

communities near planned highways did not have lower home values, though they were more likely to fall within redlined areas.

I then compare the built and planned networks directly by running a horse race between these two routes. Appendix Figure B.4 plots the difference between the coefficients for the built and the planned network, with 95% confidence intervals. The two routings differ in their proximity to Black communities, while the differences in home values, income, and redlining status are smaller and not statistically significant. Together, the two exercises indicate that highways were built closer to where Black individuals lived in 1950 than the engineering plans alone would predict.

The previous analysis cannot disentangle the role of each of these factors in highway placement, as they are likely correlated. To address this and incorporate geographic controls such as terrain slope, I study which census tract characteristics in 1950 predict future highway construction. Columns (1) to (5) of Appendix Table B.1 show that highways were more likely to be built through neighborhoods with a larger Black population share, lower land values, greater proximity to the city center, and a planned Yellow Book route. The relationship between Black population share and highway construction remains positive after accounting for land values, distance to the city center, and local geography. As in the individual-level evidence, column (6) shows that planned routes are not predicted by the racial composition of the tract. More details on this analysis are available in Appendix Section B.2.

Taken together, these analyses indicate that Interstate placement disproportionately affected low-income and Black communities, through both displacement and the bisection of their neighborhoods. Since planned routes show no such pattern, the evidence suggests that deviation from the federal plans played a role in the disproportionate effect on these communities.

5. EMPIRICAL STRATEGY

My goal is to estimate the causal effect of highway construction on the long-run outcomes of displaced individuals and their neighbors. The results of Section 4 and the historical evidence indicate that, while the neighborhoods where highways were placed differed from the rest of the city, location decisions within those neighborhoods were driven by the availability of large roads that could be converted into highways. Hence, the preferred empirical strategy compares individuals affected by highway construction with those in the same neighborhood who are not affected, conditional on 1950 characteristics. The intuition behind my strategy is that individuals within these neighborhoods were similar to one another, thus displacement varied according to idiosyncratic factors conditional on the decision to live in that neighborhood.

The research design, therefore, needs to define a distance threshold beyond which individuals are considered unaffected by their proximity to the highway. I follow a data-driven approach to select this cutoff. Appendix Section C.1 studies how long-run outcomes differ with distance to the highway. The analysis shows that those living within 100 meters of the highway die at a younger age and are less likely to be in the same community at the time of death. These findings motivate including them as a second treatment group in all analyses, which I refer to as those adjacent to the highway. The results also suggest that spillovers are highly localized, with no significant effect beyond 100 meters. Therefore, I use individuals living 100 to 200 meters from the newly constructed highway as the comparison group. These individuals chose to live in the affected communities and, for idiosyncratic reasons, were located farther from the right-of-way.²² The final sample consists of individuals observed in the 1950 census who were either displaced by highway construction or living within 200 meters of these projects, linked to administrative mortality records from 1988 to 2007.

The comparison requires controlling for the pre-construction characteristics that could otherwise confound the effects between displaced individuals and their neighbors. In all specifications, I control for race, gender, birth year, metropolitan area of residence, and the household head's occupational score. Ideally, I would also control for earnings and wealth, such as income, homeownership, and property values. However, these variables are not systematically available in the 1950 census. Income was recorded for only one in six individuals, and housing characteristics are not available at all. I therefore proxy a household's socioeconomic status with the household head's occupational score, which is available for everyone and is positively correlated with income and wealth. Appendix Figure C.10 shows that the occupational score is positively correlated with income, homeownership, and property values in the 1940 census.

I estimate the impacts for those affected by highway construction by estimating the following specification:

$$y_i = \beta_1 \text{Displaced}_i + \beta_2 \text{Adjacent}_i + \mathbf{X}_i' \Gamma + \varepsilon_i. \quad (2)$$

In equation 2, y_i is an individual long-term outcome. The treatment status, Displaced_i and Adjacent_i , is determined by the home the individual resided in at the time of the 1950 census, not by where they were living when construction began. The coefficients of interest are β_1 and β_2 , which capture the effect of displacement and of adjacency on long-run outcomes relative to unaffected neighbors living 100 to 200 meters away. Because treatment is assigned by 1950 residence regardless of whether an individual remained until construction, these coefficients recover an intention-to-treat effect. Measurement

²²Appendix Figure C.2 shows that the results are robust to using individuals living 500 to 600 meters and 1,000 to 1,100 meters away as the control group.

error arising from out-migration before construction would bias the estimates toward zero, making the estimates a lower bound of the *true* effect. Appendix Section A.7 formalizes this attenuation and develops a correction, and Appendix Section A.6.1 bounds the true effect between the estimate and 3.05 times its value.

The vector X_i includes race, gender, and birth year dummies, the household head's occupational score, and the metropolitan area fixed effects. Identification comes from comparing neighbors within the same metropolitan area who were the same race, gender, and age. Standard errors are always clustered at the metropolitan area level.²³

Equation 2 identifies the causal effects of highway construction on displaced and adjacent individuals under two assumptions, conditional independence and the stable unit treatment value assumption (SUTVA). Conditional independence requires that, among individuals in the same neighborhood, displacement is independent of potential outcomes conditional on observables. This assumption accounts for the otherwise idiosyncratic assignment process arising from routing decisions through low-income and Black communities, and it is necessary because long-run outcomes are observed only in the post-period.

The results from Appendix Section C.1 provide evidence that SUTVA likely holds, as spillovers are highly local and the control group is unaffected by highway construction across all studied outcomes. General equilibrium spillovers from changes in market access remain possible under a counterfactual network, but they are unlikely to affect the treatment and control groups differentially. Most counterfactual networks would have connected these metropolitan areas regardless, as they were large manufacturing hubs, so any within-metropolitan-area differences in market access would have been negligible.

To assess the plausibility of the conditional independence assumption, I conduct balance tests by estimating equation 2 with pre-construction characteristics as outcomes. I standardize continuous characteristics to have mean zero and standard deviation one, and I report the estimates for discrete variables as percentage changes ($\hat{\beta}$ divided by the sample mean). Figure 3 plots the differences between treatment and control individuals across a set of pre-construction characteristics. Before adjusting for any covariates, I find that displaced individuals differ from their unaffected neighbors on two dimensions: they are more likely to be Black and to live in households with lower occupational scores. This within-neighborhood gap mirrors the routing I document in Section 4, where Interstates were disproportionately built through Black and lower-income communities. Conditional on controls, displaced and adjacent individuals resemble their unaffected neighbors across the full set of pre-treatment characteristics. Two small differences remain. Displaced individuals have around 8% lower educational attainment and live 0.1 standard deviations closer to the city center than their unaffected neighbors. Both gaps are economically small,

²³Appendix Figure D.1 shows that the results are robust to clustering at the state level.

and Appendix Figure D.1 shows that the estimates are unchanged when I control for these variables.

I also test for pre-trends by linking the 1950 and 1940 censuses using the crosswalk created by Ruggles et al. (2025).²⁴ As in the balance analysis, I estimate equation 2 using cross-decade changes in economic and demographic outcomes. I restrict the sample to those older than 10 years in 1950 for demographic variables, and those between 24 and 65 years old for labor variables. Appendix Figure C.3 shows that both displaced individuals and those adjacent are balanced in pre-trends. Displaced individuals are less likely to live in the same neighborhood in both 1940 and 1950. Consistent with the identification assumptions, I find no systematic evidence of differential economic trends between the treatment groups and the control group. Together, these results lend support to the conditional independence assumption underlying the design.

Although the two groups are balanced on observable characteristics in 1950, omitted variables may still threaten conditional independence. Major roads were both a source of pollution and more likely to be converted into highways, so the longevity of those living next to these roads could differ from that of the control group even in the absence of construction. Highways were also more likely to be routed through redlined neighborhoods, which may independently shape long-run outcomes. Many possible omitted variables, including food insecurity, crime, and access to health care, vary across neighborhoods rather than within them. Because treated and control individuals lived within a few hundred meters of each other, these neighborhood-level omitted variables would be common across treatment groups. What remains is a narrower form of selection on unobserved individual characteristics such as baseline health or wealth. If individuals in worse underlying health sorted into the blocks later taken for construction, conditional independence would not hold. As shown in Appendix Figure D.4, these groups did not differ on their predicted mortality based on pre-existing characteristics.

To lend credibility to the estimates, I present two complementary research designs that address the remaining threats to identification. The first compares displaced individuals to observationally similar individuals living far from any highway, which addresses both selection on unobservables within neighborhoods and the possibility that the nearby control group was itself affected by construction. The second uses the Yellow Book planned routes as a counterfactual, comparing those living along highways that were constructed to those living along routes that were planned but never built.

Matching. One concern is that unobserved spillovers may still affect the control group. For example, highway construction reduced the housing stock in affected neighborhoods, particularly for low-income households, which may have raised rents in the neighborhood.

²⁴Appendix Table A.3 studies the representativeness of the linked sample and shows that its differences with the full sample arise from the age restriction, as the linked sample only includes individuals older than 10 years in 1950.

To address this concern, I use propensity score matching to pair displaced and adjacent individuals with observationally similar individuals living in the same city but more than 2,000 meters from any constructed highway, matching on individual, household, and neighborhood characteristics.²⁵ The 2,000-meter threshold is based on the evidence that the effect of highway pollution on newborns dissipates above this distance (Currie and Walker, 2011). The matching algorithm is able to match 56 percent of the treated group. Appendix Figure C.4 compares matched and unmatched treated individuals, with the largest differences in redlining status and distance to the city center. The matched sample is balanced both in its pre-construction characteristics and in its trends (Appendix Figures C.5 and C.6).

Yellow Book. Another concern is that, even conditional on the neighborhood, displaced individuals differ from those living more than 100 meters away in ways not captured by the balance test. Highways were typically built downtown along primary roads, which were later upgraded to Interstates, so individuals living along these corridors were already exposed to traffic and pollution that fell off sharply within a few hundred meters. I use the highway routes planned in the Yellow Book as a third empirical strategy to address this concern. These maps covered 100 cities, were drawn to minimize construction costs, and typically followed the primary roads of the time. Therefore, the strategy uses as the control group individuals who would have been displaced had these routes been built.

A concern with this strategy is that living in downtown areas or along primary roads may have, by itself, an effect on the outcomes of interest. I test this possibility by running a placebo test comparing those who would have been displaced by a planned highway, or who lived within 100 meters of one, to those living 100 to 200 meters away from the planned route.²⁶ Appendix Figure C.9 shows coefficients close to zero and not statistically significant at conventional levels, indicating no effect of proximity to a planned highway on long-term outcomes. More details about this strategy are provided in Appendix Section C.4.

The three different research designs rely on different identifying assumptions. For an unobserved factor to overturn the results, it would have to vary within the neighborhood, survive a matched comparison to controls far from any highway, and differ between residents of built and planned but unbuilt routes. Together, these strategies leave little room for selection on unobservables to drive the estimates, lending credibility to a causal interpretation of the effects of displacement.

²⁵See Appendix Section C.3 for more details about the matching algorithm.

²⁶Planned highways consist of a unidirectional segment of the highway. I assume that planned highways had a right-of-way width of 150 feet (45.7 meters).

6. LONG-RUN CONSEQUENCES OF INTERSTATE CONSTRUCTION

This section studies the long-run effects of Interstate construction on individuals whose homes were destroyed and on those living within 100 meters of the newly built highway.

6.1 Effects on longevity

How did Interstate construction affect the wellbeing of displaced residents and their neighbors? I answer this question by studying longevity, one of the largest components of wellbeing (Jones and Klenow, 2016). The sign is not obvious *ex ante*. Given the lack of protection provided for those displaced, one would expect adverse effects. Yet, Deryugina and Molitor (2020) find that individuals forced to move after Hurricane Katrina who relocated to cities with healthier behaviors lived longer. The effect of Interstate construction on longevity is therefore an empirical question.

Highway construction reduced longevity for those displaced and for those living within 100 meters of the construction site. Figure 4 plots the estimates for displaced and adjacent individuals across the three empirical strategies, with age at death as the dependent variable in the top row. Across the three strategies, displaced and adjacent individuals die younger, and the coefficients are similar in magnitude. The similarity between the neighbors strategy and the two alternative designs suggests that the results are not driven by omitted variables or by unobserved spillovers. Because the three strategies produce similar estimates, in the rest of the paper I focus on the neighbor-comparison design. On average, displaced individuals die 0.092 years earlier than their unaffected neighbors, or about one month. For those adjacent, longevity falls by 0.066 years. These magnitudes are economically meaningful. Chetty et al. (2016b) find a clear positive relationship between longevity and income percentile in the US. The estimate for those displaced is equivalent to the difference in life expectancy between households at the 48th and 50th income percentiles, or approximately \$2,726 in annual income. This average masks a sharp difference by race. The pooled effect is driven by white individuals, and I find no reduction in longevity for displaced Black individuals (Figure 7). Section 6.4 digs deeper into the racial difference in the results.

The bottom two rows of the figure report results for survival beyond ages 75 and 80. The effect is concentrated at the older threshold, with no difference between groups at age 75. This result alleviates concerns that construction affects the probability of appearing in the linked sample. If that were the case, I would expect the same differences to appear at age 75.²⁷ The overall decrease in life expectancy comes primarily from lower survival

²⁷I find similar results when estimating a Cox Proportional Hazard Model in Appendix Section D.3. One advantage of estimating hazard rates is that it enables direct comparison with estimates for other traumatic life experiences. The estimated hazard rate for displacement is similar in magnitude to the effects the

rates beyond age 80 among displaced and adjacent individuals. The estimated effects are also economically significant. Relative to a baseline survival probability of 30.33 percent, I estimate that the probability of living to age 80 falls by 2.95 percent of the mean among those displaced and by 1.66 percent among those adjacent to the highway.

Predicted mortality. The results on mortality reflect the causal effect of displacement and proximity to the highway on individual mortality under the assumption that baseline mortality risk among those affected is uncorrelated with factors that determined highway placement. This identification assumption would be violated if individuals with higher latent mortality risk systematically sorted into the areas where highways were later built, such as those near major roads.

As a direct test of differential underlying mortality risk, I estimate how the predicted mortality risk of affected individuals varies with treatment status. To predict mortality, I estimate a saturated OLS model using 1950 characteristics of individuals living more than 2,000 meters away from any highway built after 1956.²⁸ The fitted model is then used to generate out-of-sample mortality predictions for individuals in the estimating sample (i.e., those living within 200 meters of the highway). The predicted values are highly correlated with realized longevity (see Appendix Figure D.3).

Appendix Figure D.4 reports the results of estimating equation 2 using predicted mortality as the dependent variable. The estimates are small and statistically insignificant for both displaced and adjacent individuals. The limited degree of sorting on observable mortality risk provides strong evidence that the observed effects on actual mortality are not driven by systematic construction of highways in areas with higher underlying mortality risk but rather reflect the causal impact of displacement.

Recovering the *true* costs of displacement. One concern is that residence is measured in 1950 while construction took place between 1956 and 1970, so some individuals had left their homes before construction started. The estimates therefore recover an intention-to-treat effect, with the estimates representing a lower bound of the *true* effects. Appendix Section A.7.2 shows that scaling the estimates by the inverse of the probability of remaining in the same home recovers the *true* effect on those actually displaced.

Appendix Figure D.5 reports the corrected estimates. The effect for displaced individuals moves from -0.092 to -0.250 , implying that those whose homes were destroyed died roughly three months earlier than their neighbors. The effect for those adjacent also grows in magnitude, from -0.066 to -0.204 , equivalent to around two and a half months. The correction leaves the ranking of the two groups unchanged, with the largest effect falling on those displaced. Throughout the paper I report the uncorrected estimates, which

literature has found for divorce (Sbarra et al., 2011), equal to one-third of the mortality differences between homeless and housed populations (Meyer et al., 2023), and equal to 3 percent of the mortality effect for parents who lose a child (Song et al., 2019).

²⁸The variables used are a combination of demographic and economic variables. More details can be found in Appendix Section D.4.

should be read as a lower bound on the cost borne by those actually displaced.

Taken together, the estimates in this section indicate high non-pecuniary costs of highway construction, not only for individuals whose homes were destroyed, but also for those living within 100 meters of the new road.

6.2 Effects on spatial mobility

Section 4 shows that highways were built in lower-income and redlined neighborhoods. When moving is costly, a forced move can resolve spatial mismatch (Nakamura et al., 2021), and in modern settings it has sent displaced individuals to cities with stronger labor markets (Deryugina et al., 2018). Whether highway construction did the same is an empirical question. To answer it, I use the nine-digit ZIP code of residence at death to infer spatial mobility.

Figure 5 documents substantial changes in mobility for those displaced by or living next to the newly built highway.²⁹ The top row shows that both displaced and adjacent individuals are, respectively, 2.30 percentage points and 0.65 percentage points less likely to be living at death in the same neighborhood as in the 1950 census. These effects are also sizable, equivalent to 59% and 17% of the sample mean, respectively. Construction, however, pushed individuals within the city rather than out of it. Displaced individuals are 1.51 percentage points more likely to remain in the same city, and those adjacent 0.56 percentage points.

One candidate explanation is that highways opened the suburbs to individuals otherwise confined to the city center (Connolly, 2014). If that were the case, we would expect displaced individuals to live at death farther away from the city center into the suburbs. I find no sign of it (Figure 5, last two rows). Neither displaced nor adjacent individuals are more likely to die in a suburban neighborhood, and neither moves farther from the city center.

There are several reasons why, in the context of highway construction, displaced individuals do not move outside the city. The previous literature has focused on natural disasters, which cause greater destruction in affected cities and have large general equilibrium effects. For example, employers could relocate from the affected city if they anticipate that environmental hazards will recur. Additionally, most individuals displaced by highway construction did not receive any relocation payment to cover moving costs or secure new housing, which may have prevented them from relocating farther. Finally, unlike in the other settings, displacement in this context arises from a policymaker's decision rather than an environmental catastrophe. I am open to the possibility that the consequences differ when the source is not an environmental catastrophe, given the vast anecdotal evidence that highway-induced displacement took a substantial psychological toll on those

²⁹The figure includes the results of the preferred neighbor-comparison design. Appendix Table D.2 shows the results for the three strategies.

affected (Caro, 1974; Archer, 2020; Rose and Mohl, 2012).

Thus far, I have shown that those affected by highway construction moved out of their neighborhoods and relocated within the same city. This reorganization of urban dynamics raises the question of where these individuals relocated, given ample evidence that the neighborhood of residence affects long-run outcomes (Chetty et al., 2016a).

6.3 Neighborhood characteristics at death

The previous results show that individuals affected by highway construction leave their neighborhoods and relocate within the same city. In this section, I use the last recorded residence to infer neighborhood proxies for wealth, income, and education. A vast literature in economics shows that relocating individuals from struggling neighborhoods has a positive effect on their long-run earnings (Chetty et al., 2016a; Chyn, 2018). However, given anecdotal accounts of displaced individuals moving to economically disadvantaged areas and the increased mortality found in Section 6.1, one would expect displaced individuals to end up in neighborhoods with lower socioeconomic indicators.

Figure 6 presents suggestive evidence that displaced individuals end up in neighborhoods with worse economic characteristics. Each outcome corresponds to the year 2000, the median year in the mortality sample, and has been normalized to have a mean of zero and a standard deviation of one. I find that displaced individuals live in neighborhoods with unemployment rates 0.018 standard deviations higher, or 0.1 percentage points on a sample mean of 5.74%. I also find that they live in neighborhoods with a college graduate share 0.028 standard deviations lower. I find no significant effect on neighborhood income, home values, rents, or homeownership rates. For those adjacent, the pattern is weaker. I find that those adjacent move to places with unemployment rates 0.010 standard deviations higher, with no differences for the other variables studied.

How do these results translate into the *quality* of the neighborhood where individuals live? To answer this question, I construct a quality index from the first component of a principal component analysis of the neighborhood characteristics analyzed above, normalized to have a mean of zero and a standard deviation of one. Details appear in Appendix Section D.6. The bottom row of Figure 6 shows that displaced individuals relocate to neighborhoods with quality 0.020 standard deviations lower than the control group. In contrast, I find no effect on neighborhood quality among those adjacent to newly constructed highways.

6.4 Racial differences in the cost of displacement

Although highway construction disproportionately affected Black neighborhoods, I document that most of the non-pecuniary costs of displacement are borne by white individuals.

Contrary to the anecdotal record, Figure 7 shows a positive, but statistically insignificant, effect for displaced Black individuals. The overall effect of displacement is driven by white individuals, who die 0.11 years younger than their unaffected neighbors, or about six weeks. The same pattern holds for those living adjacent to the highway, where I find no effect for Black individuals and a loss of 0.075 years for white individuals. These differences do not arise from a lack of mobility (panels (b) and (c)), nor from the quality of the neighborhood in which individuals die.

Why does the effect vary by race? I find that differences in pre-construction characteristics played no role in these differences. In Appendix Figure D.7 I reweight the sample of white individuals to match the distribution of pre-construction characteristics of Black individuals, using DiNardo et al.-style (1996) weights.³⁰ Equalizing pre-construction characteristics between white and Black individuals barely narrows the racial gap in the displacement effect. The gap remains close to its baseline magnitude even when I match on city of residence, so observed pre-construction differences account for only a small share of it.

In contrast, I find that the redlined status of the neighborhood did play a role in shaping the cost of displacement (Appendix Figure D.8). Most of the effect for white individuals comes from those living in non-redlined areas, and among Black individuals displaced from redlined areas I find a positive effect, although it is too imprecisely estimated to be distinguishable from zero. Mobility follows the same line. Black individuals displaced from redlined neighborhoods move out and relocate to other parts of the city, whereas those displaced from non-redlined neighborhoods, and Black adjacent individuals in redlined neighborhoods, stay. Together, these patterns suggest that, while displacement hurt white individuals living in non-redlined neighborhoods, it helped Black household to move out from redlined communities.

6.5 Robustness

Appendix Figure D.1 presents additional robustness tests of the estimates. The results remain stable when controlling for every covariate that was unbalanced in the balance test. Controlling for year of construction fixed effects does not change the estimates, alleviating concerns that the results are driven by time to construction. Moreover, these results are similar when the exercise is repeated using the 1940 census, which alleviates concerns about selective outmigration between the census and highway construction.³¹ Likewise, the results are robust to clustering the standard errors at the state level. When

³⁰The weights are constructed based on cells defined by metropolitan area, sex, quintile of the household head's occupational score, five-year birth cohort, and homeownership status in 1940, where individuals in households that I do not link to the 1940 census form their own tenure category.

³¹Given the longer period between the observed residence and treatment, the effects are smaller in magnitude.

several treatments are estimated at once, regression coefficients can be contaminated by treatment effects from other treatments (Goldsmith-Pinkham et al., 2024). The results remain unchanged when applying their proposed correction for this contamination bias (Appendix Figure D.2).

7. POSSIBLE MECHANISMS FOR THE MORTALITY RESULTS

Having shown that highway construction reduced longevity for white individuals displaced by and living next to a highway, I now explore the mechanisms explaining these relationships. I first consider the possibility that the effects were mediated by the origin or destination neighborhoods and the labor market conditions faced after construction. Then I look at the most likely explanation: the erosion of housing wealth for those living next to the highway and the financial and emotional costs for those displaced.

7.1 Neighborhoods and labor markets

The first explanation for the decrease in longevity for displaced and adjacent individuals is that the conditions of the neighborhood they moved to took a toll on their longevity. Chetty et al. (2016a) and Chyn and Katz (2021) are recent examples of how neighborhoods affect residents' long-run outcomes. Yet, I find no evidence that neighborhood of residence at death explains these results. The neighborhood quality differences found in the previous section are economically small, amounting to less than 0.02 standard deviations for those displaced and 0.01 for those adjacent. These destination neighborhoods were not *worse* to begin with, as Appendix Figure E.1 shows they declined in quality only in later decades. After imputing the characteristics of the neighborhood when each individual turned 40, 50, or 60 years old, I find no differences in the neighborhood quality across treatments (Appendix Figure E.2). Also, I do not find evidence that worse air pollution levels in destination neighborhoods explained the mortality gap (Appendix Table E.1, column 1). While displaced individuals did reside in communities with marginally worse levels of PM_{2.5} (0.01 standard deviations, p-value = 0.094), the difference is economically small, with no effect for those adjacent. The outmigration from their origin neighborhood, which was by then adjacent to a polluting highway, likely shielded those affected from the hazardous effects of highway pollution (Currie and Walker, 2011).

A second possibility is that differences in the conditions of the origin neighborhood explained these results. Many policymakers at the time saw highway construction as a tool to revitalize neighborhoods in disarray.³² I find this possibility unlikely for three reasons.

³²For example, when studying the construction of I-95 through the neighborhood of Overtown, Miami, N.D.B. Connolly argues that "What's largely forgotten are the deplorable housing conditions [of Overtown]" (Connolly, 2014, pg. 282).

First, the preferred research design compares individuals with neighbors living up to 200 meters away, equivalent to two blocks. Thus, by construction, the empirical strategy compares individuals who faced the same neighborhood conditions in 1950. Second, the results hold even after controlling for the quality index of the origin neighborhood (Appendix Figure E.5). If within-treatment differences in pre-move neighborhood quality drove the results, we would expect the effects to vanish once I control for that quality. Finally, there is no clear pattern when I split the sample across different measures of neighborhood quality. In Appendix Figure E.6, I find that the estimated effects are larger for non-redlined neighborhoods, stable across income quartiles, and larger for neighborhoods in the lowest half of the unemployment distribution. The lack of a clear pattern suggests that the longevity effect is unlikely to be mediated by pre-move neighborhood characteristics.

A third explanation for the documented mortality gap is that those affected faced worse labor market conditions after construction. Access to employment could have declined either because individuals relocated to neighborhoods with fewer job opportunities or because construction destroyed local businesses. Figure 6 shows that affected individuals did move to places with higher unemployment rates, but the differences are small, amounting to 0.02 standard deviations. They also emerge in later decades. Estimating neighborhood unemployment and median income by decade, I find no differences before 1980, when most of the individuals in the sample were likely already out of the labor force (Appendix Figure E.3). Measuring neighborhood conditions at the ages of 40, 50, and 60, I find no differences across treatments (Appendix Figure E.4). Also, I find no differences in the distance from affected individuals' post-displacement residences to the city center, a common proxy for job access (Couture and Handbury, 2020), as shown in column 2 of Appendix Table E.1. Taken together, these results suggest that differential access to employment does not account for the estimated mortality gap.

I find suggestive evidence that racial segregation shaped the effects of displacement for Black individuals (Ananat, 2011; Chyn et al., 2026). Appendix Figure E.7 divides cities into terciles based on the dissimilarity index, a common proxy for racial segregation (Boustan, 2012). If racial segregation imposed additional constraints on Black individuals, we would expect the costs for Black individuals, relative to white individuals, to be higher in high-segregation cities. I find that the costs of displacement are larger for white individuals than for Black individuals across all levels of segregation. However, the gap between the estimates decreases in high-segregation cities, although these differences are not statistically significant at the usual levels. The reduction comes from a larger negative effect for Black individuals in high-segregation cities.

7.2 Erosion of housing wealth

After exploring the role of neighborhood and labor market conditions, I turn to the erosion of housing wealth as a mediating mechanism. Housing is the most valuable asset for most American households (Akbar et al., 2025). Highway construction eroded the value of homes, whether through eminent domain compensation set below market rates (Lewis, 2013, pg. 142; Rose and Mohl, 2012, pg. 128) or through the negative externalities of proximity to the new roads (Brinkman and Lin, 2024; Moretti and Wheeler, 2025). If this channel mediates the longevity results, we should expect the effects to be concentrated among the homeowners who bore these losses.³³

I find that the erosion of housing wealth explains the results for those adjacent to the highway but plays no role for those directly displaced. Figure 8 shows that the mortality effect for those living adjacent to the newly constructed highway is driven entirely by homeowners. For those displaced, however, the pattern reverses: the effect comes from renters, with no effect among those who owned their residence at the time. Consistent with the estimates reported in Section 6.4, both effects are driven by white individuals, with displaced Black homeowners showing an increase in longevity (Appendix Figure E.8).

Qualitative records document the erosion of housing value near the new highways. Some lobbying groups argued that urban highway construction would stimulate “rising land values and prevent ‘the spread of blight and slums’ ” (Rose and Mohl, 2012, pg. 101), but the outcome was the opposite. Lewis (2013, pg. 296) concludes that construction “removed property from the tax rolls and left the ground and adjacent buildings in shadow and economically depressed.” This account implies that the losses would be largest where there was value to lose. Appendix Figure E.9 shows that the effect for those adjacent to the highway is concentrated among homeowners in non-redlined areas, with no effect in redlined neighborhoods, where property values were already low (Hynsjö and Perdoni, 2022). The pattern is consistent with qualitative records and with housing wealth as the channel mediating the decrease in longevity for those adjacent to the highway.

7.3 The cost of displacement

After exploring what explains the results for those adjacent to the highway, I argue that the most likely explanation for the decrease in longevity associated with displacement is the lack of relocation assistance. I call on both qualitative and quantitative evidence to

³³Housing information is not available in the 1950 census. I assign homeownership status using the linkage between the 1940 and 1950 censuses. An individual is classified as a homeowner if anyone in their 1950 household was a homeowner in 1940. Because the 1940 census assigned homeownership at the household level, this definition classifies as a homeowner every individual in a 1950 household containing a member who belonged to an owner-occupied household in 1940. The homeownership rate in 1950 is 44.8 percent under this definition. The results are robust to tighter and looser definitions of homeownership.

support this claim. Displaced individuals received no payment at the time of moving to help them bear the cost of relocation. Even small financial burdens can have long-lasting consequences when households have poor access to financial markets.

First, historians argue that displacement imposed substantial costs on affected families, both in the value of what was taken and in the manner in which it was carried out, with consequences that extended over the remainder of their lives. Fred Salvucci, later the Massachusetts Secretary of Transportation, describes what happened when the state agency came for the home where his grandmother had lived: “They knocked on the door [...], gave her a dollar and said ‘The house is ours.’ She had thirty days to move seven decades of possessions and memories” (Lewis, 2013, pg. 297). Such treatment was not exceptional, and for most families the costs extended well beyond the value of the property. One resident of Overtown, Miami, recalls that the highway “destroyed the cultural, the spiritual, the educational concept of the entire community. We lost our neighbors. We lost our friends” (Connolly, 2014, pg. 282). Rose and Mohl (2012, pg. 97) document that throughout most of the expressway-building era little was done to connect highway construction to housing programs or to provide relocation assistance for the displaced families, businesses, and community institutions in its path.

Second, I document that the introduction of relocation payments for displaced individuals offset the negative mortality effects. Beginning in the early 1960s, states gradually adopted relocation assistance programs that provided payments to displaced tenants, but adoption was uneven: by 1963, only 11 states offered such assistance (Highway Research Board, 1963, pg. 43). I compile a new dataset on the enactment year of each state’s first relocation assistance law and use it to estimate the effect of relocation payments.³⁴ Details on the construction of this dataset are provided in Appendix Section F.

I exploit variation in the timing of enactment across states to compare outcomes for individuals displaced before and after their state enacted assistance, following:

$$y_i = \gamma_1 \text{Displaced}_i + \gamma_2 \text{Displaced}_i \times R_i + \delta_1 \text{Adjacent}_i + \delta_2 \text{Adjacent}_i \times R_i + \lambda_{t(i)} + \theta R_i + \mathbf{X}'_i \Gamma + \epsilon_i \quad (3)$$

where R_i indicates whether construction of the highway nearest to individual i ’s home began after the state enacted relocation assistance.³⁵ The coefficient γ_2 captures the effect of

³⁴By using enactment timing rather than actual receipt of payments, I avoid concerns about selection into receipt. Many eligible households likely never received payments, as most states and cities lacked the administrative infrastructure to facilitate disbursement. Incomplete take-up is common in U.S. social insurance programs: Moving to Opportunity vouchers had a take-up rate of 43 percent (Chetty et al., 2016a), and roughly half of eligible workers apply for Unemployment Insurance benefits (Moore and McQuillan, 2025).

³⁵The introduction of relocation payments could have altered where or what types of projects were built (Cordes and Weisbrod, 1979). I find limited evidence of such selection. Appendix Figure F.3 shows that those displaced while eligible for payments were less likely to reside in redlined areas but were balanced on almost every other variable. Even where they are not balanced, the disparities are economically small.

relocation assistance on displaced individuals, while δ_2 captures any differential effects for adjacent individuals. Since only displaced individuals were entitled to compensation, the effect for those adjacent, δ_2 , acts as a placebo test. I restrict the sample to individuals living in states that enacted relocation assistance between 1956 and 1969 and that experienced displacement both before and after enactment. Given the time elapsed between when I observe individuals and the enactment of relocation assistance laws, I include year-of-construction fixed effects $\lambda_{t(i)}$. Thus, identification comes from assistance per se rather than from the date of construction.³⁶

Figure 9 shows that relocation assistance payments offset the negative mortality effects of displacement. Individuals displaced after the enactment of relocation assistance ($\hat{\gamma}_2$) experienced a reversal of the effect, with an increase in longevity of 0.15 years (panel (a)). A natural concern is that this reversal reflects not the payments themselves but the characteristics of later construction, either because later highways were routed through different types of neighborhoods or because residents had more time to leave before construction began. If these characteristics drove this result, we would also expect to see an improvement for those adjacent to the highway. Yet, I do not find that those adjacent saw their mortality effect reverse after relocation payments were enacted. Because this group was not entitled to compensation, the difference points to the payments themselves rather than to differences in other characteristics. Including state-year fixed effect does little to the estimates, ruling out that the results are driven by within-state changes in acquisition practices or housing availability (Appendix Figure F.2). By running an event study, I show that most of the longevity gain emerges within two years of enactment (Appendix Section F.2).

How did relocation payments mediate the mortality findings? Relocation assistance combined two distinct components: a cash payment to the displaced household and a requirement that qualifying housing be available before acquisition could proceed. Historians have noted that relocation assistance laws required states to ensure the availability of “decent, safe, and sanitary” housing before acquiring property for highway routes (Rose and Mohl, 2012, pg. 145), a requirement often satisfied through public housing.³⁷ In the context of urban renewal, Kushner and Werner (1976, pg. 752) document that two-thirds of displaced families in D.C. who received assistance were relocated into public housing. Although the evidence is not causal, I find a strong relationship between displacement and moving into public housing in 1973 and 1974, years when all states had already adopted relocation payments. The 1973 and 1974 waves of the American Housing Survey list “displacement by highway construction” among the reasons for moving in the previous

³⁶A causal interpretation of γ_2 requires the additional assumption that, absent assistance, the displaced-control gap would not vary systematically with adoption status. Although I cannot test this directly, the lack of pre-trends when running an event-study supports this assumption (Appendix Section F.2).

³⁷In a speech at a Highway Research Board Conference in 1968, the director of the Federal Highway Administration, Lowell Bridwell, claimed: “If we can’t find housing, we can’t build highways” (Rose and Mohl, 2012, pg. 146).

twelve months. I find that those displaced are 194 percent more likely to move to public housing than the average mover (Appendix Table E.2). This pattern is consistent with housing security being the underlying mechanism behind the longevity reductions for those displaced. Placement into public housing also explains why relocation assistance did not alter mobility patterns or the quality of destination neighborhoods (Figure 9, panels (b)–(d)).

8. CONCLUSIONS

This paper studies the non-pecuniary costs of Interstate Highway construction for those whose homes were destroyed and for those living in the communities where the highways were built. Because no systematic records of the displaced exist, I develop a new method to identify them in historical censuses and link them to administrative mortality records. More than 30,000 individuals lost their homes each year between 1956 and 1970, most of them without any relocation assistance. Construction fell disproportionately on vulnerable communities, whose residents were more likely to be Black, to earn less, and to live in redlined areas. Comparing the built network with the routes planned by the federal government, I find that these disparities came from deviations from those plans rather than from engineering constraints alone.

The main analysis compares individuals displaced by construction, or living within 100 meters of it, to their neighbors between 100 and 200 meters away. Displaced individuals die about one month earlier than these neighbors, and nearly all leave their 1950 neighborhood for lower-quality areas in the same city. A back-of-the-envelope calculation puts the mortality cost of displacement at roughly \$11,400 per individual in 2010 dollars, or more than \$5.2 billion in total. I also find evidence of negative spillovers for residents of the communities where these projects were built. Those adjacent to the highway die about twenty days earlier and are also more likely to leave their neighborhood, but they do not end up in worse ones. The results hold across alternative specifications and identification strategies. Although construction disproportionately affected Black communities, the long-run costs fall almost entirely on white residents. The erosion of housing wealth explains the effect for adjacent residents, while the absence of relocation assistance explains the effect for those displaced.

This paper has limitations that future research may address. Because of data constraints, the analysis focuses on the long-run consequences of highway construction for individuals. Recent advances in linking restricted-access census data to tax records could enable a more comprehensive analysis of the short- and medium-run effects of forced relocation, particularly on labor market outcomes. Future work should also consider the general equilibrium effects of displacing large numbers of individuals on the communities that

receive them.

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9. FIGURES

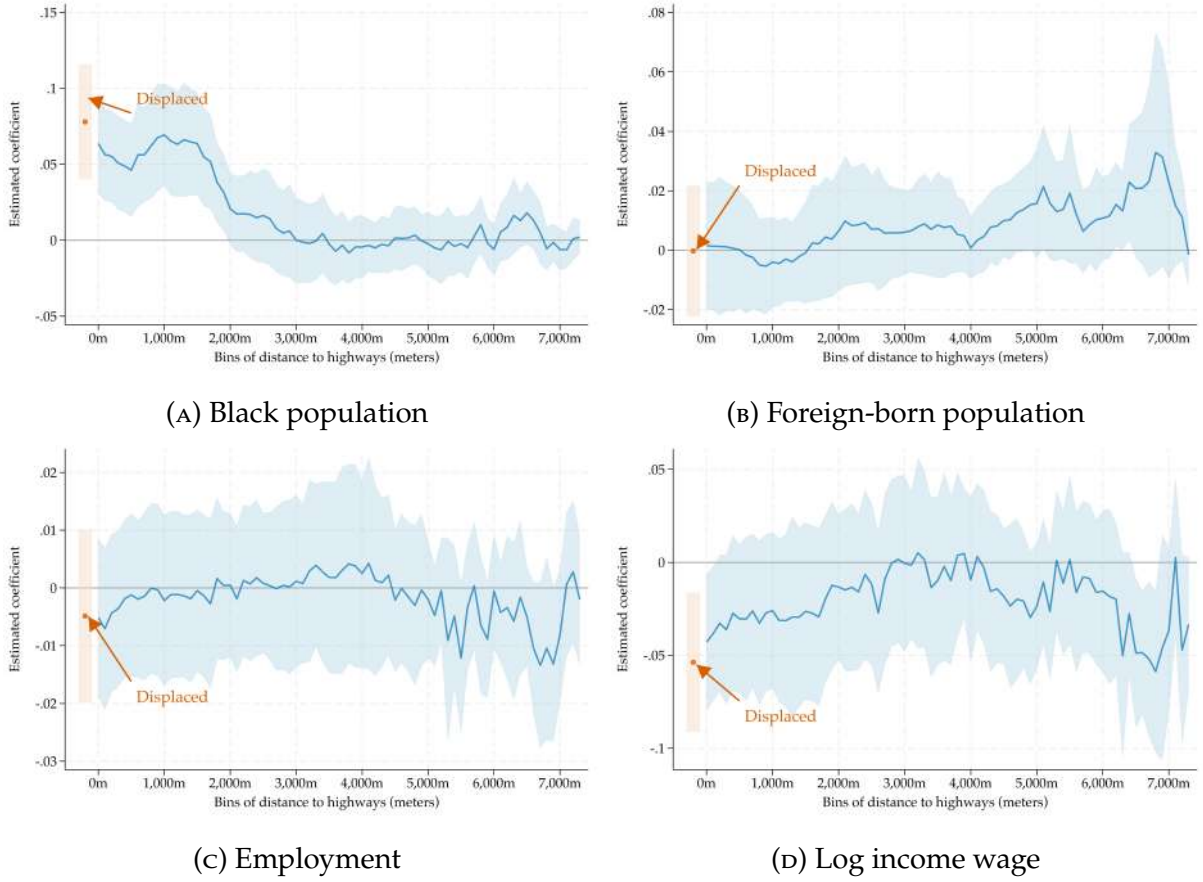
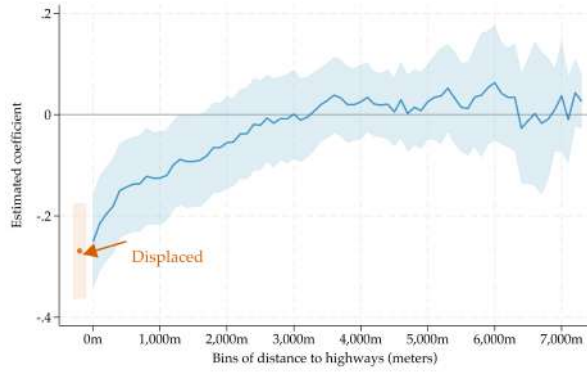
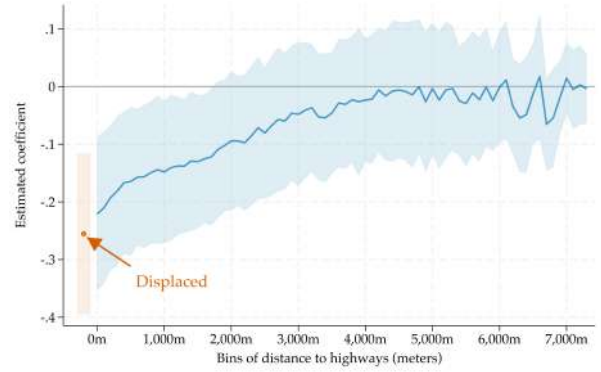


FIGURE 1: Pre-construction characteristics by distance to future Interstates.

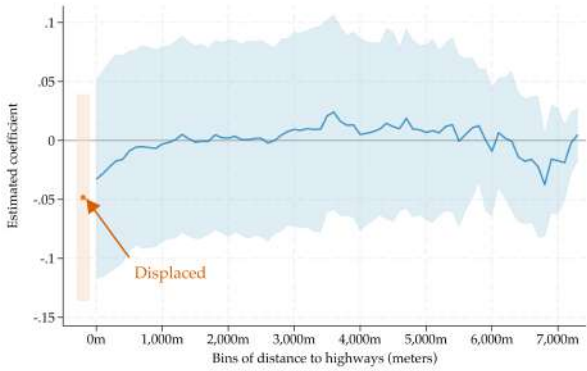
Notes: The figures present the estimated coefficients and confidence intervals of equation 1. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. The sample consists of individuals in metropolitan areas who lived within 7,500 meters of a future Interstate in 1950. The comparison group corresponds to those living between 7,400 and 7,500 meters from the future Interstate. The dependent variables are reported in the name of each panel. All regressions include metropolitan area fixed effects and distance to the city center. Appendix Figure B.2 reports the same estimates controlling only for metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.



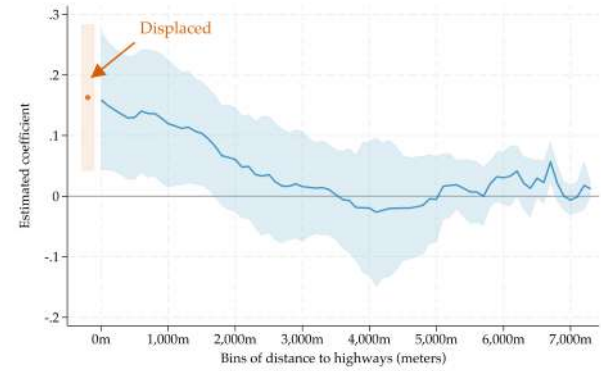
(A) Log home value in 1940 (2010 dollars)



(B) Log rent in 1940 (2010 dollars)



(C) Homeownership rate in 1940



(D) Redlined status

FIGURE 2: Pre-construction housing characteristics by distance to future Interstates.

Notes: The figures present the estimated coefficients and confidence intervals of equation 1. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. The sample consists of individuals in metropolitan areas who lived within 7,500 meters of a future Interstate in 1950. Panels (a)–(c) report the coefficients for the sample of individuals linked to the 1940 Census who lived in the same property in both years. Census-to-census linkages are provided by Ruggles et al. (2025). The comparison group corresponds to those living between 7,400 and 7,500 meters from the future Interstate. The dependent variables are reported in the name of each panel. All regressions include metropolitan area fixed effects and distance to the city center. Appendix Figure B.2 reports the same estimates controlling only for metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

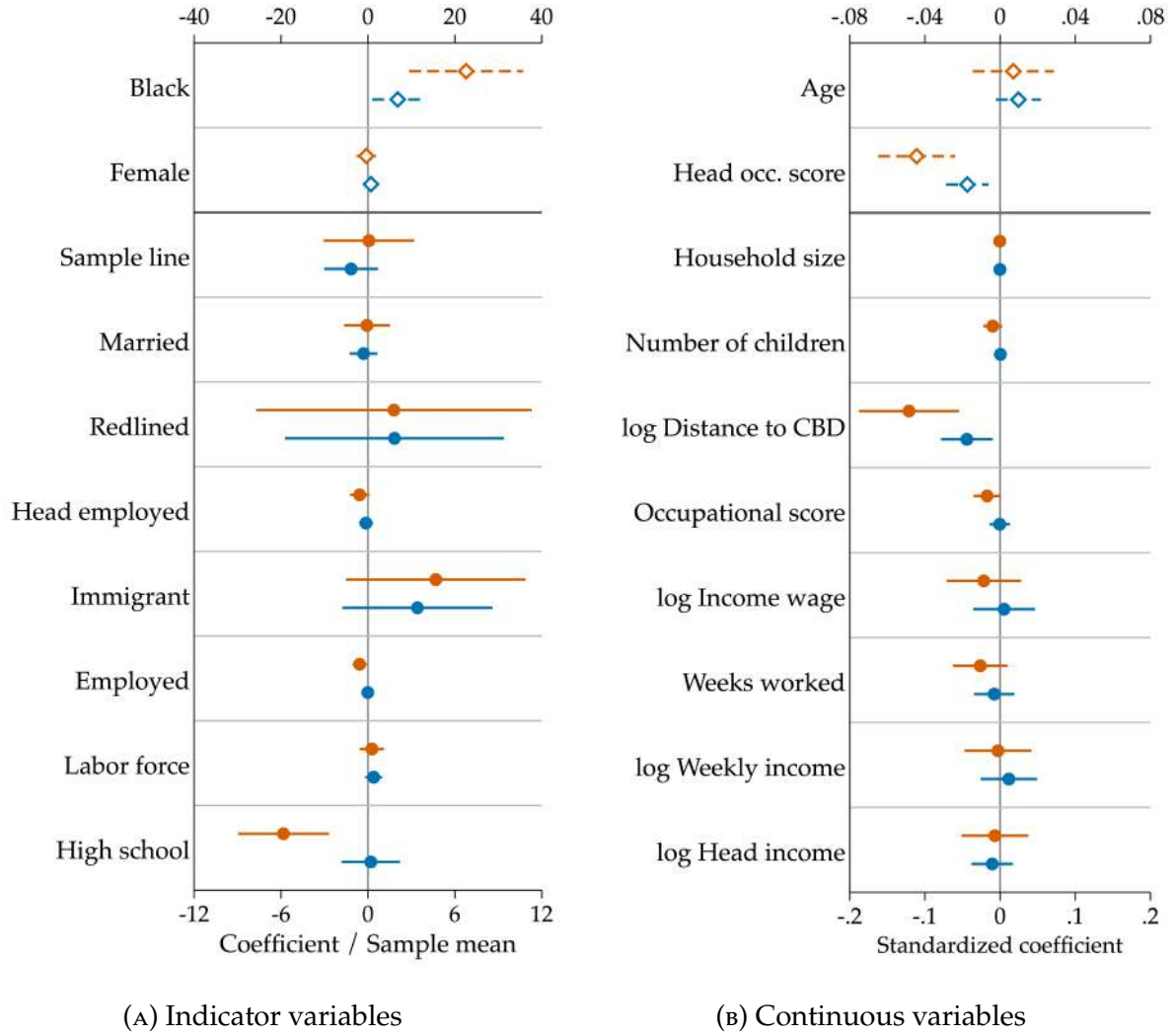


FIGURE 3: Pre-construction balance.

Notes: The figures display estimates from regressions of pre-construction characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. In both panels, the top two coefficients, shown as hollow diamonds, control only for metropolitan area fixed effects and are plotted against the top x-axis. The remaining coefficients, shown as solid circles, control for the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects, and are plotted against the bottom x-axis. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (b) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.

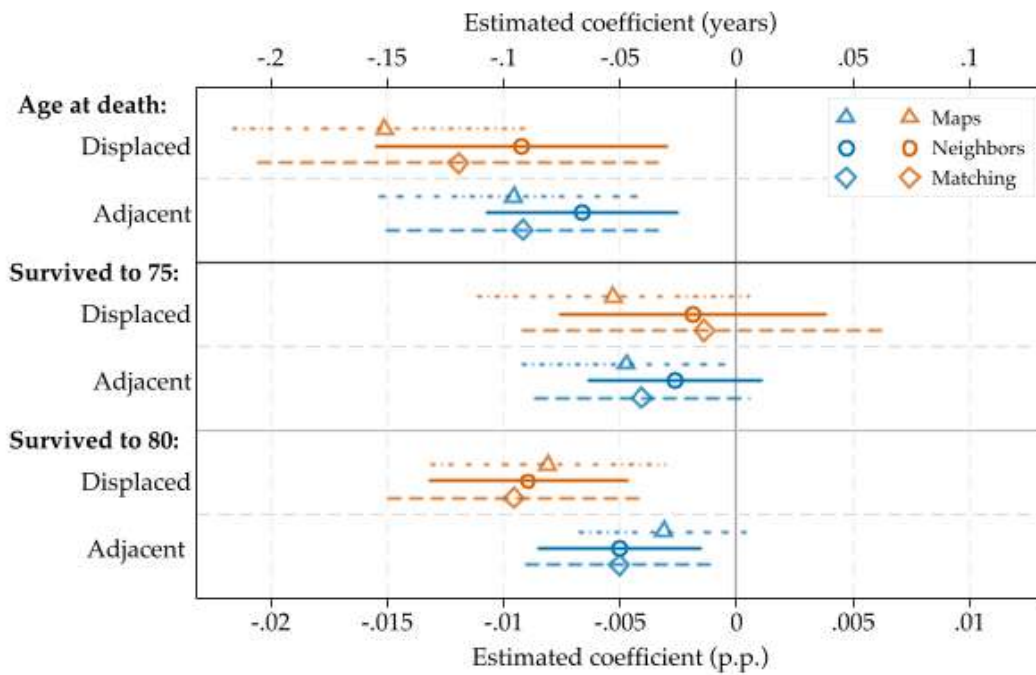


FIGURE 4: Effects of Interstate construction on longevity.

Notes: The figure displays estimates from regressions of longevity characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, using the three research designs described in Section 5 and standard errors clustered at the metropolitan area level. The strategy denoted as “Neighbors” compares treated individuals to their unaffected neighbors living between 100 and 200 meters from the highway; strategy “Matching” compares treated individuals to matched individuals living in the same city, and strategy “Maps” compares treated individuals to those who would have been displaced by the Yellow Book routes. The top section uses age at death as the dependent variable, while the middle and bottom sections use the probability of surviving past the ages of 75 and 80 as the dependent variable, respectively. The estimates are plotted against two different horizontal axes to improve readability. All regressions include the household head’s occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Appendix Table D.1 reports the numerical values of the estimates and standard errors.

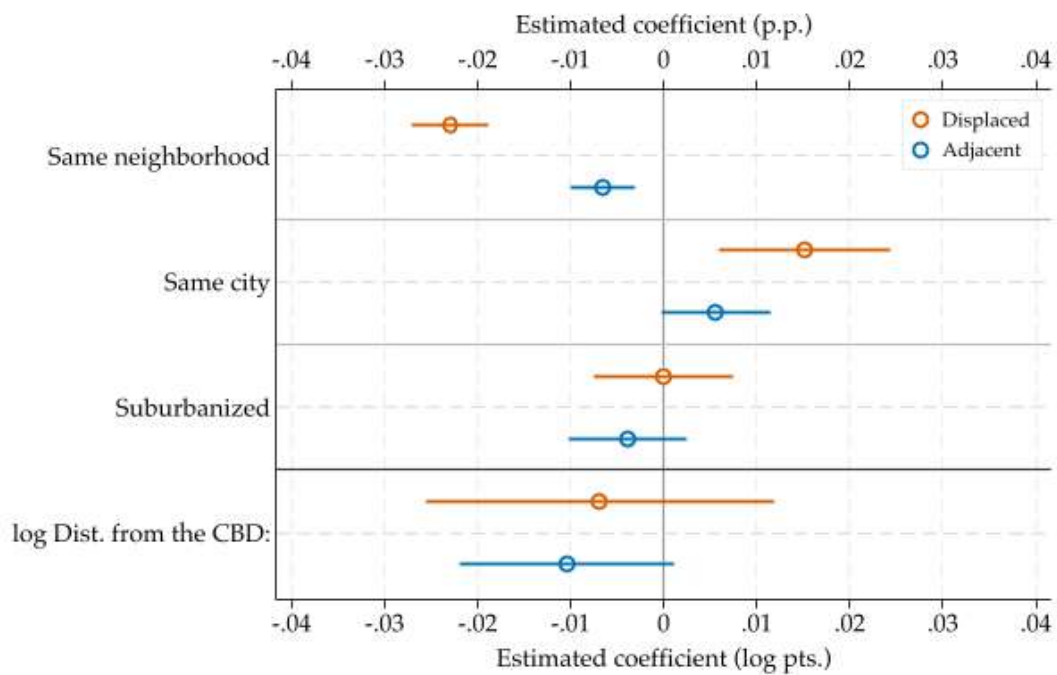


FIGURE 5: Effects of Interstate construction on spatial mobility.

Notes: The figure displays estimates from regressions of spatial mobility characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variables are reported on the y-axis, and the estimates are plotted against two different horizontal axes to improve readability. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. Appendix Table D.2 reports the numerical values of the estimates and standard errors.

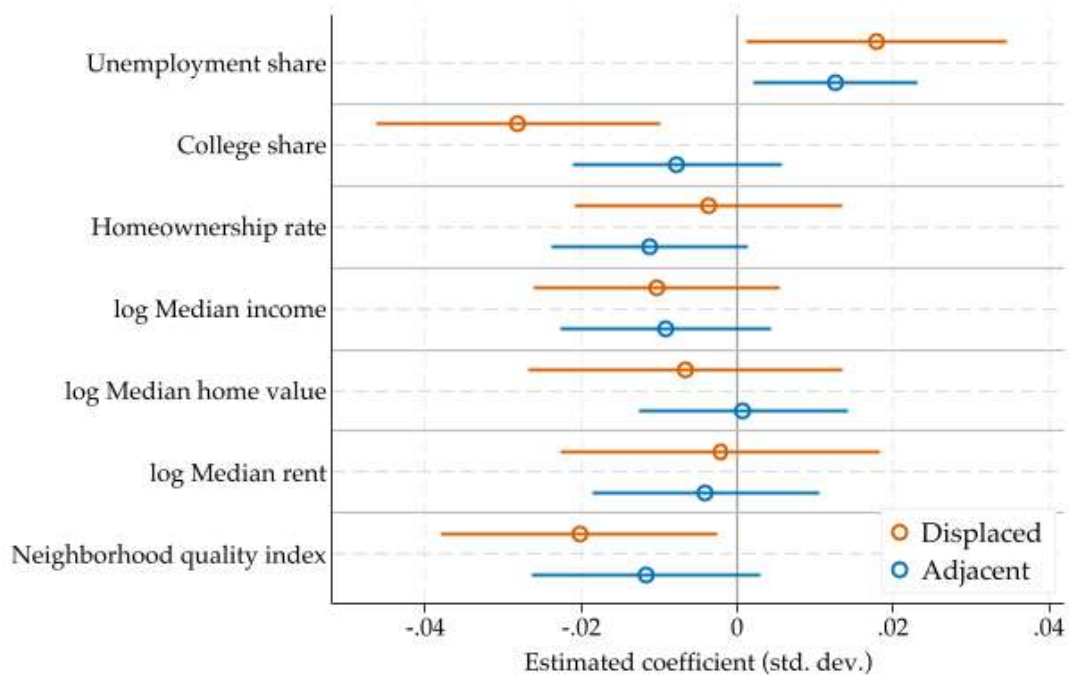


FIGURE 6: Effects of Interstate construction on neighborhood characteristics.

Notes: The figure displays estimates from regressions of neighborhood characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variables are reported on the y-axis, and they are normalized to have a mean of zero and a standard deviation of one. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. Appendix Table D.3 reports the numerical values of the estimates and standard errors.

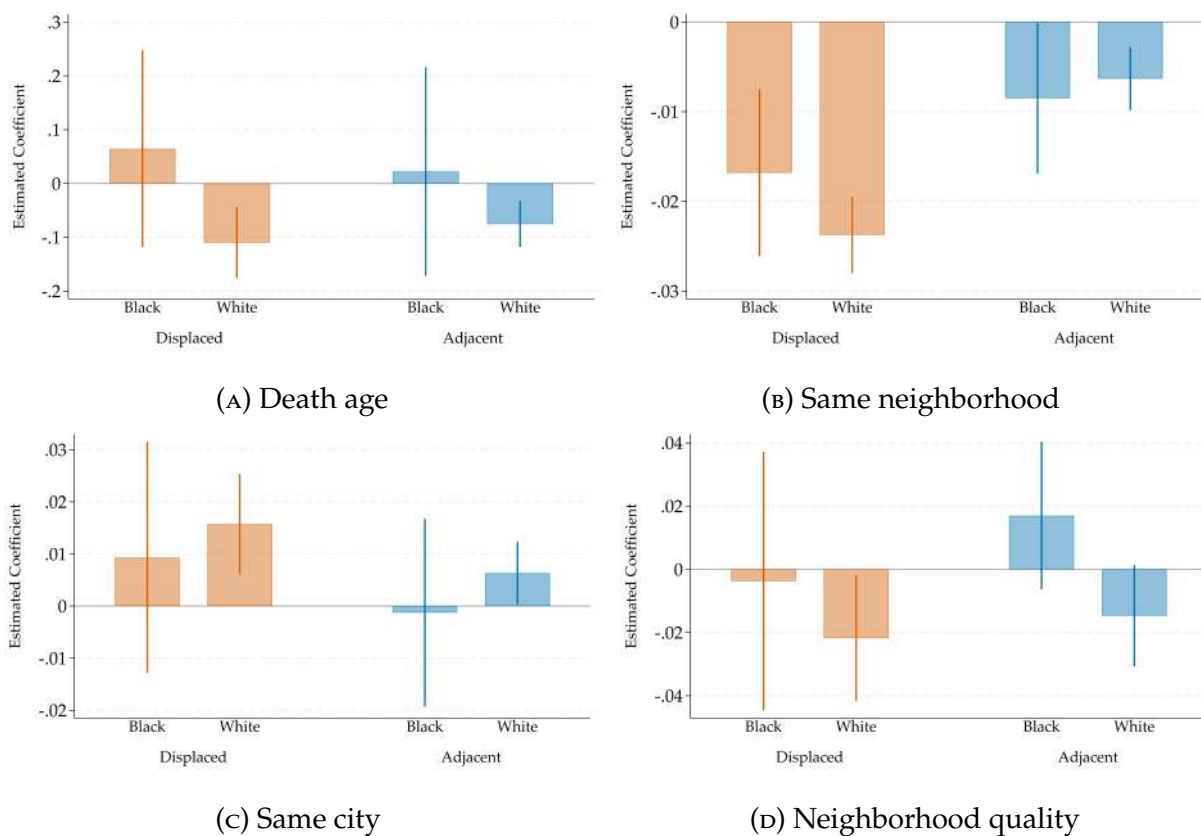


FIGURE 7: Heterogeneity by race.

Notes: The figures display estimates for the outcomes named in each panel, from regressions in which the sample is split by race. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each bar represents an estimate along with 95% confidence intervals. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

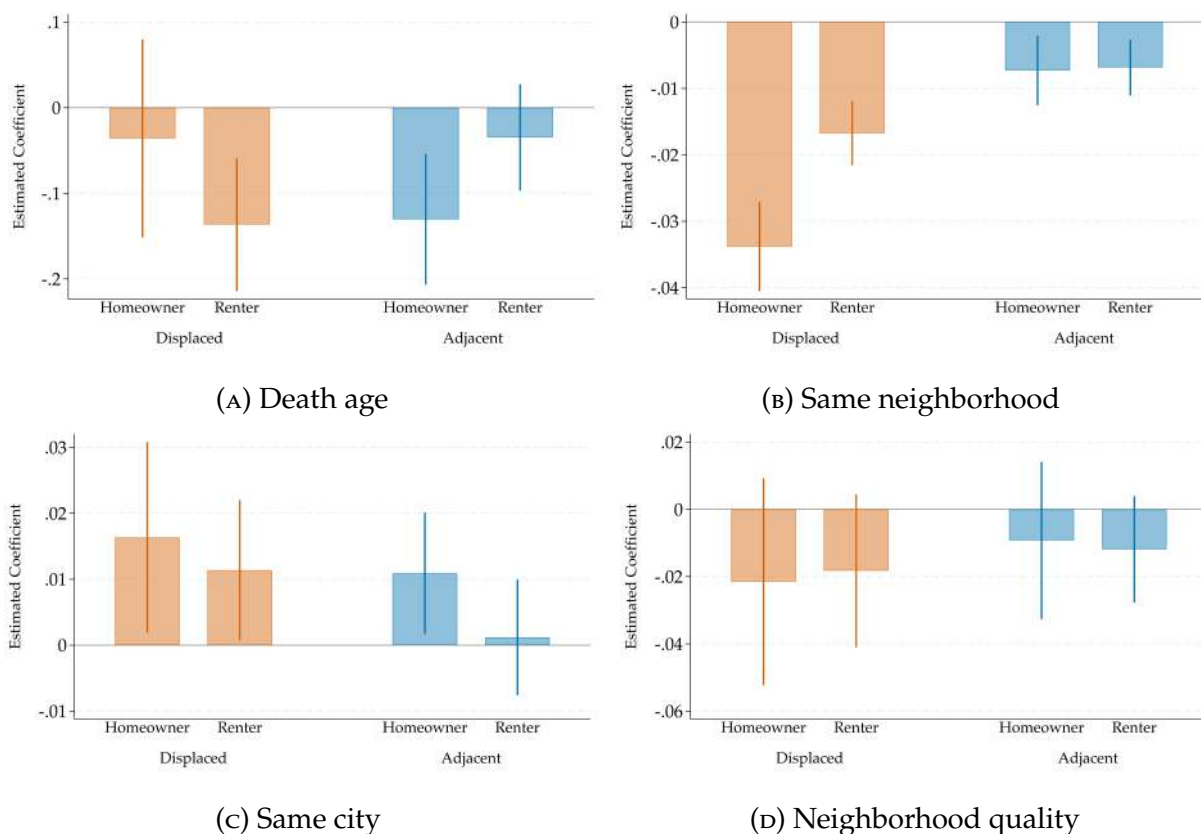


FIGURE 8: Heterogeneity by homeownership status.

Notes: The figures display estimates for the outcomes named in each panel, from regressions in which the sample is split by homeownership status. Homeownership is measured using linked 1940 and 1950 census data. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each bar represents an estimate along with 95% confidence intervals. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

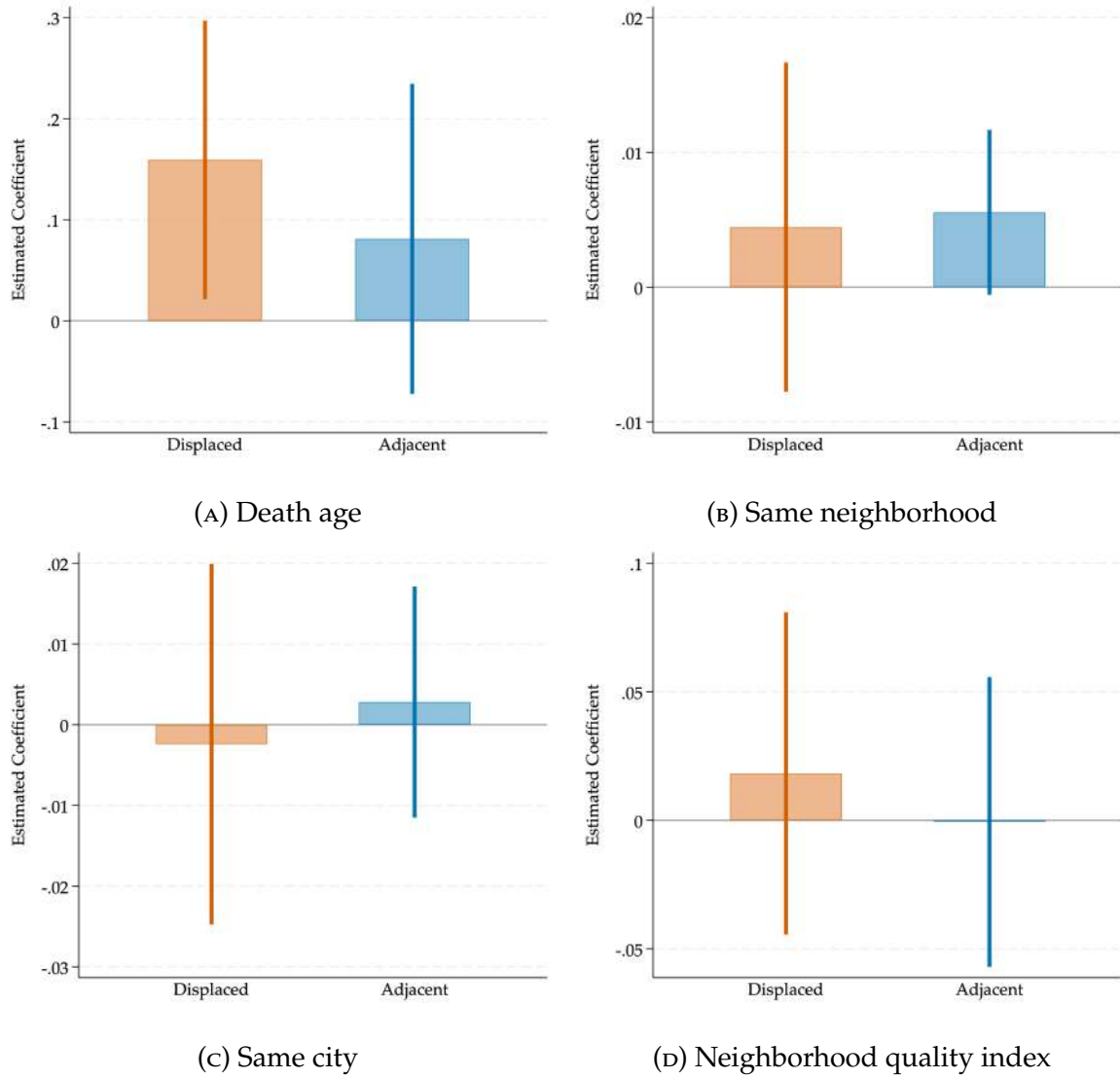


FIGURE 9: Relocation assistance.

Notes: The figure compares the changes in the treatment effect of Interstate highway construction for displaced individuals and those residing within 100 meters of the highway, with and without relocation assistance laws in place. In particular, it plots the estimates of γ_2 and δ_2 from equation 3 for each outcome. The dependent variable in Panel (a) is the age at which the individual died. The dependent variable in Panel (b) is an indicator for whether the individual lived in the same neighborhood (census tract) at the time of the census and at the time of death. The dependent variable in Panel (c) is an indicator for whether the individual lived in the same city (metropolitan area) at the time of the census and at the time of death. The dependent variable in Panel (d) is an index of neighborhood quality, constructed as the first principal component of the neighborhood characteristics described in Section 6.3. The comparison group consists of unaffected neighbors living between 100 and 200 meters from the highway. All regressions control for the household head's occupational score and post-enactment dummy and include fixed effects for year of construction, gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan area level.

APPENDIX AND SUPPLEMENTARY MATERIAL

A. DATA APPENDIX

A.1 Cleaning historical addresses

I geocode the restricted access full-count censuses from 1940 to 1950 and focus on the 285 counties that contained one of the 168 Standard Metropolitan Areas in 1950 (Ruggles et al., 2024).³⁸ Rural addresses often lack street names and house numbers, which leads to inaccurate geocoding. For this reason, I focus only on urban areas in these counties. The sample includes 74.28% of the urban households in the U.S. in 1950. To clean historical addresses, I follow the procedures recommended by Logan and Zhang (2018). For many observations, the address is either incomplete or carries extra text that has to be cleaned. In particular, I start by parsing raw addresses and numerations.

To parse the addresses, I assume that enumerators stayed on the same side of the street when moving from building to building, and I then apply the algorithm described below:

1. *Dwelling IDs*: I work with the serial ID provided by IPUMS, as at the moment of writing there is a coding issue with IPUMS' *dwelling ID* variable in which every household is incorrectly coded as a unique dwelling (Ager et al., 2023). Street name and numeration are constant across serial IDs. I also keep the state, county, city, and enumeration district.
2. *Extract the street name*: Sometimes it contains additional information. For example, it may include a word like "Cont." or "Rear". In these cases, I consider the street name to be the same as the previous record. In some cases, the street name contains the house number, for example, in large apartment complexes, hotels, or hospitals. In those cases, I store the house number by parsing the street name in the search for street numbers in addition to other keywords such as apartment, hospital, and hotel.
3. *Carrying forward a street name*: Some addresses have a valid house number but no street name. To fill in the missing information, I carry forward the street name from the previous record under two conditions. First, the two records should be on the same enumeration page. Second, the adjacent records should not have a skip in the house number larger than 6, taking into consideration if the numeration is odd or even.
4. *Cleaning house numbers*: There is considerable variation in the way house numbers are recorded. To standardize the records, I set to "missing" any record that includes the following fields:

³⁸Standard Metropolitan Areas are the equivalent of Metropolitan Statistical Areas in the 1950 census.

- (a) A continuation of the previous house number: 'cont', 'con't', 'contd', 'cont'd', etc.
 - (b) A location relative to the previous house number: 'rear', 'basement', '1/2', 'back', 'front'. Unless the house number is within the text (5147rear), these records are set to missing.
 - (c) A house number that indicates a floor, indicated by 'floor' or 'fl' in the text.
 - (d) Apartment numbers indicated by 'apt' in the text.
 - (e) A combination of numbers and letters that does not follow the standard format of a house number, like '[0-9][][a-zA-Z]' or '[0-9][-][a-zA-Z]'. For example: 7-B. These cases are most likely the rooms in a hotel.
5. *Dealing with missing house numbers:* I interpolate missing house numbers in the same way as missing street names. The only difference is that I take into consideration the side of the street. Some interpolations result in suspicious numbers and are cataloged as missing. For example, the interpolation may result in house numbers far outside the logical range of house numbers. To identify these anomalies, I compare the interpolated house number to the range of house numbers on the same side of the street in the same ED. If the interpolated house number is outside the range or has a skip larger than 6, I set it to missing.

Once the historical addresses are standardized, I proceed to clean them by using the historical street names for each decade available in [SteveMorse.org](https://www.stevemorse.org). This website includes the street names contained in each enumeration district in 1910, 1920, 1930, 1940, and 1950. This allows me to overcome possible errors in the OCR process of the historical street names or during the standardization of the data. To do so, I perform a probabilistic match between the standardized street name and the street name in SM's records for each ED. I do a Jaro-Winkler distance match between the standardized street name and the street name in SM's records, with a similarity tolerance greater than 0.8. I use the street name in SM's records for those records with a match. If there is no match, either because the similarity is too low or because the ED has no candidate street, I use the street name in the census records.

Once the addresses are cleaned and standardized, I proceed to geocode them. Given that the confidentiality agreement signed with Ruggles et al. (2024) does not allow using cloud geocoding, I rely on ArcGIS Street Map Premium. The software does the geocoding on the local machine, circumventing the use of external geocoders. I drop all observations with suspicious geocoding. In particular, I set the geocode to missing for records matched to a different city, county, or state, even if the matched addresses belong to the same metropolitan areas.³⁹ I also drop the geocoding for those records with a *match score* lower

³⁹For example, the address 24 SW 3rd Ave in Miami was geocoded to 24 SW 3rd Ave in the city of Boca Ratón. These types of matches are not in the final sample.

than 85 and those without a unique match, as in Hynsjö and Perdoni (2022).⁴⁰ I also drop those matches that do not allow me to identify an address, such as those that only match the street. The cleaning and geocoding procedure leaves me with 12,309,022 geocoded households in the 1950 census, which accounts for 58.53% of urban households in the counties of study. Appendix Table A.1 shows a breakdown of the number of geocoded dwellings by state.

A.2 Highway data

Highway information comes from OpenStreetMap (OSM). I download the actual network of highways and their exits from OpenStreetMap (2017) and then link it to the PR-511 database from Baum-Snow (2007) to get the starting and opening date of each highway segment financed by the Highways Act. Since I am interested in the displacement effect of highway construction, I include exits as part of the highway database. I only consider the highway segments opened between 1956 and 1970 to use segments close in time to the 1950 census.

The network in OSM is recorded as *Polylines* with negligible width, whereas in reality, highways are *Polygons*. To estimate the right-of-way acquisition for highway construction, I assume that a two-lane highway had a width of 150 feet (45.7 meters), the minimum width used in the 1950s (Weingroff, 2017).⁴¹ For segments with more than two lanes, I add 12 feet (3.66 meters) per additional lane, which is equal to the average lane width at the time (Federal Highway Administration, 2007).⁴² For ramps, I set the right-of-way buffer so a one-lane ramp has a total width of 75 feet, the figure prescribed by the AASHTO Green Book (American Association of State Highway and Transportation Officials, 2018). Panel (b) of Appendix Figure A.13 shows the highway network in Cleveland, Ohio, in 1950. I assign planned Interstate routes from the 1955 Yellow Book (Bureau of Public Roads, US, 1955) a buffer of 150 feet for each carriageway. Using these width assumptions, I construct a buffer around each highway segment representing the right-of-way area.

⁴⁰The match score of a candidate address ranges from 0 to 100. A score of 100 corresponds to a perfect match. The score is penalized according to the number of changes the geocoder needs to make to match the address.

⁴¹The modal and median number of lanes in the sample is two. When the lane count is missing for a segment, I impute it with the state median lane count.

⁴²If actual right-of-way widths exceeded my assumptions, some displaced individuals would be misclassified as adjacent. Therefore, the spillover effects on adjacent individuals could be driven by misclassification of displaced individuals. However, the results are robust to dropping individuals living within 82 feet (25 meters) of the highway, which would be the case if the right-of-way width was larger than my assumptions (Appendix Figure A.1).

A.3 Identifying displacement

I classify an individual as displaced if their 1950 residence falls within the right-of-way of a highway segment opened between 1956 and 1970. For all households, I calculate the distance to the nearest highway segment. Thus, for each household, I can determine whether it was displaced by highway construction, the distance to the nearest highway segment, and whether it was within the planned route of the highway system.

Modern geocoders interpolate addresses from the range of house numbers along their street segment. When highway construction destroyed a street, the geocoder still places the household on the segment closest to its original location, and the dwelling is still classified as displaced. Appendix Section A.4 benchmarks this procedure against historical street grids.

A.4 Benchmarking geocoding accuracy

Geocoding historical addresses with modern geocoders could be problematic because street names and numerations may change over time. These concerns, however, do not invalidate the results. First, I study only those individuals living in dwellings that the geocoder was able to match and whose match passed all the filters described above. If an address is not matched, either because the street was destroyed or because it had changed its name, the record will not be considered in the analysis. This leads to fewer observations to work with, but an accurate geocoding. Second, modern geocoders are equipped to handle missing numeration. This is particularly helpful when the reason behind the missing numeration is highway construction. In this case, the geocoder will match the address to the street segment that is closest to the original address. The geocoder flags this type of match as *StreetAddress* and penalizes the match score accordingly. Thus, these observations' geometry will come from an interpolation at the block level, minimizing the location error. In other words, these observations will be located in the correct block, and their exact location within this block will come from a linear interpolation based on their enumeration. As a robustness check, and to minimize measurement error, I re-estimate the main specification using only observations with a high match score. These observations are the ones that, in addition to passing all the aforementioned filters, have a match score over 98. Appendix Figure A.1 shows that the main results remain unchanged when I use this more restrictive sample.

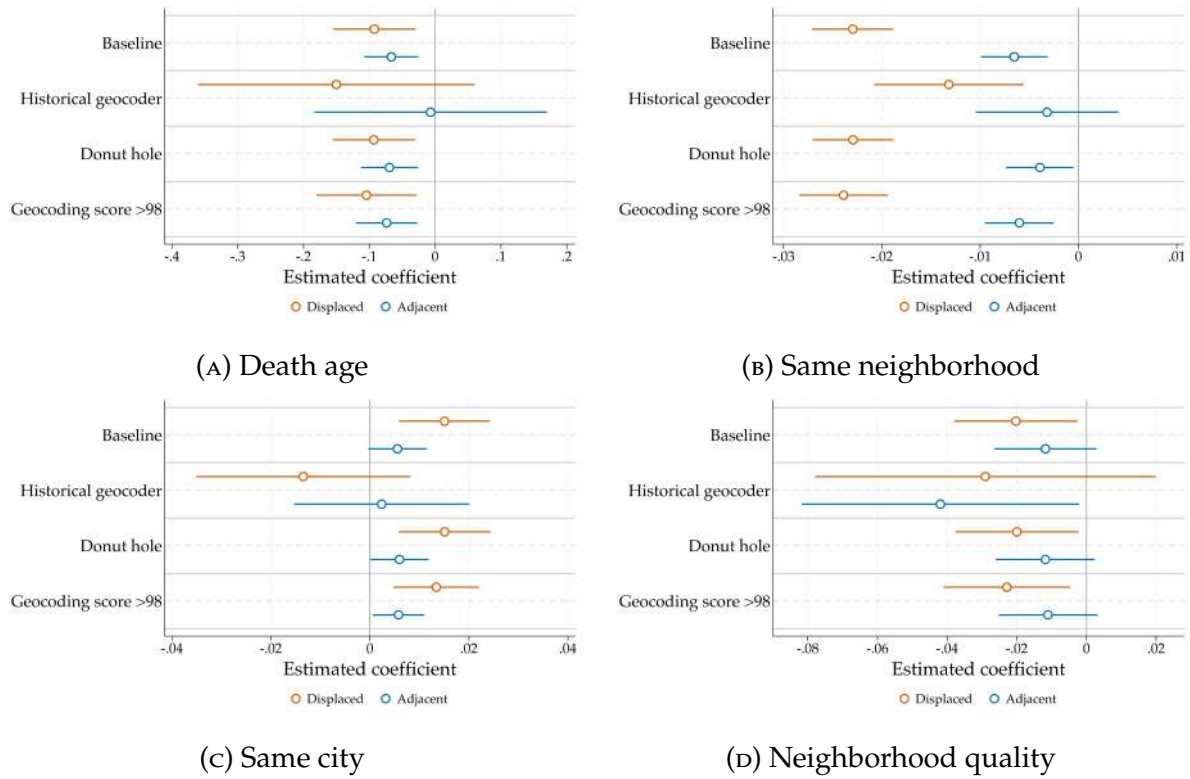


FIGURE A.1: Robustness to different geocoding strategies.

Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top panel corresponds to the baseline results. *Historical geocoder* includes the sample of individuals in the 9 cities with historical geocoders. *Donut-hole* drops individuals within 25 meters of the highway from the adjacent group. *Geocoding score ≥ 98* restricts to individuals with a geocoding score greater than 98.

Finally, using historical street grids from 1940, I construct a locator for nine cities. These cities are: Albany, NY; Buffalo, NY; Cincinnati, OH; Cleveland, OH; Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; and St. Louis, MO. The historical street grids were created by the *Urban Transition Historical GIS Project* and are available at <https://s4.ad.brown.edu/Projects/UTP2/citymaps.htm>.⁴³ These grids predate highway construction and thus provide the “ground truth” against which to benchmark modern geocoding accuracy.

The two geocoders yield highly correlated locations, as shown in Appendix Figure A.2. More than 65% of the locations fall within 40 meters, which is considered within the same parcel. Appendix Figure A.3 shows the CDF of the distance between the two coordinates. When the coordinates do not match, there is no directional bias in the difference between the matched locations, as shown in Appendix Figure A.4, suggesting random measurement error. Also, in Appendix Figure A.1, I use the historical geocoder results to re-estimate the

⁴³Accessed February 10, 2026.

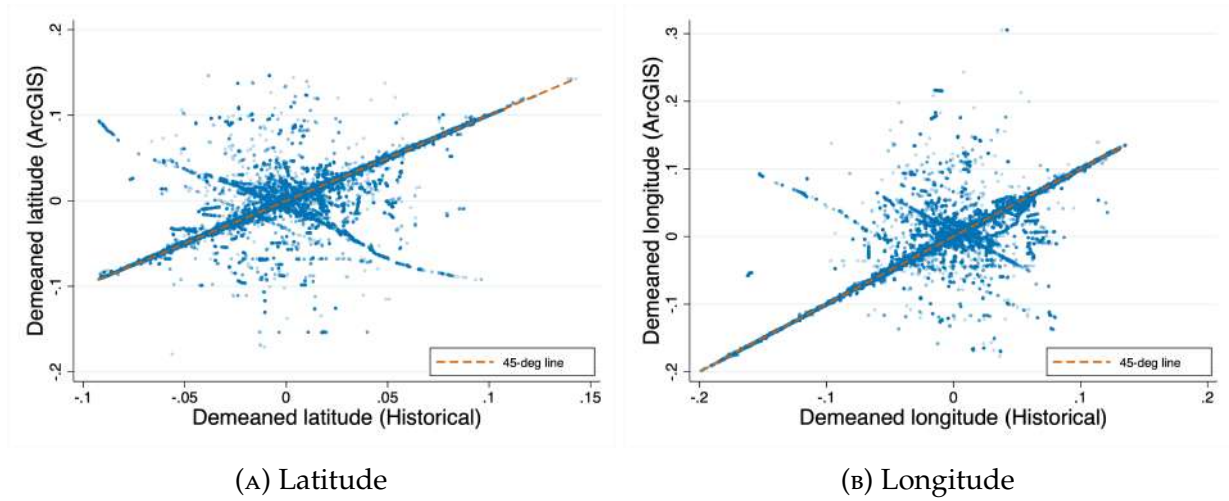


FIGURE A.2: Correlation between modern and historical geocoding results.

Notes: Each point corresponds to a household in the 1950 census. The y-axis corresponds to the coordinates obtained with the modern geocoder, and the x-axis corresponds to the coordinates obtained with the historical geocoder. The coordinates are demeaned by city average coordinates to make the figure more readable. The cities included in the figure are: Albany, NY; Buffalo, NY; Cincinnati, OH; Cleveland, OH; Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; and St. Louis, MO.

paper's main findings, which, if anything, become stronger.

Finally, Appendix Figure A.5 presents column-conditional confusion matrices comparing treatment classifications produced by the modern and historical geocoders. Each cell reports the share of individuals assigned to a given historical-geocoder category (column) who fall into each modern-geocoder category (row). Panel (a) presents results for the full sample. The two geocoders agree closely across distance bands, with diagonal shares typically exceeding 70 percent. Agreement is weaker near the displacement boundary: among individuals classified as displaced by the historical geocoder, 38 percent are classified into the adjacent 0–100 meter band by the modern geocoder. This discordance reflects the difficulty of distinguishing displacement from near-adjacency when geocoded locations fall close to the right-of-way edge, as modern geocoders assign addresses on street segments razed by highway construction to the closest intersection of existing roads. The conservative right-of-way width assumptions discussed in the previous section may further contribute, leading some displaced individuals to be classified as adjacent. Given the focus on displacement, I err on the side of caution by classifying these individuals as adjacent, which may attenuate the estimated displacement gap and inflate the spillover effects on adjacent individuals, rather than the opposite. Panel (b) excludes individuals geocoded within 25 meters of band boundaries, after which the diagonal share for displaced individuals rises to 64.7 percent and off-diagonal mass declines throughout. Appendix Figure A.1 shows that the main estimates remain virtually unchanged when dropping these observations, indicating that classification uncertainty at the margin does not drive the results.

In sum, modern geocoders *may* miss or misclassify some addresses, but the ones they match are accurate and reliable.

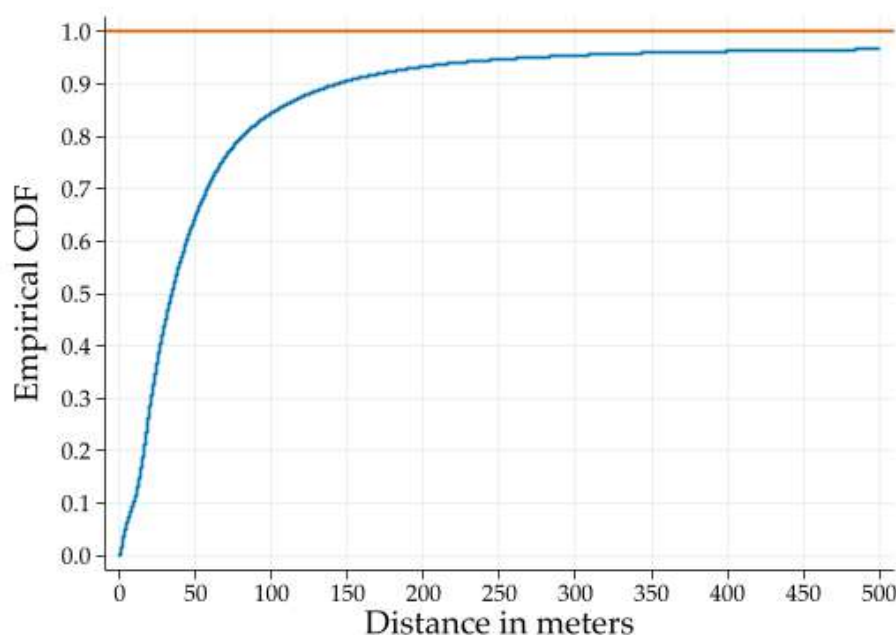


FIGURE A.3: CDF of distance between the coordinates obtained with both geocoders.

Notes: CDF of the distance between the coordinates obtained with the modern geocoder and the historical geocoder. Distance is measured in meters and presented on the x-axis. The y-axis corresponds to the share of observations with a distance smaller than the value in the x-axis. The cities included in the figure are: Albany, NY; Buffalo, NY; Cincinnati, OH; Cleveland, OH; Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; and St. Louis, MO.

A.5 Selective outmigration

The estimates recover the causal effect of displacement and proximity under the assumption that outmigration before 1950 is uncorrelated with the factors that determined highway placement. The timing of the program makes anticipatory moves unlikely, as I discuss in Section 3.1.

I also test for differential outmigration directly. I predict the probability of remaining in the same home at three different time horizons: 5, 10, and 20 years. The model is estimated on the 1960 5% census, which includes a question on whether the individual lived in the same house five years earlier and the year in which the individual moved into the current residence (binned).⁴⁴ I then project each fitted model onto the estimation sample and estimate equation 2 with the predicted probability as the outcome, using those living between 100 and 200 meters from a highway as the control group. Across the three horizons, displaced and adjacent individuals do not differ from their unaffected neighbors,

⁴⁴The probit covariates are a cubic polynomial in age, race, and gender, as well as labor outcomes, interacted with the demographic covariates.

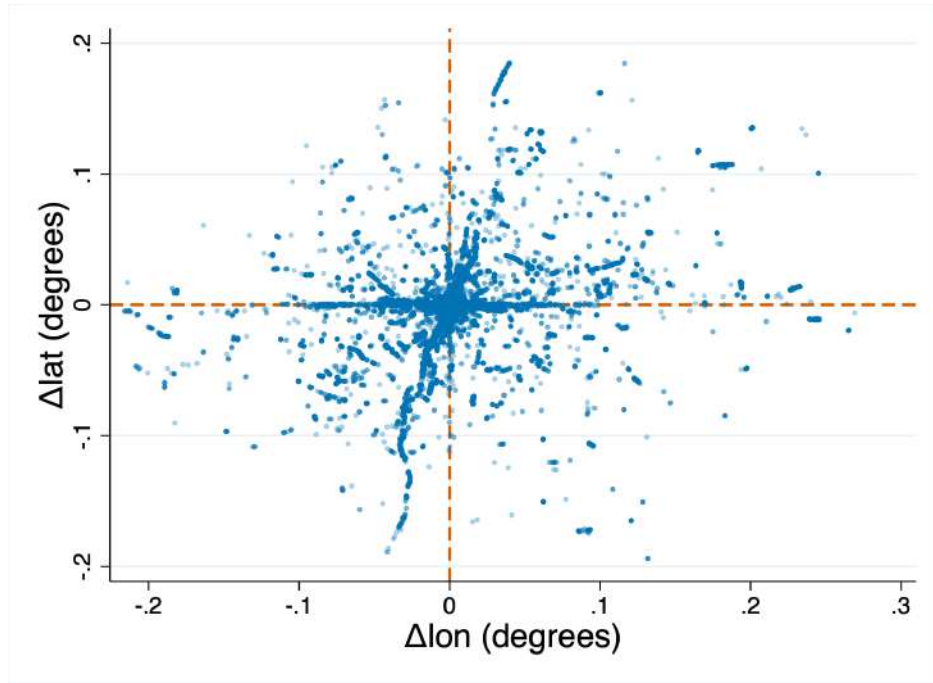


FIGURE A.4: Directional bias between modern and historical geocoding results.

Notes: Each point corresponds to a household in the 1950 census. The y-axis corresponds to the difference in latitude between the modern and historical geocoding results. The x-axis corresponds to the difference in longitude between the modern and historical geocoding results. The cities included in the figure are: Albany, NY; Buffalo, NY; Cincinnati, OH; Cleveland, OH; Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; and St. Louis, MO.

with coefficients that are small and statistically indistinguishable from zero (Appendix Figure A.6). The distribution of the predicted probability of staying is also common to the displaced, adjacent, and control groups (Appendix Figure A.7). I find no evidence that pre-construction mobility differed across the groups I compare.

A.6 Outmigration from the 1950 residence

I classify an individual as displaced based on the location of their 1950 residence, but highway construction did not begin until 1956. An individual who left their house between 1950 and 1956 is therefore recorded as displaced even though they were never displaced. This misclassification is one-sided. An individual recorded outside the right-of-way in 1950 cannot have been displaced, because their home was never taken, so the error only moves truly displaced individuals out of the displaced group. One-sided misclassification of a binary treatment attenuates the estimated effect toward zero. In this section I study how large this attenuation can be and bound the true displacement effect. I proceed in three steps. First, I construct bounds on the *true* effects using observed census-to-census mobility between 1940 and 1950. Second, I formalize the attenuation bias introduced by the nature of my data. Finally, I develop the theory that uses the predicted probability of outmigrating to retrieve the *true* coefficients.

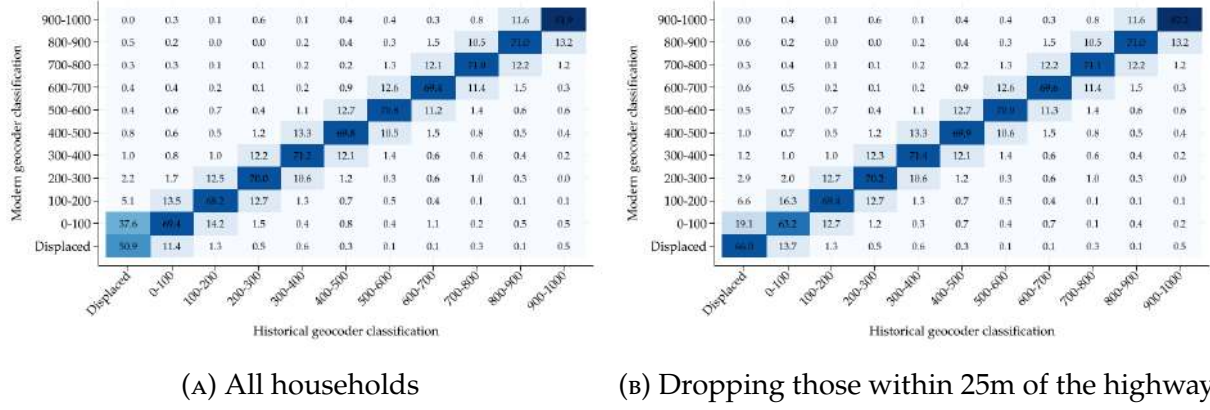


FIGURE A.5: Heatmap of correlation between modern and historical geocoding results.

Notes: Panel (a) shows the full sample; Panel (b) excludes individuals whose modern-geocoder distance to the highway is between 0 and 25 meters. Each cell reports the column-conditional percentage of individuals classified into a given modern-geocoder distance bin (y-axis) among those assigned to a particular historical-geocoder distance bin (x-axis). Distance bins are measured in meters from the highway network built before 1970. The modern geocoder uses ArcGIS address matching to current-day coordinates; the historical geocoder uses parcel-level geocoding to period-appropriate coordinates. The sample includes individuals from Albany, NY; Buffalo, NY; Cincinnati, OH; Cleveland, OH; Dayton, OH; Louisville, KY; Spokane, WA; Springfield, MA; and St. Louis, MO, restricted to those within 1,000 meters of a highway. “Displaced” denotes individuals whose residence falls within the highway right-of-way. Panel (b) removes near-boundary cases where small geocoding discrepancies are most likely to alter treatment classification.

A.6.1 Bounding exercise

The displacement gap captures differences between individuals who lived in a house that was destroyed by a highway and their neighbors who lived in a house that was not destroyed. Because highway construction started in 1956, individuals could have moved out of their houses before the highway construction started. Therefore, some individuals classified as displaced may have left before construction. In this section, I will bound the *true* displacement gap by exploiting the mobility of individuals between 1950 and 1956.

Without loss of generality, I will focus on the displacement effects. The effect for individuals living adjacent to the highway is analogous. The derivation abstracts from the controls in the main specification.⁴⁵ The estimated displacement gap comes from

$$y_i = \alpha + \beta D_i^{50} + \epsilon_i, \quad (\text{A.1})$$

where D_i^{50} equals one if individual i lived in 1950 in a house later destroyed by highway construction. Displacement, however, depends on where the individual lived when construction began, which I denote D_i^{56} , so the true effect comes from

$$y_i = a + b D_i^{56} + e_i, \quad (\text{A.2})$$

⁴⁵With controls the bias takes a slightly more complicated form. After partialling out the controls by Frisch-Waugh-Lovell the treatment indicator is no longer binary, but the logic is unchanged.

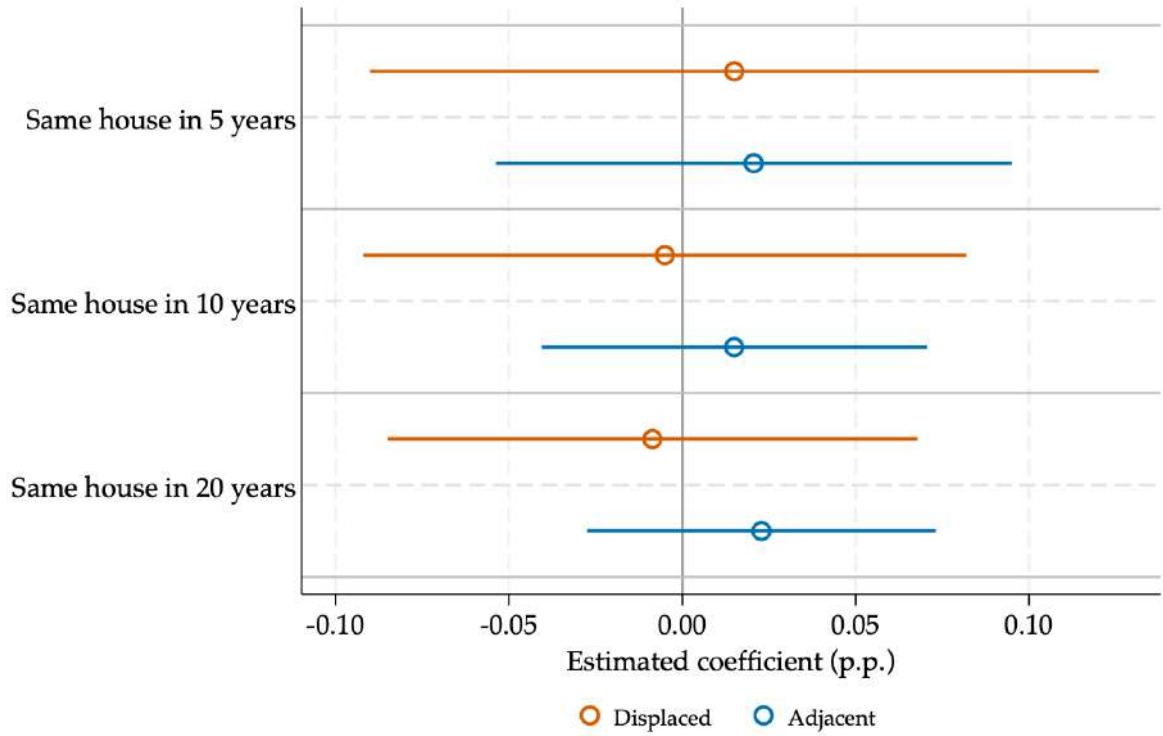


FIGURE A.6: Results on the probability of staying in the same residence

Notes: The figure plots the displaced (β_1 , orange) and adjacent (β_2 , blue) coefficients from estimating Equation 2 with the predicted probability of staying in the 1950 residence at different time horizons as the dependent variable, together with 95% confidence intervals. The probability of staying in the same home is estimated by a probit and projected out of sample onto the main dataset. The probit covariates are a cubic polynomial in age, race, and gender, as well as labor outcomes, interacted with the demographic covariates. Each row uses a different time horizon in the 1960 5% census. The estimation sample includes individuals within 200 meters of a highway built between 1956 and 1970. All regressions include the household head's occupational score and gender, race, birth-year, and metropolitan-area fixed effects. Standard errors are clustered at the metropolitan-area level.

where b corresponds to the *true* parameter.

Then, the probability limit of the estimated displacement effects, $\hat{\beta}$, is

$$\hat{\beta} \xrightarrow{p} \mathbb{E}[y_i \mid D_i^{50} = 1] - \mathbb{E}[y_i \mid D_i^{50} = 0] = b \left(\mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 1) - \mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 0) \right)$$

which depends on the probability of *actual* displacement based on my *observed* displacement. If no one moved between 1950 and construction, every recorded-displaced individual was truly displaced, $\left(\mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 1) - \mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 0) \right) = 1$, and $\hat{\beta} \xrightarrow{p} b$. Assuming the probability of being displaced in 1956 is strictly higher for those who lived within the right-of-way in 1950 than for those who did not, the magnitude of the true effect lies within:^{46,47}

⁴⁶An individual already living in a house slated for destruction is mechanically more likely to be displaced than one who is not. This assumption makes the true effect larger than the estimated effect and keeps the upper bound finite.

⁴⁷In case the estimated coefficient is negative, the bounds should be flipped.

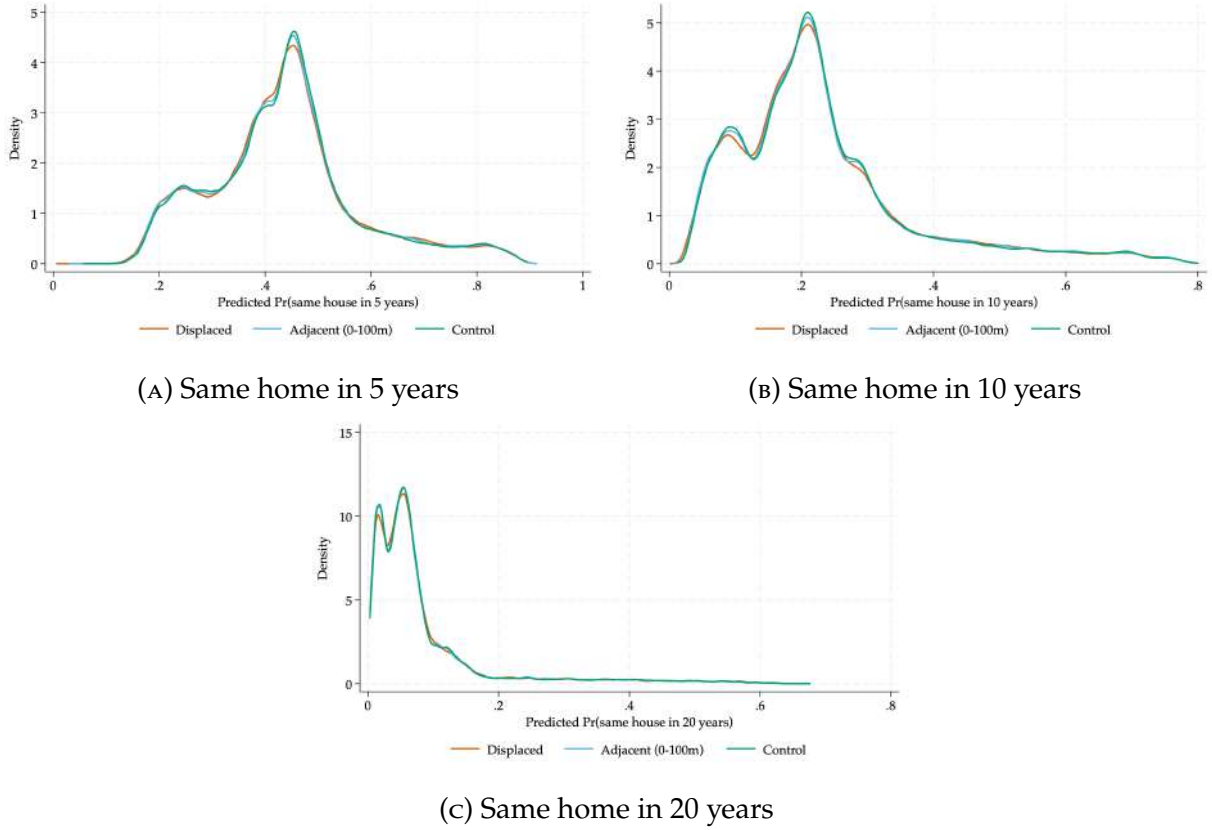


FIGURE A.7: Distribution of the probability of staying in the same home across treatments

Notes: The figure includes the probability density function of the probability of staying in the same home across different treatment groups: displaced, adjacent, and control (100–200 meters). All probabilities are the fitted values of a probit regression of the probability of staying in the same home at different time horizons using the 1960 5% census sample. Panel (a) plots the density of the probability of staying in the same home in 5 years, panel (b) in 10 years, and panel (c) in 20 years.

$$b \in \left[\hat{\beta}, \frac{\hat{\beta}}{\mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 1) - \mathbb{P}(D_i^{56} = 1 \mid D_i^{50} = 0)} \right]. \quad (\text{A.3})$$

The two probabilities in equation A.3 depend on the residence at the time of construction, which I do not observe. I therefore use the 1940-to-1950 transition as their empirical counterpart. I proxy the probability of still living there at construction given one lived there in 1950 with the probability of still living in a doomed house in 1950 given one lived there in 1940. The probability of moving into a doomed house between 1940 and 1950 stands in for moving into one between 1950 and construction. The linked 1940–1950 census is the only source in which I observe the same household at two dates that includes their displacement status in both periods.⁴⁸ To estimate these moments, I geocode the 1940

⁴⁸This is a noisy measure of remaining in place, because neighborhood trends between 1940 and 1950 need not match those between 1950 and 1956. The exercise also relies on the representativeness of the linked sample. Helgertz et al. (2022) documents the linkage method, and Appendix Table A.4 compares the linked and full samples. Linked individuals are older, more likely to be male and white, and more likely to be in the labor force, though most of these gaps follow mechanically from the requirement that linked individuals be at least 10 years old in 1950. For employment, occupational score, and household-head occupational

census and use the linkage of Ruggles et al. (2025) to match individuals from 1950 back to 1940, treating a person as having stayed if their 1950 residence lies within 30 meters of their 1940 residence. The estimated transition probabilities are

$$\begin{aligned}\hat{\mathbb{P}}(D_i^{50} = 1 \mid D_i^{40} = 1) &= 0.334, & \hat{\mathbb{P}}(D_i^{50} = 1 \mid D_i^{40} = 0) &= 0.006, \\ \hat{\mathbb{P}}(A_i^{50} = 1 \mid A_i^{40} = 1) &= 0.400, & \hat{\mathbb{P}}(A_i^{50} = 1 \mid A_i^{40} = 0) &= 0.017,\end{aligned}$$

where A_i^{40} and A_i^{50} are the analogous indicators for the adjacent group.

The estimated difference in probability is 0.328 for displaced individuals and 0.383 for those adjacent to the highway. These results imply that the displacement coefficient lies in $b^D \in [\hat{\beta}^D, 3.05\hat{\beta}^D]$ and the adjacency coefficient in $b^A \in [\hat{\beta}^A, 2.61\hat{\beta}^A]$.

A.7 Attenuation bias

Treatment in the main analysis is assigned from the address at which an individual is observed in the 1950 census. Highway construction, however, takes place over the following two decades, so an individual who left the 1950 address before construction is coded as treated although she had already left. The resulting misclassification is one-sided.⁴⁹ Every individual who remained until construction is correctly classified, while some of those coded as treated have true exposure of zero. In this section, I derive the consequences of this measurement-error structure for the bias.

Following the notation of Appendix Section A.6.1, D_i^{50} denotes the treatment assigned to individual i from her 1950 address and D_i^{56} denotes true exposure at the time of construction. The parameter defined by equation A.1 is:

$$\beta = \frac{\text{cov}(y_i, D_i^{50})}{\text{var}(D_i^{50})}, \quad (\text{A.4})$$

whereas the parameter of interest is b , defined as:

$$b = \frac{\text{cov}(y_i, D_i^{56})}{\text{var}(D_i^{56})}, \quad (\text{A.5})$$

Let S_i be an indicator that takes the value of one if individual i remained in their 1950 residence until construction: $D_i^{56} = D_i^{50} S_i$. Then I can define the probability of an individual staying in their home from 1950 until construction as,

score, the linked and full samples are very similar.

⁴⁹The previous section shows that the probability of moving into a soon-to-be destroyed house is small. This estimate is likely overstating that probability, as it is calculated before the 1956 Federal-Aid Highway Act. Once enacted, the likelihood an individual moved into a home doomed for construction is lower (Lin et al., 2023).

$$s_i \equiv \widehat{\Pr}(S_i = 1 \mid W_i) = 1 - p_i, \quad (\text{A.6})$$

where p_i is the predicted probability of moving and W_i corresponds to the individual characteristics used to estimate the probability. The probability in equation A.6 is unconditional on displacement status. It estimates the baseline residential mobility for an individual with characteristics W_i and is estimated for the entire sample.

I can retrieve the *true* coefficient under the following assumptions.

1. **Exogenous assigned treatment.** $E[e_i \mid D_i^{50}, W_i] = 0$. Destruction of the 1950 address is as good as randomly assigned. This assumption is likely to follow from the discussion in Section 5.
2. **No effect for movers.** Destruction of the 1950 address does not affect the outcomes of individuals who had already left.
3. **No anticipation.** $S_i \perp D_i^{50} \mid W_i$. Residential mobility out of the 1950 address is independent of whether that address was later destroyed. This assumption is supported by the results in Appendix Section A.5.
4. **One-sided error.** There is no in-migration after 1950 to homes that will be destroyed by highway construction.

A.7.1 Attenuation of the estimated coefficients

How are b and β related? By plugging equation A.2 into equation A.4, and using the relationship $D_i^{56} = S_i D_i^{50}$ and assumption 1, we can express β as,

$$\beta = \frac{\text{cov}(S_i D_i^{50}, D_i^{50})}{\text{var}(D_i^{50})} = b \cdot \frac{E[D_i^{50} S_i]}{E[D_i^{50}]} = b \cdot \Pr(S_i = 1 \mid D_i^{50} = 1) \leq b \quad (\text{A.7})$$

The coefficient recovered from the naive regression, β , equals the parameter of interest scaled by the share of displaced individuals who stayed in their homes. The structure of equation A.7 is that of an intent-to-treat estimate. Regressing outcomes on the treatment status defined by the 1950 census recovers the *true* parameter times the probability of “compliance”.

A.7.2 The corrected estimator

Let me define $\tilde{D}_i$ as the expected treatment status conditional on observed treatment status and covariates. Then we can re-write it as,

$$\tilde{D}_i \equiv E [D_i^{56} | D_i^{50}, W_i] = D_i^{50} E [S_i | D_i^{50}, W_i] = D_i^{50} s_i, \quad (\text{A.8})$$

where the final equality uses the assumption that the probability of staying is orthogonal to the observed treatment status (Assumption 3). Then, the corrected treatment, i.e., $\tilde{D}_i$, is the observed treatment scaled by the probability of staying. The same construction applies to both treatments, since displacement and adjacency share the same structure.

We can decompose the *true* exposure around its corrected value,

$$D_i^{56} = \tilde{D}_i + v_i, \quad v_i = D_i^{50} (S_i - E [S_i | D_i^{50}, W_i]). \quad (\text{A.9})$$

By construction $E [v_i | D_i^{50}, W_i] = 0$, so v_i is orthogonal to any function of the observed data. Since $\tilde{D}_i$ is itself such a function of D_i^{50} and W_i , by the law of iterated expectations:

$$E [\tilde{D}_i v_i] = E [\tilde{D}_i E [v_i | D_i^{50}, W_i]] = 0. \quad (\text{A.10})$$

Then, substituting equation A.9 into equation A.2 yields an equation I can take into the data.

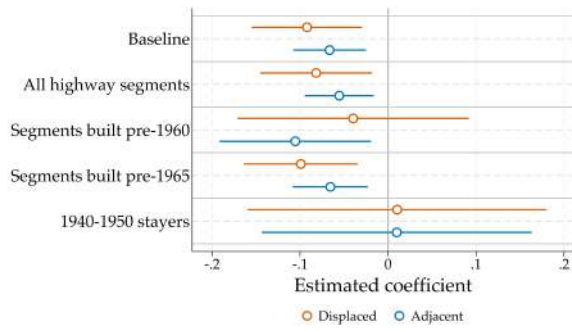
$$y_i = a + b \tilde{D}_i + u_i, \quad u_i = b v_i + e_i. \quad (\text{A.11})$$

Consistency requires that $E [\tilde{D}_i u_i] = 0$. Notice that the two components of the composite error vanish. The first term vanishes mechanically by equation A.10. Assumptions 1 and 2 guarantee that $E [\tilde{D}_i e_i] = 0$. Then, it follows that the least squares estimator of equation A.11 satisfies

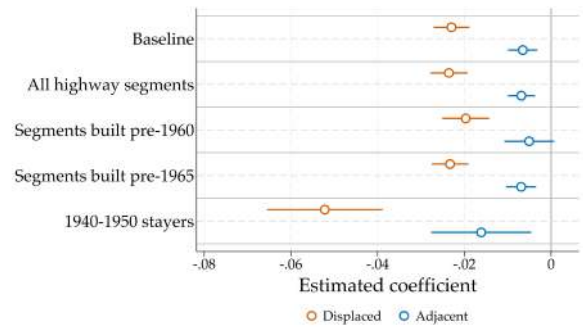
$$\text{plim } \hat{b} = b. \quad (\text{A.12})$$

A.8 Outmigration, other results

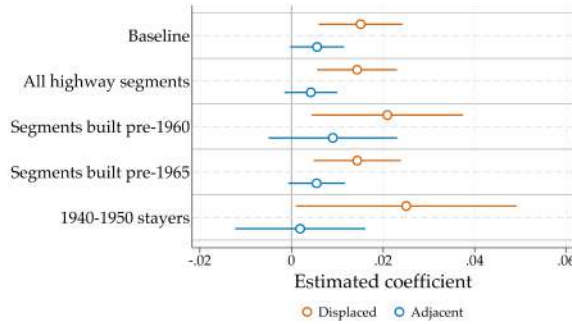
The conclusion is robust to several alternatives. Restricting the sample to individuals affected by early highway segments or those observed in the same home in both 1940 and 1950, for whom misclassification is least likely, leaves the estimates intact (Appendix Figure A.8). The effects are larger for older individuals, who are less likely to move out of their home (Appendix Figure A.9). And the estimates are stronger for individuals displaced closer to the 1950 census, when measurement error is smallest (Appendix Figure A.10).



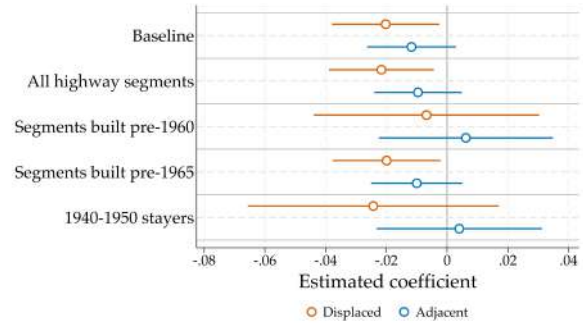
(A) Death age



(B) Same neighborhood



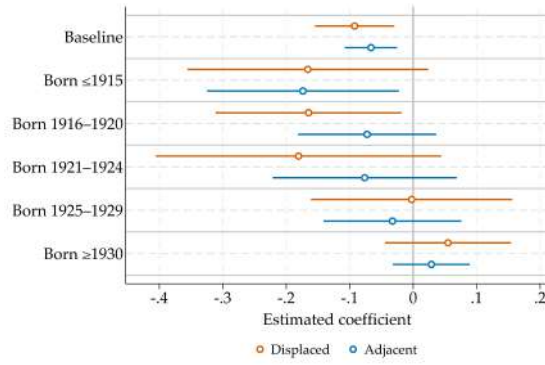
(C) Same city



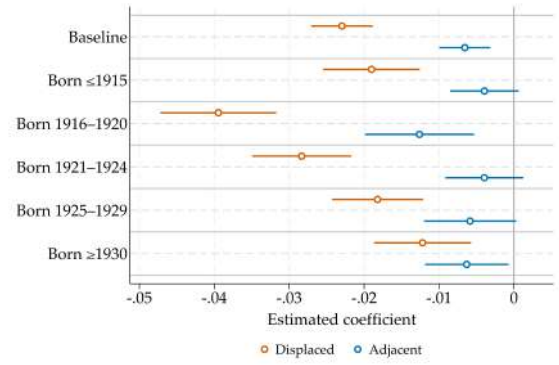
(D) Neighborhood quality

FIGURE A.8: Robustness to different outmigration probabilities.

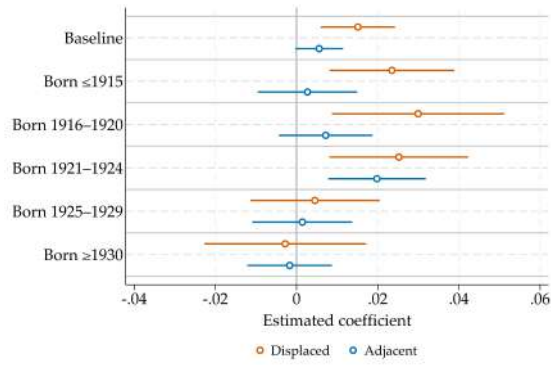
Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top panel corresponds to the baseline results. *All highway segments* includes all segments in the Interstate Highway System. *Built pre-1960* restricts to segments completed before 1960. *Built pre-1965* restricts to segments completed before 1965. *1940-1950 stayers* restricts to the sample of (linked) individuals who were living in the same home between 1940 and 1950.



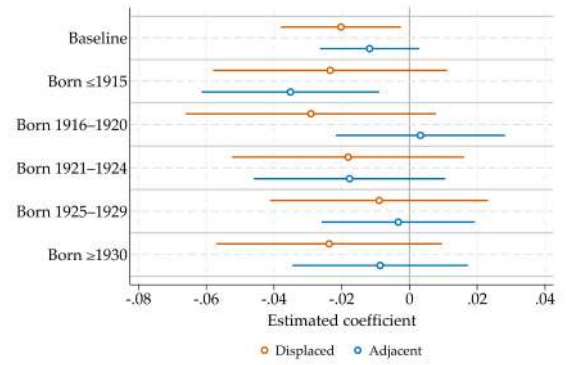
(A) Death age



(B) Same neighborhood



(C) Same city



(D) Neighborhood quality

FIGURE A.9: Effects by different age cohorts.

Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals by different age groups. The sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top panel corresponds to the baseline results.

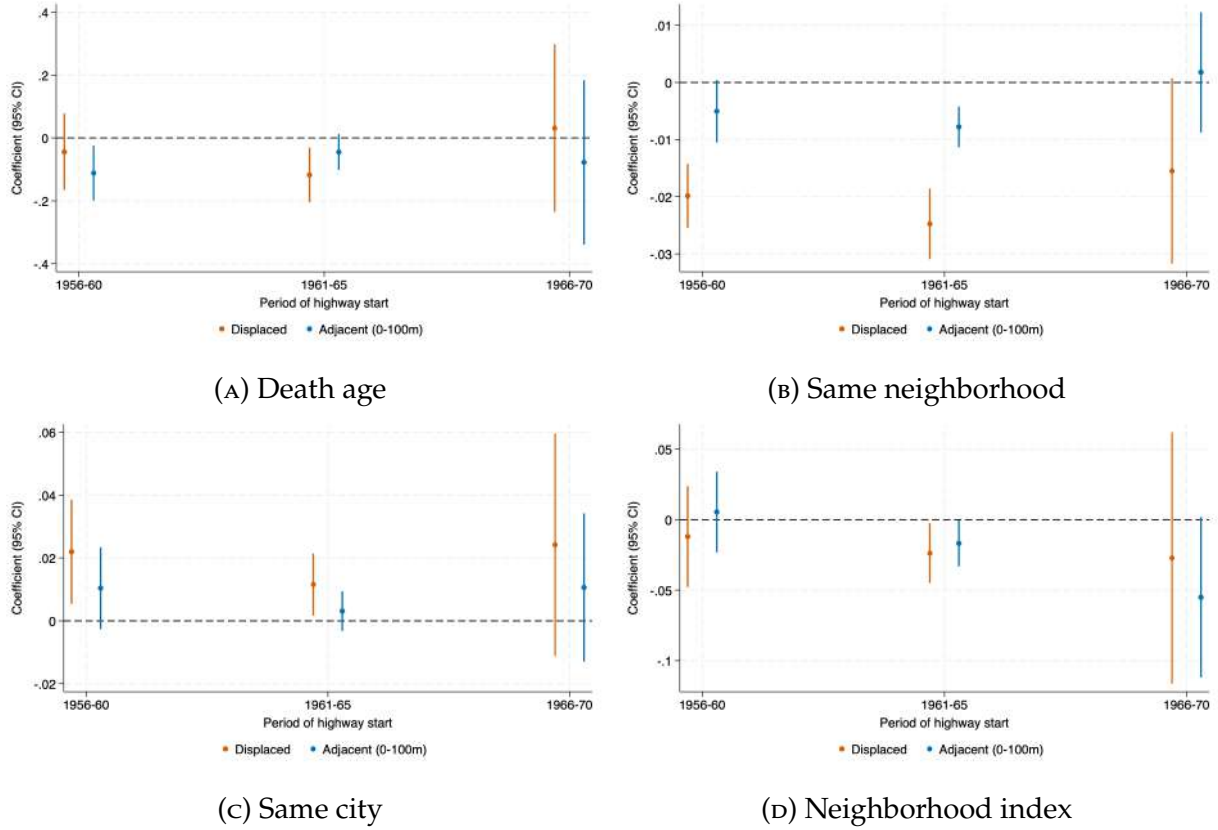


FIGURE A.10: Results by year of displacement

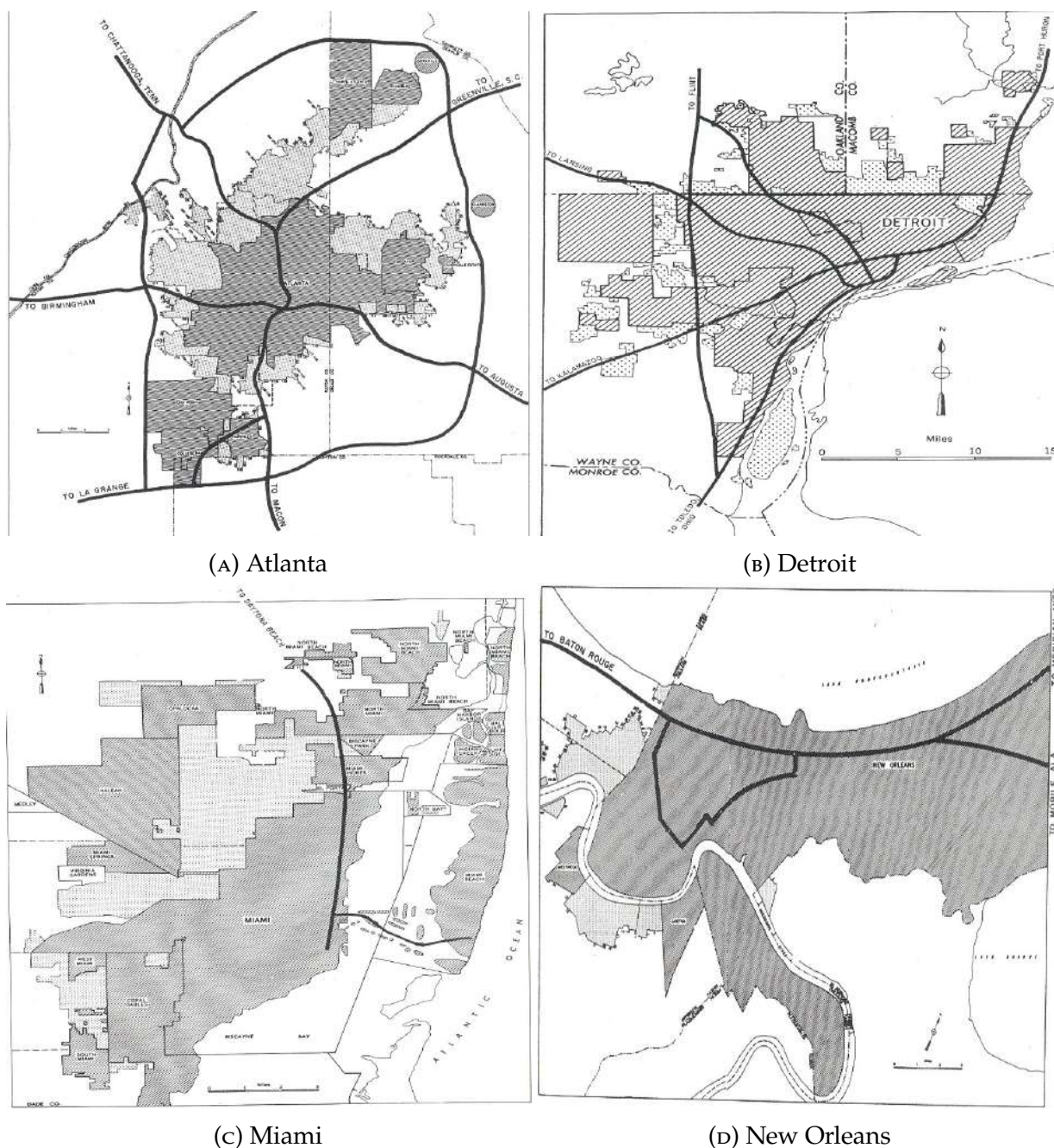
Notes: The figure plots event-study coefficients from estimating Equation 2 separately by the year in which displacement occurred. All regressions control for household-head occupational score and race, gender, and birth-year dummies, as well as metropolitan-area fixed effects. Standard errors are clustered at the metropolitan-area level. Panel (a) presents the results for death age, Panels (b) and (c) for mobility away from the neighborhood and the city, respectively, and Panel (d) for the neighborhood quality at death.

A.9 Linkage between the 1950 census and the NUMIDENT records

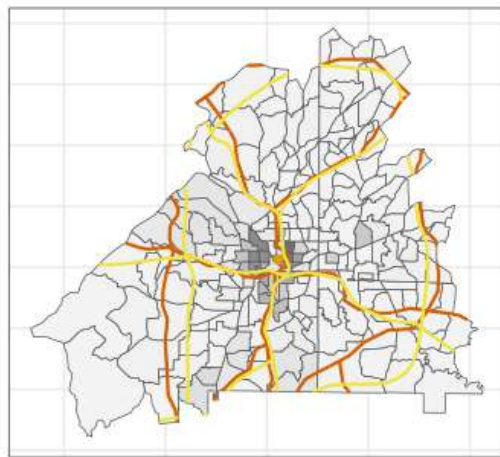
To maximize the linkage between the 1950 census and the NUMIDENT records, I combine two linkage methods. The first, the IPUMS Multigenerational Longitudinal Panel version 2.0 (Ruggles et al., 2025), uses machine learning to create links across databases. The second uses the “ABE-EI” algorithm developed by the Census Linking Project (CLP) (Abramitzky et al., 2022, 2024), which matches individuals based on time-invariant characteristics and uses extra information to resolve ties. For individuals who cannot be linked directly from the 1950 census to the NUMIDENT, I use Ruggles et al.’s (2025) census-to-census crosswalk to link them to the 1940 census and then match the resulting 1940 records to the NUMIDENT dataset. When both methods link the same Social Security Number to different individuals in the 1950 census, I use the direct 1950-NUMIDENT link.

I combine the linked data as follows: of the 1950-NUMIDENT links identified by either MLP or CLP, 1.1% were identified in both MLP and CLP crosswalks, 81.6% were in MLP only, and 17.3% were in CLP only. For the cases where the MLP and CLP links differed, I

used the MLP link, although the results are robust to using CLP links instead. Appendix Figure A.15 shows that the results do not change when using only MLP links or CLP links, but the sample size decreases, increasing standard errors, and some results are not significant at the usual levels.



Notes: The figure includes the maps in the Yellow Book for the cities of Atlanta, Detroit, Miami, and New Orleans.



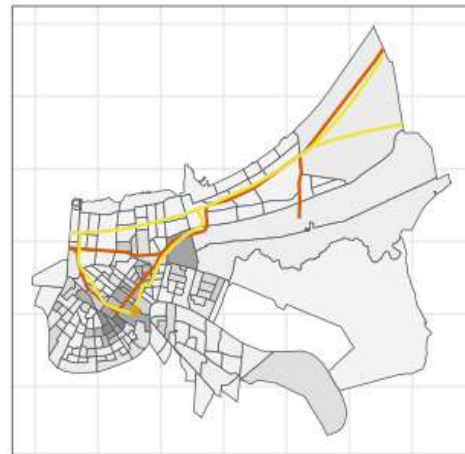
(A) Atlanta



(B) Detroit



(c) Miami



(D) New Orleans

FIGURE A.12: Racial Distribution, Highways, and Planned Routes

Notes: The figure includes maps for Atlanta, Detroit, Miami, and New Orleans. Each observation is a census tract, and its filling corresponds to the number of Black residents in the tract in 1950. Depicted in red is the highway network that was built. The network planned in the Yellow Book is presented in yellow. Finally, the city center is plotted in orange.

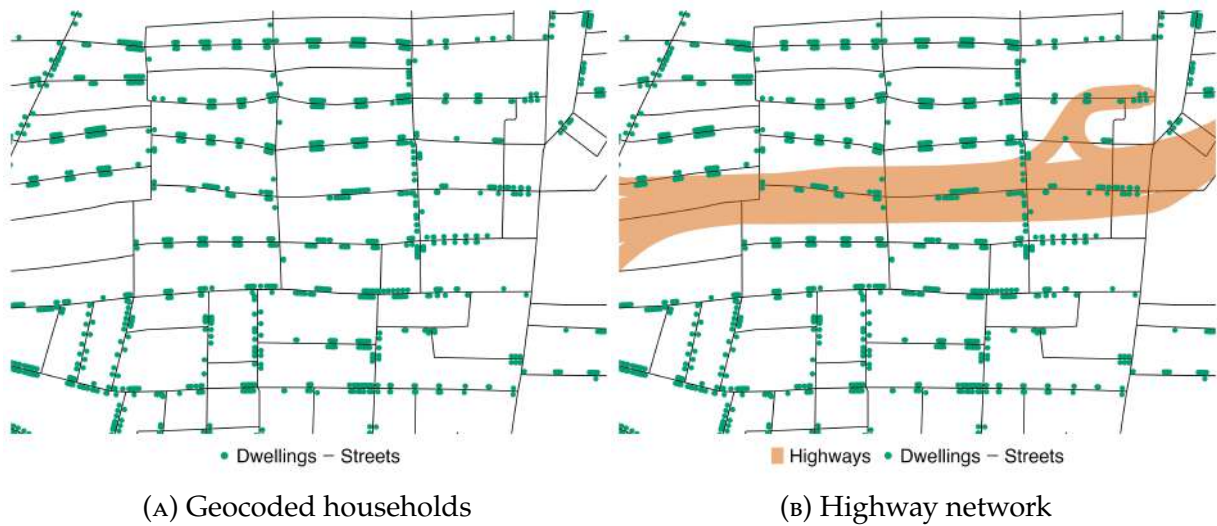


FIGURE A.13: Example of geocoded households in the 1950 census.

Notes: Each green point corresponds to a household in the 1950 census. Panel (a) shows the geocoded households in Cleveland, OH, and panel (b) shows the highway network later built.

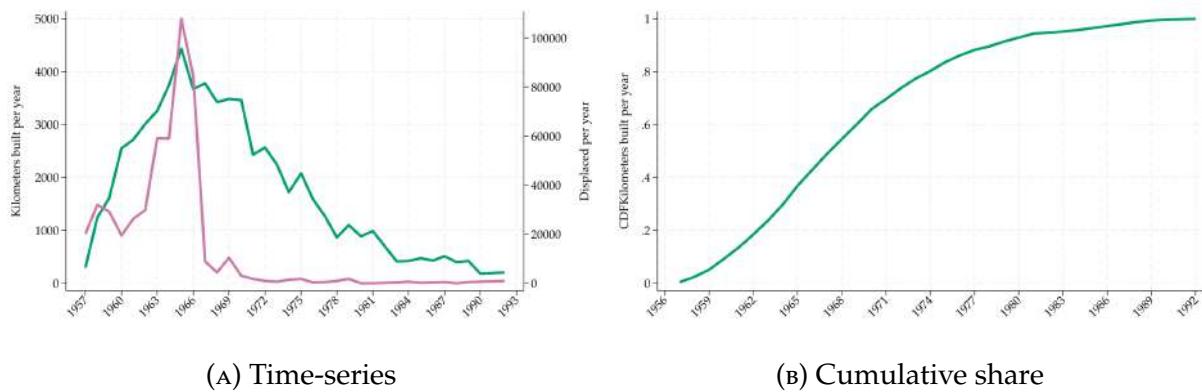


FIGURE A.14: Kilometers of Interstate highways built per year.

Notes: The data in this figure come from Brooks and Liscow (2023) and my own analysis. Panel (a) shows the kilometers of Interstate highways built and the number of individuals displaced per year, and panel (b) shows the share of total kilometers built up until each year.

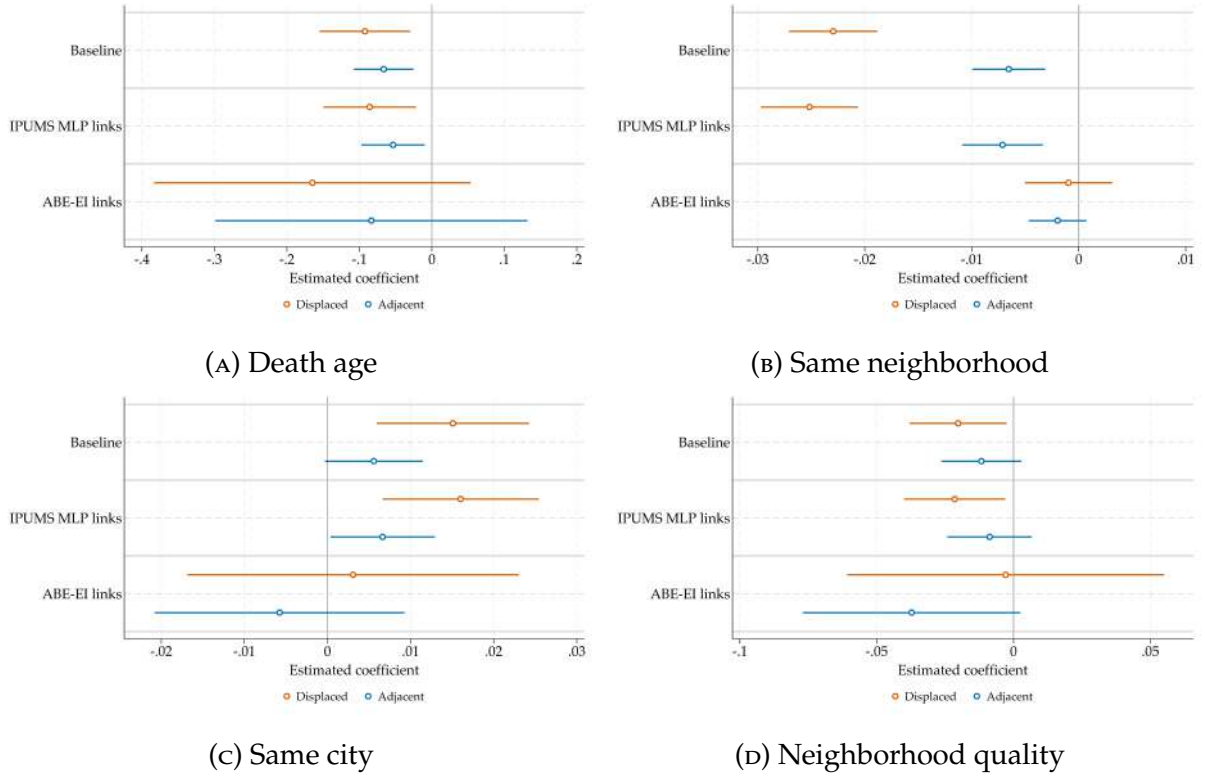
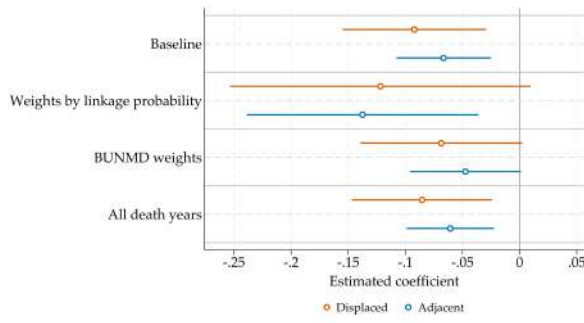
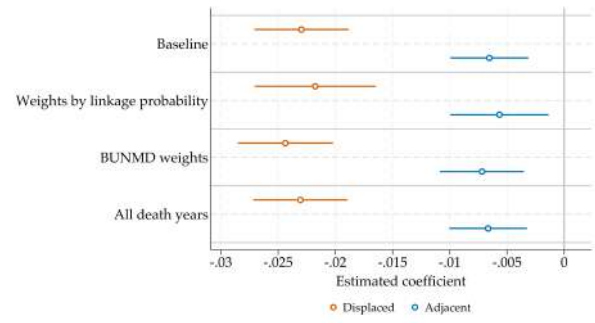


FIGURE A.15: Robustness linkage method.

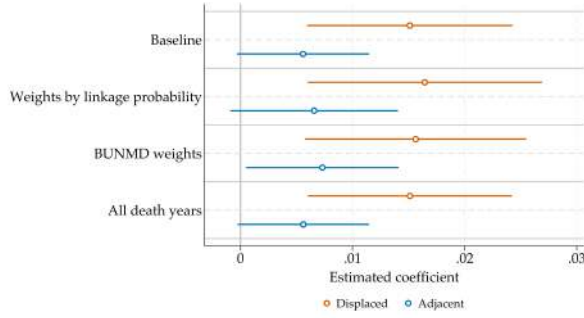
Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. The sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top panel corresponds to the baseline results. The second row uses only those individuals in the 1950 census linked to the mortality records using Helgertz et al. (2022) links. The last row uses only those individuals linked using the Abramitzky et al. (2024) algorithm.



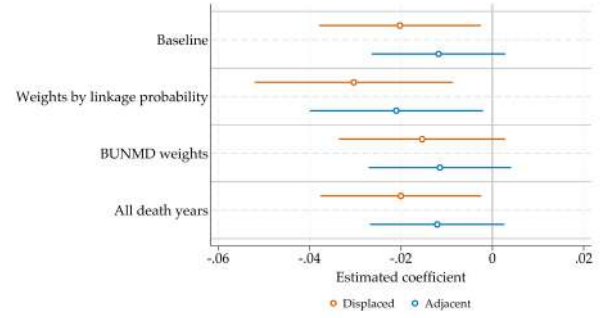
(A) Death age



(B) Same neighborhood



(C) Same city



(D) Neighborhood quality

FIGURE A.16: Robustness data quality.

Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. The sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top panel corresponds to the baseline results. The second row of each panel weights each observation by the inverse of their probability of being included in the sample of geocoded-and-linked individuals. The third row uses Breen et al. (2023) weights, which correct for unrepresentativeness in the NUMIDENT dataset across cohorts-race-gender. The last row extends the mortality years prior to 1988, when the mortality records were not representative.

A.11 Additional tables

TABLE A.1: Geocoded households by state

State	Households	Geocoded	Share
Alabama	219,546	118,025	53.76%
Arizona	52,000	34,155	65.68%
Arkansas	48,697	30,294	62.21%
California	2,260,200	1,419,042	62.78%
Colorado	175,096	98,000	55.97%
Connecticut	318,914	194,638	61.03%
Delaware	38,251	22,571	59.01%
District of Columbia	261,726	170,954	65.32%
Florida	300,689	215,877	71.79%
Georgia	239,220	13,275	5.55%
Illinois	1,754,758	1,044,632	59.53%
Indiana	427,419	294,591	68.92%
Iowa	179,858	130,924	72.79%
Kansas	125,027	79,805	63.83%
Kentucky	194,775	124,547	63.94%
Louisiana	263,824	183,648	69.61%
Maine	37,961	22,628	59.61%
Maryland	311,664	207,771	66.67%
Massachusetts	1,001,544	537,184	53.64%
Michigan	994,475	624,279	62.77%
Minnesota	371,015	215,124	57.98%
Mississippi	29,770	16,057	53.94%
Missouri	582,270	352,632	60.56%
Nebraska	114,856	79,519	69.23%
New Hampshire	37,098	18,176	48.99%
New Jersey	1,008,146	596,849	59.20%
New Mexico	36,024	18,811	52.22%
New York	3,451,076	1,788,624	51.83%
North Carolina	166,316	70,066	42.13%
Ohio	1,327,072	739,360	55.71%
Oklahoma	154,549	92,802	60.05%
Oregon	147,216	112,643	76.52%
Pennsylvania	1,722,267	1,044,917	60.67%
Rhode Island	183,745	120,681	65.68%
South Carolina	95,112	37,228	39.14%
South Dakota	17,086	8,054	47.14%
Tennessee	272,132	148,219	54.47%
Texas	960,195	587,464	61.18%
Utah	85,641	59,098	69.01%
Virginia	290,152	152,936	52.71%
Washington	318,449	224,738	70.57%
West Virginia	105,695	57,976	54.85%
Wisconsin	347,823	200,208	57.56%
Total	21,029,348	12,309,022	58.53%

TABLE A.2: Probability of linkage to NUMIDENT by treatment status

	(1)	(2)	(3)	(4)	(5)	(6)
Displaced	-0.00174 ^b (0.00080)	-0.00124 ^c (0.00070)	-0.00067 (0.00064)	-0.00082 (0.00057)	-0.00088 (0.00056)	-0.00096 ^c (0.00056)
Adjacent	-0.00019 (0.00047)	0.00003 (0.00039)	0.00019 (0.00038)	0.00010 (0.00035)	0.00016 (0.00036)	0.00006 (0.00036)
Observations	2,874,666	2,874,666	2,874,666	2,874,661	2,874,661	2,861,346
Adjusted R-squared	0.000	0.001	0.001	0.044	0.063	0.063
Mean dependent variable	0.089	0.089	0.089	0.089	0.089	0.089
Std. dev. dependent variable	0.285	0.285	0.285	0.285	0.285	0.285
MSA FE	No	Yes	Yes	Yes	Yes	Yes
Race FE	No	No	Yes	Yes	Yes	Yes
Birth year FE	No	No	No	Yes	Yes	Yes
Gender FE	No	No	No	No	Yes	Yes
HH's occ. score	No	No	No	No	No	Yes

Note: OLS estimates are reported. Coefficients are reported with standard errors, clustered at the MSA level, in parentheses. The unit of observation is an individual in the 1950 census. The dependent variable is an indicator of whether the individual was linked to a NUMIDENT record in the 1950 census. The sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

TABLE A.3: Representativeness of the sample

	All	Valid address	Geocoded	Linked NUMIDENT
Age	33.090 (20.849)	33.046 (20.932)	33.219 (20.957)	26.625 (15.755)
Female	0.516 (0.500)	0.516 (0.500)	0.517 (0.500)	0.316 (0.465)
Black	0.106 (0.307)	0.102 (0.303)	0.100 (0.300)	0.081 (0.273)
Employed	0.949 (0.220)	0.949 (0.221)	0.949 (0.221)	0.944 (0.230)
Labor Force	0.430 (0.495)	0.429 (0.495)	0.432 (0.495)	0.567 (0.495)
Occupational score	10.552 (14.252)	10.660 (14.302)	10.748 (14.321)	14.441 (15.017)
Household head occscore	22.282 (14.817)	22.649 (14.707)	22.639 (14.712)	23.006 (14.319)
Observations	66,746,327	49,071,153	40,204,341	3,521,085

Note: The unit of observation is an individual in the 1950 census living in metropolitan areas. Column (1) reports the values for the full sample, whereas column (2) reports them for the sample of individuals with a valid address in the 1950 census. Column (3) reports the values for the sample of individuals geocoded in the 1950 census. Finally, column (4) reports the values for the sample of geocoded individuals linked to a NUMIDENT record.

TABLE A.4: Representativeness of the sample linked to the 1940 census

	Full sample	Linked to 1940	Difference
Age	33.106 (21.136)	39.763 (18.591)	13.229 ^a (0.006)
Female	0.517 (0.500)	0.492 (0.500)	-0.050 ^a (0.000)
Black	0.095 (0.293)	0.061 (0.239)	-0.068 ^a (0.000)
Employed	0.949 (0.219)	0.953 (0.212)	0.009 ^a (0.000)
Labor Force	0.425 (0.494)	0.514 (0.500)	0.176 ^a (0.000)
Occupational score	10.536 (14.253)	13.146 (15.073)	5.187 ^a (0.004)
Household head occscore	22.471 (14.760)	22.768 (14.976)	0.591 ^a (0.004)
Observations	51,546,156	25,605,766	

Note: The unit of observation is an individual in the 1950 census living in metropolitan areas. Column (1) reports the values for the full sample, whereas column (2) reports them for the sample of individuals linked to the 1940 census. OLS estimates of the difference between the two samples are reported in column (3). ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

B. PLACEMENT APPENDIX

B.1 Additional results using 1950 microdata

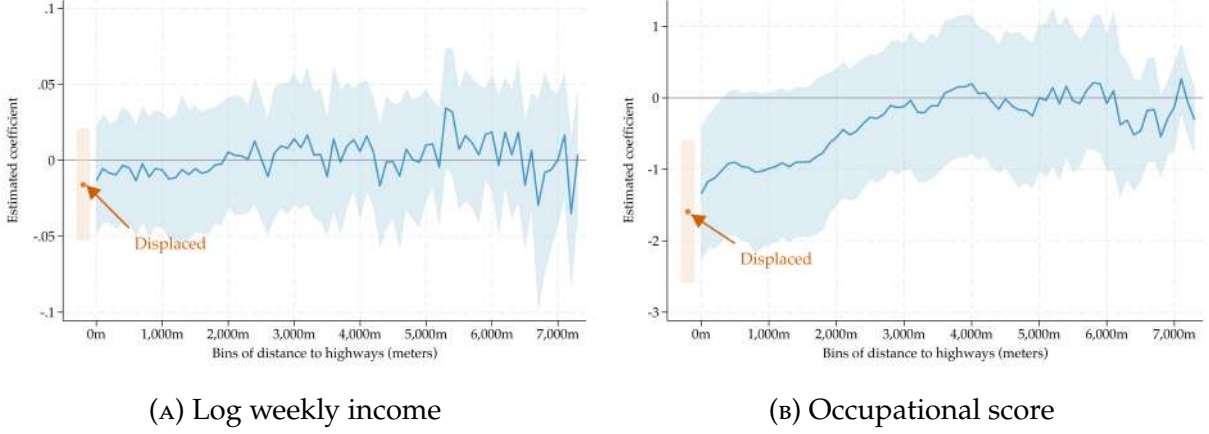


FIGURE B.1: Pre-construction labor characteristics by distance to future Interstates.

Notes: The figures present the estimated coefficients and confidence intervals of equation 1. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. The sample consists of individuals in metropolitan areas who lived within 7,500 meters of a future Interstate in 1950. The comparison group corresponds to those living between 7,400 and 7,500 meters from the future Interstate. The dependent variables are reported in each panel's title. All regressions include metropolitan area fixed effects and controls for distance to the city center. Standard errors are clustered at the metropolitan area level.

B.2 Analysis using census tracts

In this section, I present cross-sectional evidence on the relationship between a neighborhood's socioeconomic and geographic characteristics and future highway construction. To do this, I exploit variation across census tracts in 1950, the last recorded census before the 1956 Federal-Aid Highway Act that initiated highway construction. I transform the 1950 census tract data into the 2010 census tract definition using area-weighted interpolation, following Lee and Lin's (2018) crosswalk. The estimating equation follows:

$$y_n = \lambda_{c(n)} + \gamma \log(\text{DistCBD})_n + \beta \text{Black Share}_n + \Gamma'_1 \mathbf{S}_n + \Gamma'_2 \mathbf{P}_n + \Gamma'_3 \mathbf{G}_n + \epsilon_n \quad (\text{B.1})$$

The sample consists of census tracts from 62 metropolitan areas that have spatial information available for 1950. In this equation, n indexes census tracts, and $c(n)$ indexes the metropolitan area in which the census tract is located. The dependent variable, y_n , takes a value of one if a highway was built through the census tract and zero otherwise. The variable Black Share_n corresponds to the share of the city's Black population residing in

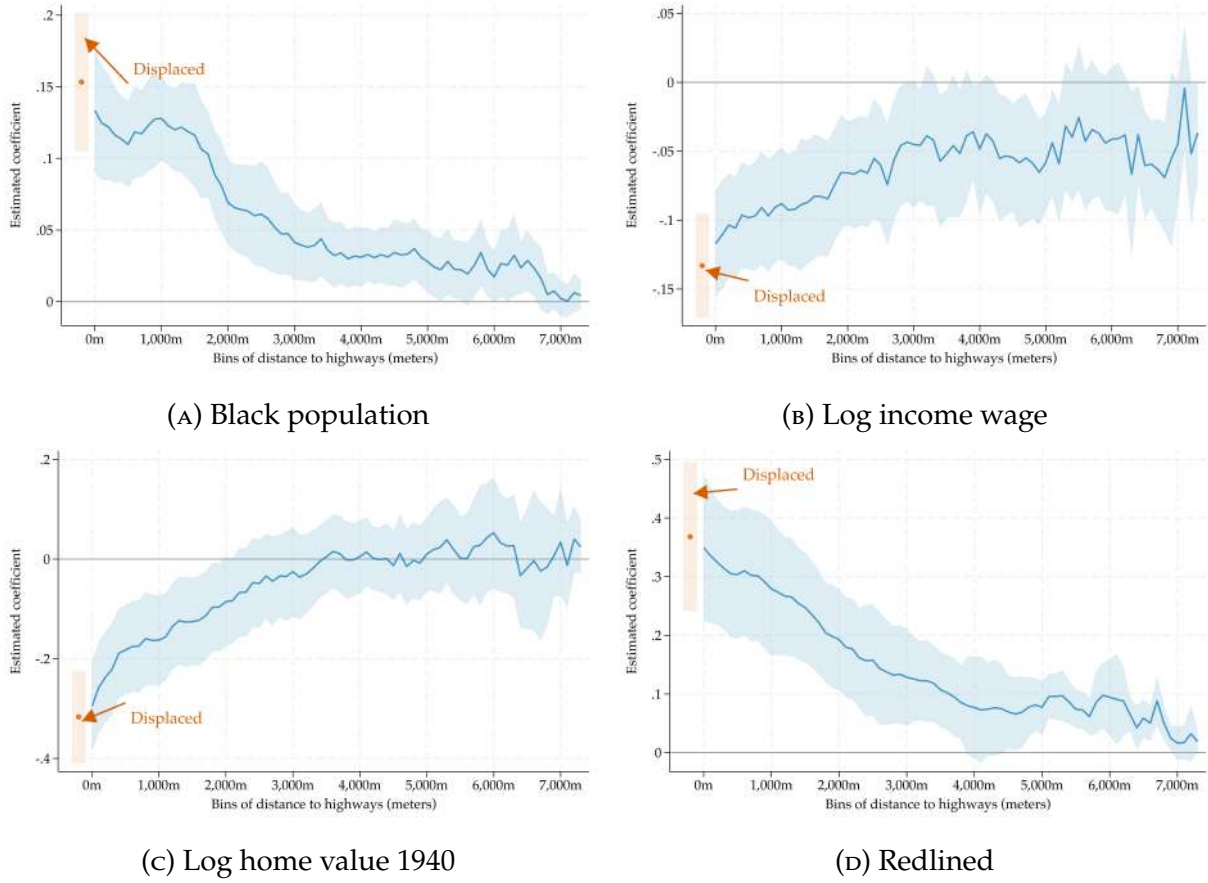


FIGURE B.2: Pre-construction characteristics by distance to future Interstates – Only MSA Fixed Effects.

Notes: The figures present the estimated coefficients and confidence intervals of equation 1, replicating the main-text figures without controlling for the individual's log distance to the city center. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. The sample consists of individuals in metropolitan areas who lived within 7,500 meters of a future Interstate in 1950. The comparison group corresponds to those living between 7,400 and 7,500 meters from the future Interstate. The dependent variables are reported in each panel's title. All regressions include metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

the tract. The variable $\log(\text{DistCBD})_n$ is the log of the distance to the central business district, which is included to account for the fact that highways were built to connect city centers and Black households sorted themselves into city centers (Boustan, 2010). The vector $\mathbf{S}_n$ contains socioeconomic characteristics of the tract, such as the median income and the share of the adult population with a high school degree. The vector $\mathbf{P}_n$ contains controls for the log median rent and the log median home value. These controls are included to account for the price of land in the neighborhood. The vector $\mathbf{G}_n$ contains geographic and state controls, including log average slope in degrees, the log area, the distance to the nearest river and to the railroad network, the number of cars per 10,000 inhabitants, and the governor's political party. $\lambda_{c(n)}$ corresponds to the city fixed effects. The regression results are weighted by the total population of tract n in 1950, and the standard errors are clustered at the city level.

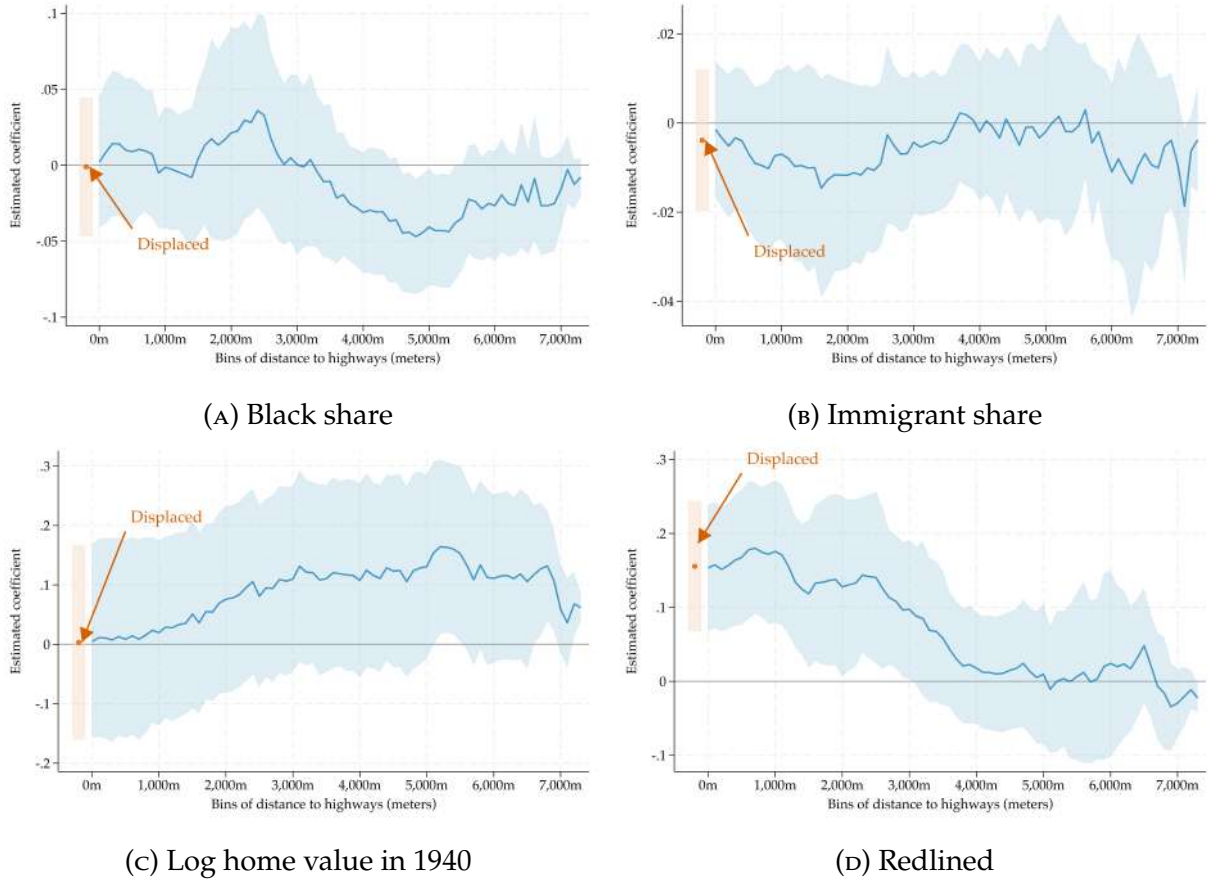


FIGURE B.3: Pre-construction characteristics by distance to the Yellow Book highways.

Notes: The figures plot estimated coefficients and confidence intervals from equation 1 for distance to planned highways. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands; the red marker indicates the coefficient for individuals who would have been displaced. The sample includes individuals residing in metropolitan areas with planned highways who lived within 7,500 meters of the planned highway in 1950, with those living between 7,400 and 7,500 meters serving as the comparison group. The dependent variable in each panel is indicated by the panel title. Panel (c) restricts the sample to individuals linked to the 1940 census who lived in the same house in 1950 and 1940. All regressions include metropolitan area fixed effects and control for log distance to the city center. Standard errors are clustered at the metropolitan area level.

The variable Black Share_n corresponds to the share of the city's Black population residing in tract n in 1950:

$$\text{Black Share}_n = \frac{\text{Black population}_n}{\text{Black population}_{c(n)}}$$

This definition differs from the conventional measure used in the urban economics literature (Weiwu, 2025), which captures the share of a tract's *total* population that is Black. The definition used here is motivated by qualitative accounts suggesting that highway builders targeted neighborhoods “where Black individuals live” (Rose and Mohl, 2012), which implies that the concentration of Black residents within a city, not their share of any given tract's population, was what led routing decisions. To illustrate, consider a city with two neighborhoods: one with 1,000 Black residents out of a total population of 2,000, and a

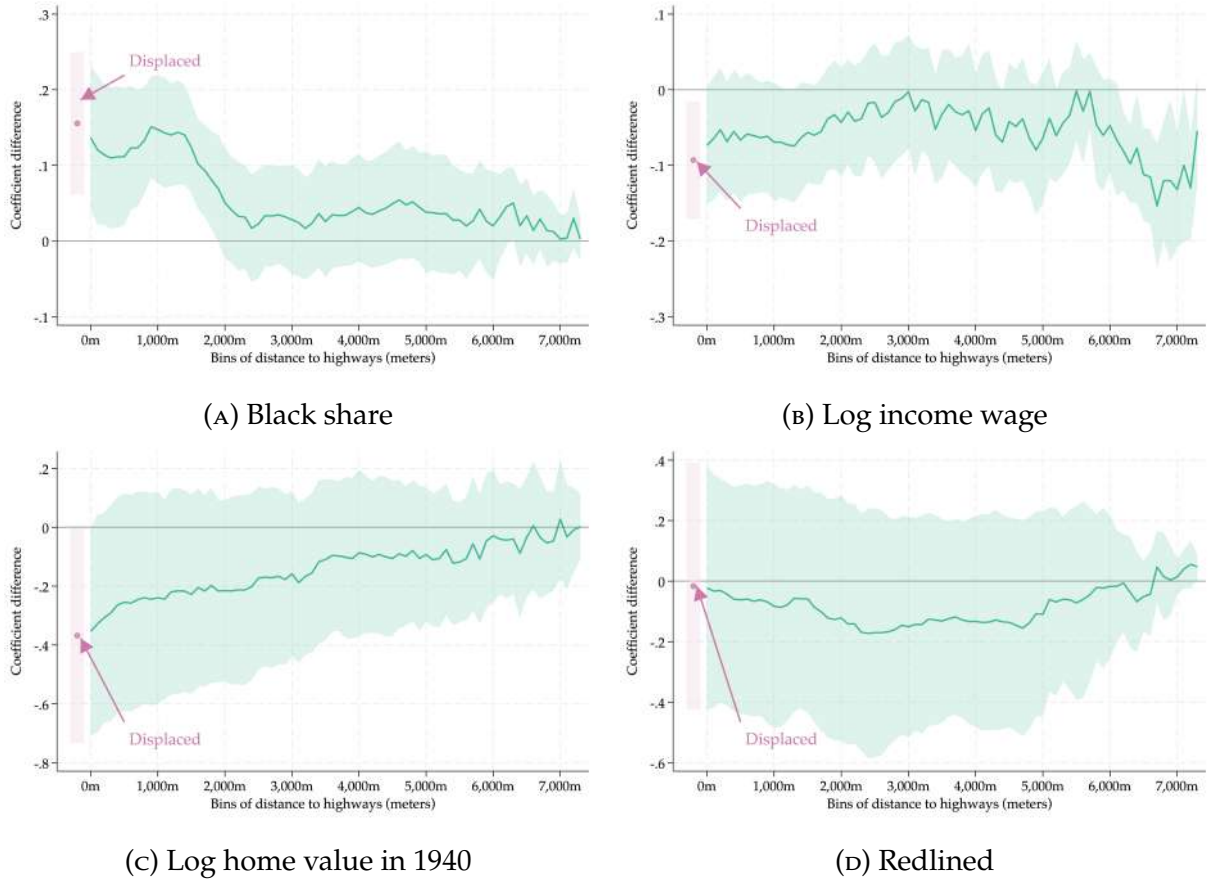


FIGURE B.4: Difference between planned and built highways at each distance.

Notes: The figures plot the estimated difference between coefficients for distance to the built highway network and the coefficient for distance to the Yellow Book maps from a regression that includes both running variables together. The green line plots the difference between the distance coefficients for the actual and planned highways $\beta_k^H - \beta_k^{YB}$ with 95% confidence intervals shown as shaded bands. The pink marker indicates the same difference for displaced individuals and for those who would have been displaced. The sample includes individuals residing in metropolitan areas with built and planned highways who lived within 7,500 meters of the planned highway in 1950, with those living between 7,400 and 7,500 meters serving as the comparison group. The dependent variable in each panel is indicated by the panel title. Panel (c) restricts the sample to individuals linked to the 1940 census who lived in the same house in 1950 and 1940. All regressions include metropolitan area fixed effects and control for log distance to the city center. Standard errors are clustered at the metropolitan area level.

second with 100 Black residents out of a total population of 100. The conventional measure assigns a Black share of 50% to the first neighborhood and 100% to the second, suggesting the second is more racially salient. Yet the first neighborhood contains roughly 90% of the city's Black population, the quantity that better reflects the targeting motive described above. Appendix Figure B.5 shows the demeaned distribution of this variable across census tracts in the sample. It shows that most census tracts had a small share of the city's Black population, with long tails on both sides of the distribution, showing the prevalent racial segregation at the time (Cutler et al., 1999).

Estimates of equation B.1 are reported in Appendix Table B.1. I find that, in equilibrium, highways were built through neighborhoods with lower land prices and a larger Black

share, and closer to the central business district. Columns (1) to (5) present the results for the network of highways built. In all specifications, Black households reside closer to future highway developments, which may partly reflect the political agendas of the builders, which led to unequal placement of highways (Trounstein, 2018). In addition, highways were built in neighborhoods with an initially lower price of land, as can be seen in the negative and significant coefficient of median home value and rent. These results are partially explained by the fact that highways were built to connect city centers, which were experiencing a decline in their economic conditions and an increase in their Black population (Boustan, 2010). However, the results are robust to the inclusion of distance to the central business district. Thus, Black individuals were, in equilibrium, more likely to be displaced by highways than their White counterparts.⁵⁰

One possible explanation for these results is that state planners followed the Federal government's dictates. In column (6) of Appendix Table B.1, I re-estimate equation B.1 using the Yellow Book maps as the dependent variable. In particular, I use an indicator that takes the value of one if a highway was planned in the neighborhood and zero otherwise. I find that the estimate for racial composition is not different from zero after including proximity to the city center and the price of land. These results suggest that the racial composition of neighborhoods played a role in the decision of state planners to deviate from the Federal plan.

These results indicate that highways were disproportionately routed through neighborhoods with higher Black population shares, even after accounting for proximity to the city center and land values. While the analysis does not establish a causal relationship between the racial composition of neighborhoods and the decision to build highways through them, it suggests that racial composition systematically influenced routing decisions, beyond what federal planning alone would predict.

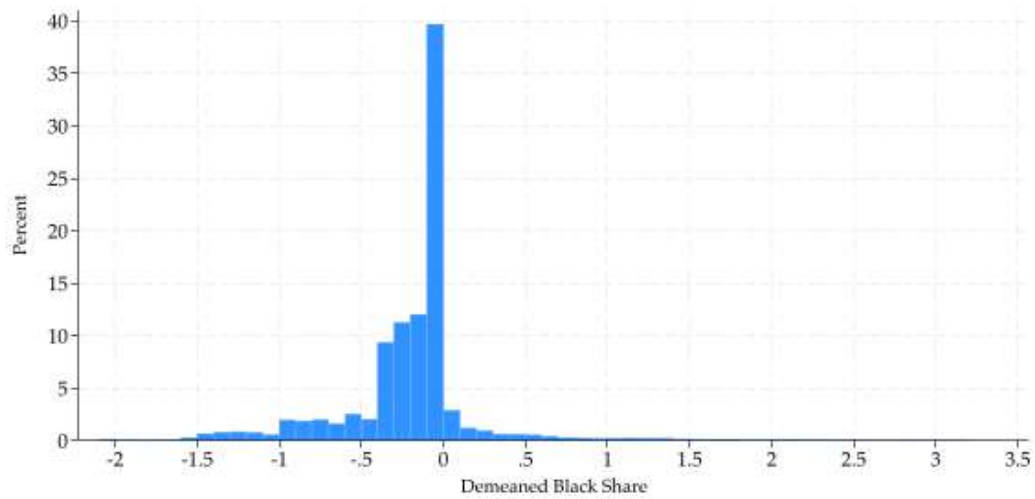


FIGURE B.5: Demeaned distribution of Black share.

Notes: The figure presents the distribution of the Black share variable, which is defined as the share of the city's Black population residing in the tract in 1950. The variable is demeaned by the city average, so the mean of the distribution is zero. The sample consists of census tracts from 62 metropolitan areas that have spatial information available for 1950, using the 2010 census tract boundaries.

B.3 Results using census tracts

TABLE B.1: Census Tract Determinants of Highway Construction

	Dep Var: Indicator if the tract is crossed by Built highway					Plan
	(1)	(2)	(3)	(4)	(5)	(6)
Black share	1.236 ^a (0.276)	1.083 ^a (0.278)	1.276 ^a (0.281)	0.855 ^a (0.254)	0.837 ^a (0.226)	0.060 (0.218)
(log) Median income	-0.009 (0.006)	-0.004 (0.006)	0.002 (0.009)	0.002 (0.008)	0.001 (0.006)	0.002 (0.006)
High school share	-0.002 ^b (0.001)	-0.000 (0.001)	-0.001 (0.001)	0.001 (0.001)	0.001 ^c (0.001)	-0.001 (0.001)
(log) Median home value		-0.058 ^b (0.028)	-0.078 ^a (0.025)	-0.074 ^a (0.024)	-0.067 ^a (0.023)	-0.022 (0.018)
(log) Median rent		-0.079 ^a (0.022)	-0.043 ^b (0.016)	-0.029 ^c (0.016)	-0.028 ^c (0.015)	-0.003 (0.019)
(log) Distance to city center				-0.108 ^a (0.015)	-0.078 ^a (0.012)	-0.095 ^a (0.014)
Highway planned					0.318 ^a (0.037)	
Mean dependent var.	0.220	0.223	0.223	0.223	0.223	0.186
Geo. controls	No	No	Yes	Yes	Yes	Yes
Obs.	18,246	17,206	17,197	17,195	17,195	17,195
R ² (Adj.)	0.049	0.055	0.098	0.117	0.199	0.088

Note: OLS estimates are reported. Coefficients are reported with standard errors, clustered at the MSA level, in parentheses. The unit of observation is a census tract in 1950, using the 2010 census tract boundaries. In columns (1) to (5), the dependent variable is an indicator for whether a highway was built through the tract. In column (6), the dependent variable is an indicator for whether a highway was planned through the tract according to the Yellow Book maps. The sample consists of census tracts from 62 metropolitan areas that have spatial information available for 1950. The regression results are weighted by the total population of the tract in 1950. All specifications include MSA fixed effects and controls for the (log) area and slope of the tract, distance to the nearest river, an indicator for whether the governor of the state was part of the Republican Party, the state (log) number of car registrations per 10k inhabitants, and the distance to the 1921 railroad network. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

⁵⁰These results contradict part of the findings of Carter (2023) and Weiwu (2025). They find that the median home value was the most significant predictor of the highway location and that the share of Black individuals did not have a substantial effect. These papers differ in scope, and they model the neighborhood Black share differently than I do. While they use the share of the tract's population that is Black, I use the share of the city's Black population residing in the tract.

C. EMPIRICAL STRATEGY APPENDIX

This section presents the analysis behind the distance cutoff, further discussion of the alternative empirical strategies used to estimate the effect of displacement on mortality, and supporting tables and figures.

C.1 Choosing the distance cutoff

Section 4 showed that highways were built through low-income and Black communities. Whether highway construction harmed or benefited residents of these communities, however, is an empirical question. On the one hand, highways impose direct costs: they expose nearby residents to elevated noise and pollution (Currie and Walker, 2011) and they erode neighborhood social capital by creating physical barriers between neighbors (Brinkman and Lin, 2024), with potentially long-lasting consequences. On the other hand, policymakers at the time viewed highway construction as a tool for urban revitalization, and the economic activity generated by highway projects could have improved local living conditions (Schwartz, 1975).

To answer this question, I study how long-run outcomes differ with distance to the highway, estimating:

$$y_i = \beta_0 \text{Displaced}_i + \sum_{k \in \mathcal{K}} \beta_k \text{Bin}_i^k + \mathbf{X}_i' \Gamma + \epsilon_i \quad (\text{C.1})$$

where y_i corresponds to life expectancy, the probability of residing in the same neighborhood, and the probability of residing in the same city at death. The variable Displaced_i indicates individuals whose homes were under the right-of-way path and Bin_i^k are indicators for 100-meter distance bins measured from the future highway. For instance, Bin_i^{500} equals one if the individual lived between 400 and 500 meters from a future Interstate in 1950. Because the effects of interest are on residents of affected neighborhoods rather than entire cities, I restrict the sample to individuals living within 2,000 meters of Interstate segments built before 1970, with those living 1,900 to 2,000 meters from the highway serving as the comparison group. Motivated by the fact that highways were disproportionately routed through low-income and Black communities, all regressions control for the household head's occupational score as well as race, gender, birth year, and metropolitan area fixed effects. Standard errors are clustered at the 1950 metropolitan area of residence.

Appendix Figure C.1 shows that the long-run effects of Interstate construction on communities are highly localized. The figure presents estimated effects on life expectancy and residential mobility for individuals in different distance bins from the highway, with the coefficient for displaced individuals shown in red. Panel (a) reports effects on life expectancy. Displaced individuals die about 0.12 years younger than neighbors living

1,900 to 2,000 meters from the highway. The effects spill over to residents within 100 meters of the highway, who also die at younger ages, but there is no significant effect beyond 100 meters. These findings complement evidence of negative effects of highway externalities on newborns living within 2,000 meters of a highway (Currie and Walker, 2011). For adults, spillovers are considerably more spatially concentrated.

Panels (b) and (c) examine spatial mobility, using indicators for whether individuals remained in the same neighborhood and the same city between 1950 and death, respectively. Panel (b) shows that displaced individuals are 3 percentage points less likely to reside in the same neighborhood at death. Residents within 100 meters of the highway, though not directly displaced, are also more likely to leave the neighborhood. These estimates are economically meaningful, as they represent 60 and 30 percent of the sample mean for displaced and adjacent individuals, respectively. The reverse pattern emerges for the probability of residing in the same city: both displaced and adjacent individuals are *more* likely to remain in the same city at death. Together, these results suggest that affected individuals relocate within the same city rather than leaving it entirely. As with mortality, spillovers fade quickly, with no systematic evidence of mobility effects beyond 100 meters.

Overall, the results show that the construction of the Interstate network had localized negative effects on individuals living near highways, which aligns with previous findings by Moretti and Wheeler (2025) and Kashner and Ross (2025) who also find that highway noise and traffic externalities are highly local.

C.2 Pre-trends for the main empirical strategy

The main empirical strategy compares individuals displaced by, and those living within 100 meters of, newly constructed highways to those living between 100 and 200 meters from the construction. To test if these individuals were trending in a similar fashion before highway construction, I use the sample of individuals linked to the 1940 Census and estimate equation 2 using changes between 1940 and 1950. The linkage to the 1940 Census is done using Ruggles et al. (2025) links.

C.3 Matching

The main empirical strategy compares individuals displaced by, and those living within 100 meters of, newly constructed highways to those living between 100 and 200 meters from the construction. Although the results in Appendix Section C.1 show that any spillover effects of highway construction are limited to individuals living within 100 meters of the highway, the estimates may still be biased if individuals living within 200 meters of the highway construction differ in unobservable characteristics that are correlated with the outcomes. For example, highway construction reduced the housing stock in affected neighborhoods,

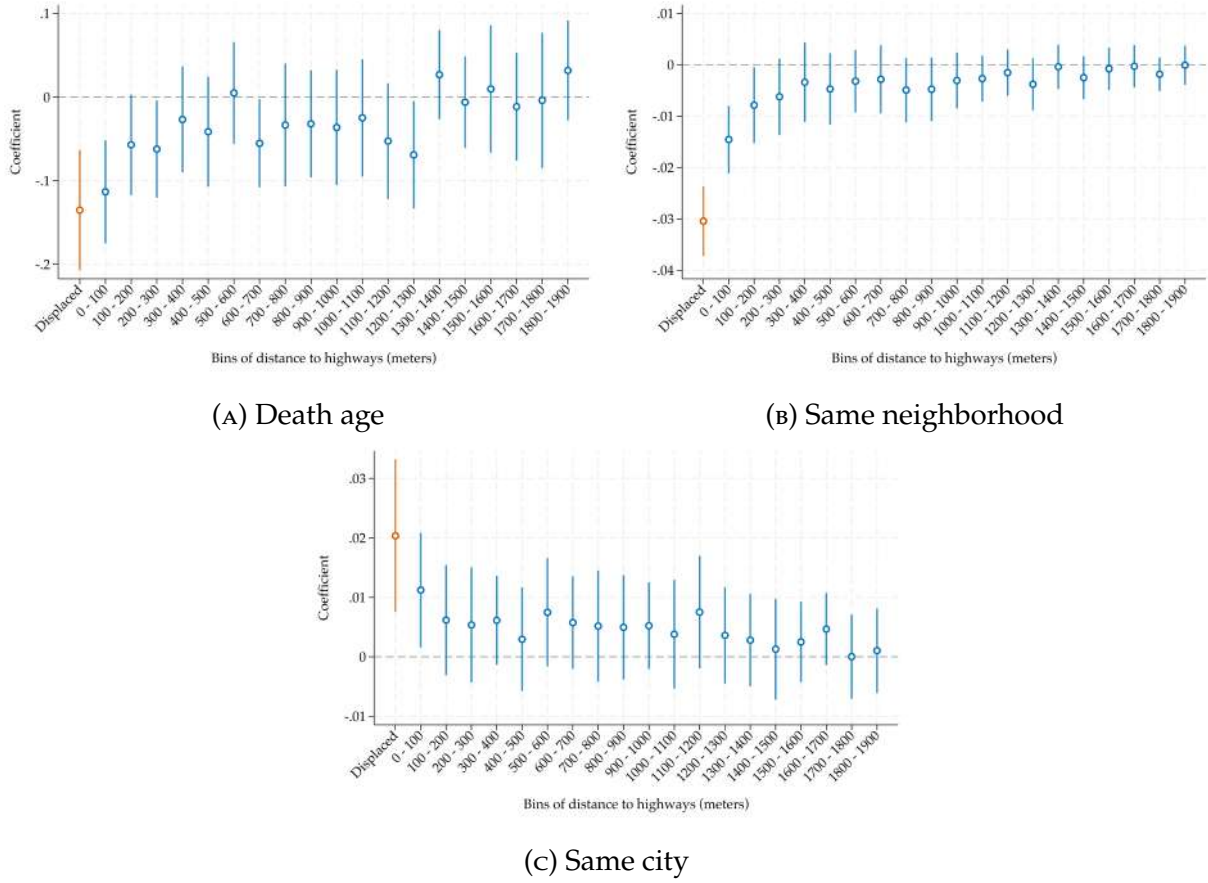


FIGURE C.1: Long-run effects of highway construction on affected communities.

Notes: The figures present the estimated coefficients and confidence intervals of equation C.1. The blue line plots the distance coefficients β_k with 95% confidence intervals shown as shaded bands, and the red marker plots the coefficient for displaced individuals. The sample consists of individuals in metropolitan areas who lived within 2,000 meters of a future Interstate in 1950, linked to administrative mortality records from 1988 to 2007. The comparison group corresponds to those living between 1,900 and 2,000 meters from the future Interstate. The dependent variables are reported in the name of each panel. All regressions include the household head's occupational score, as well as race, gender, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

particularly for low-income households, which may have raised neighborhood rents and affected the control group. Restricting the control group to individuals living further away from the highway construction may mitigate this concern, but it may also introduce bias if the control group is not comparable to the displaced individuals. To address this concern, I use propensity score matching to select a control group living in the same city and with similar observable characteristics, but living more than 2,000 meters from any highway.

I construct a matched sample as an alternative strategy to estimate the causal effects of highway displacement on long-term mortality. Matching addresses selection bias arising from non-random highway placement and systematic differences between individuals living near and far from construction sites. I define two treatment groups based on residential proximity to Interstate Highways constructed before 1970: *displaced* individuals ($D_i = 1$), whose 1950 residence fell on a future highway right-of-way (distance ≤ 0 meters),

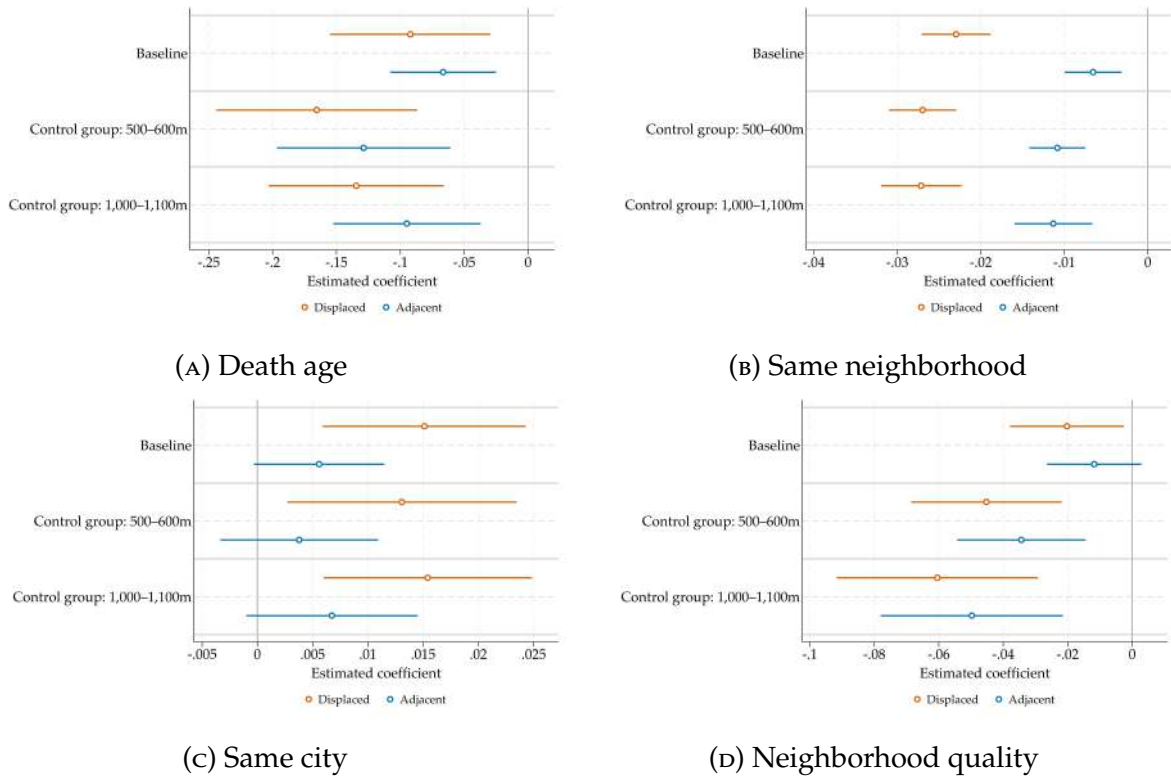


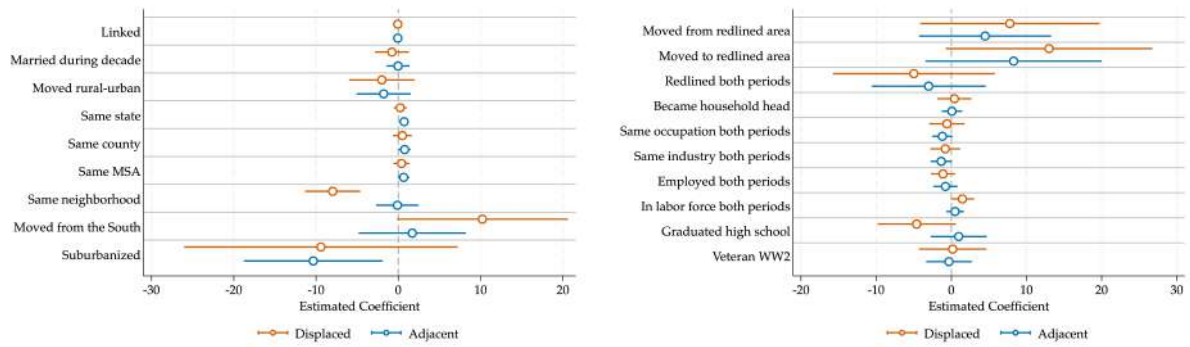
FIGURE C.2: Robustness of the results to the use of different control groups.

Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top row corresponds to the baseline results. The second uses as control groups individuals living between 500 and 600 meters. The third row uses individuals living between 1,000 and 1,100 meters.

and *adjacent* individuals ($A_i = 1$), residing 0–100 meters from a future highway segment. The control pool consists of individuals residing more than 2,000 meters from any highway segment.

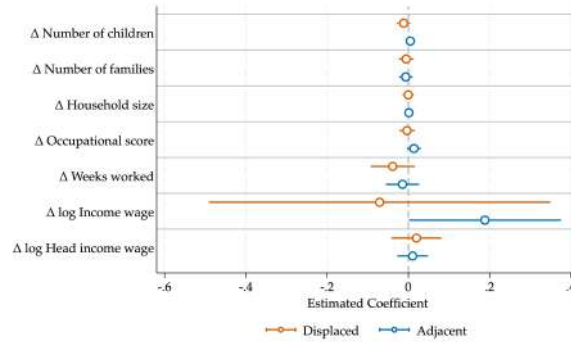
Because I use both displaced and adjacent individuals as separate treatment groups, I implement sequential matching to ensure that each control unit is matched to at most one treated observation. I first match displaced individuals to the full control pool, then match adjacent individuals to the remaining controls, so that no control unit appears in both matched samples. Within each MSA, I estimate propensity scores via probit regression using 27 pre-treatment covariates capturing demographics (race, sex, age polynomials, immigration status), socioeconomic status (employment, occupational score polynomials, education, log household income), household composition (household size, number of families), geography (log distance to CBD, redlining grade), and six interaction terms to absorb non-linear selection patterns. With the propensity score in hand, I then match each individual to their's nearest neighbor based on the estimated propensity score.

Overall, the matching process is able to match 63.1 percent of the displaced individuals. Matched individuals are different from unmatched individuals, as shown in Appendix



(A) Balance in discrete variables 1

(B) Balance in discrete variables 2



(c) Balance in continuous variables

FIGURE C.3: Pre-construction trends for the unaffected neighbors strategy.

Notes: The figures display estimates from regressions of pre-construction trends on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) and (b) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (c) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.

Figure C.4. They are more likely to be white, have higher property values, and have higher levels of education. Thus, the results should be interpreted as the effect of displacement on the matched sample, not the entire displaced population.

The identifying assumption of the matching strategy is that, conditional on the full set of individual- and neighborhood-level covariates used to construct matched pairs within each MSA, proximity to a highway is independent of potential outcomes. By matching on characteristics that capture both the official criteria for route selection and the factors governing residential sorting across neighborhoods, the design addresses selection at both the infrastructure and household level. Appendix Figures C.5 and C.6 show that the sample is balanced and shows no pre-trends, lending credibility to the strategy.

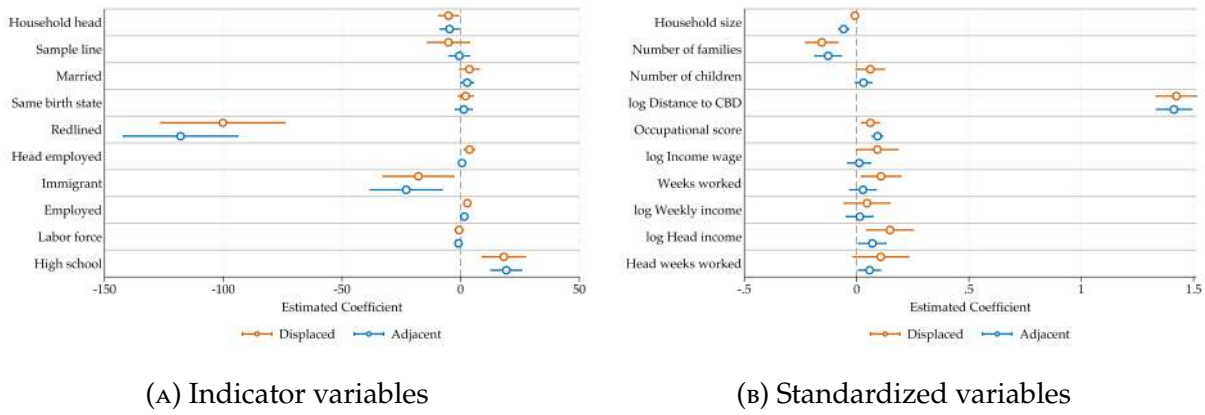


FIGURE C.4: Balance of Pre-Treatment Characteristics between matched and unmatched treated individuals.

Notes: The figure presents the coefficient and 95% confidence intervals for being matched in the matching strategy compared to unmatched treated individuals, for a set of pre-treatment characteristics. Each coefficient corresponds to a different regression and is reported with standard errors clustered at the metropolitan area level. The coefficients and confidence intervals in panel (a) are normalized by the sample mean of the dependent variable, while the dependent variables in panel (b) are standardized to have mean zero and standard deviation one. All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence.

C.4 Planned highways as potential control group

To minimize construction costs, primary roads were often converted into highways (Rose and Mohl, 2012). A natural concern is that individuals who used to live on primary roads may differ from their neighbors in unobservable characteristics that are correlated with the outcomes of interest. The routes planned in the Yellow Book, like the actual network built, were also more likely to correspond to primary roads. Thus, individuals who would have been displaced by the Yellow Book routes are similar to displaced individuals in that they were living near a planned highway, but were not displaced.

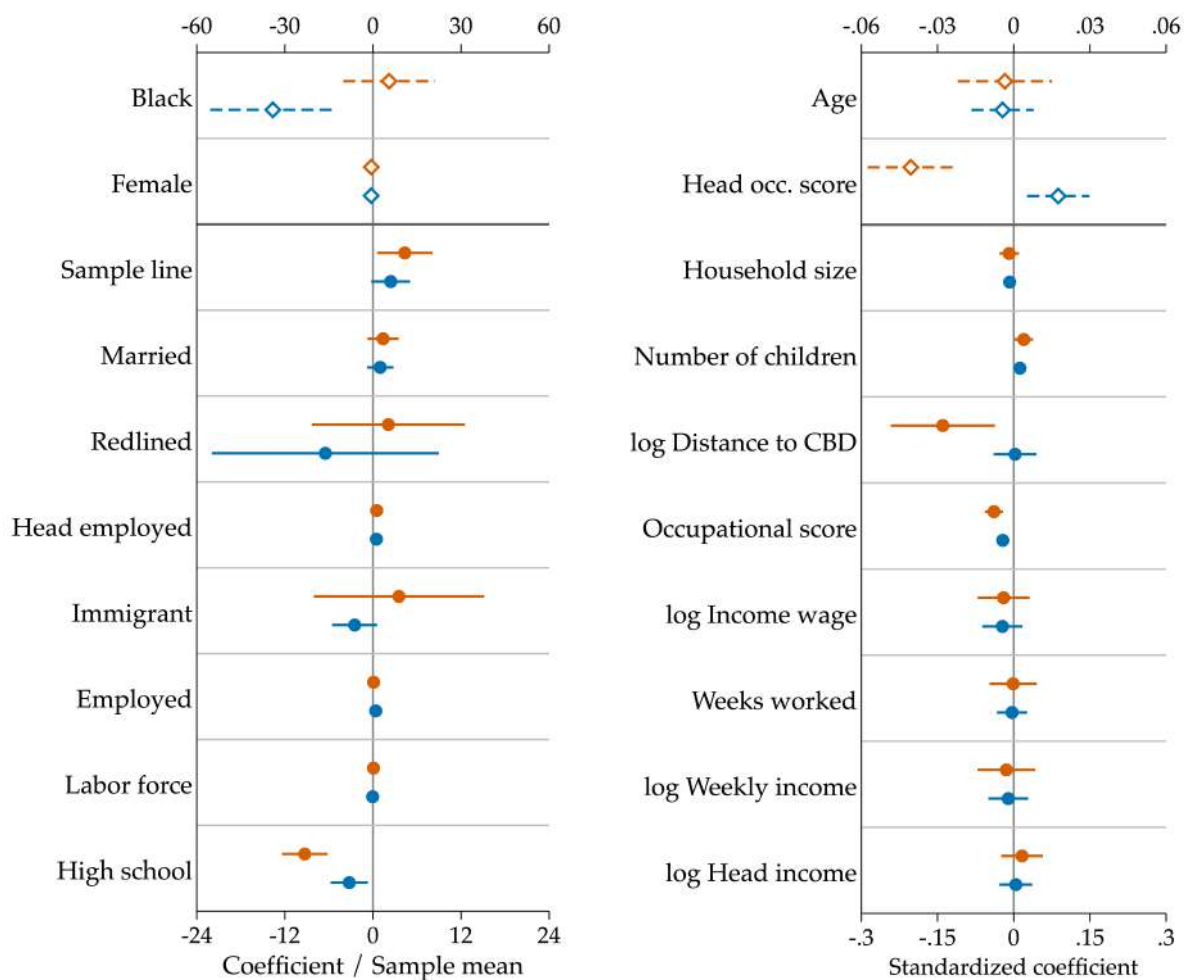
The identifying assumption of this strategy is that, conditional on the characteristics used by states and local officials to select highway routes, treatment is independent of potential outcomes between treated individuals and those who would have been displaced by the Yellow Book routes. Appendix Figure C.7 presents the balance test for this strategy, which shows that treated individuals and those who would have been displaced by the Yellow Book routes are similar in observable characteristics. Finally, Appendix Figure C.8 uses the sample of individuals linked to the 1940 Census to show that treated individuals and those who would have been displaced by the Yellow Book routes were trending in a similar fashion before highway construction.

A concern with this strategy is that living on primary roads may have, by itself, an effect on the outcomes of interest. I can test this possibility by running a placebo test using as a treatment group individuals living near these planned highways. I then repeat the exercise of comparing affected individuals to those living between 100 and 200 meters

away.⁵¹ The results show no significant effect of living close to a planned highway on individuals' long-term outcomes. Appendix Figure C.9 shows that the coefficients are very close to zero and not statistically significant at the usual levels. I interpret the lack of significant effects as evidence against the hypothesis that living near a primary road had an effect on the outcomes of interest, and thus as evidence in favor of the validity of this strategy.

C.5 Additional figures

⁵¹Planned highways consist of a unidirectional segment of the highway. I assume that planned highways had a right-of-way width of 150 feet (45.7 meters).

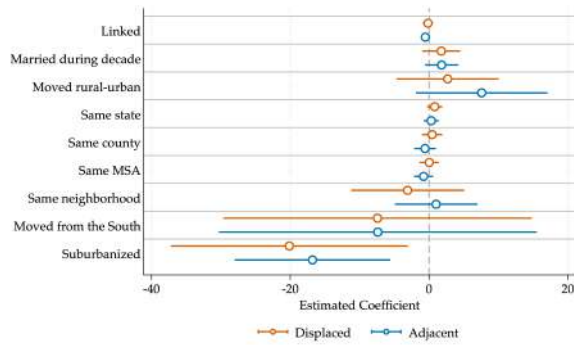


(A) Indicator variables

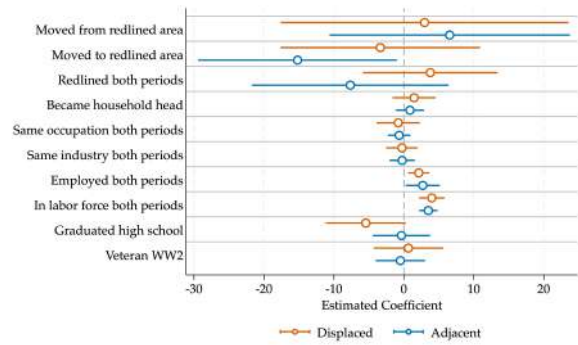
(B) Continuous variables

FIGURE C.5: Pre-construction balance for the matching strategy.

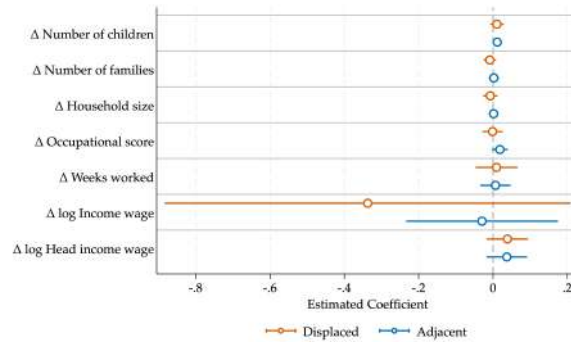
Notes: The figures display estimates from regressions of pre-construction characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals displaced and adjacent to the highway, as well as their matched comparison. In both panels, the top two coefficients, presented as hollow diamonds, control only for metropolitan area fixed effects and are plotted against the top x-axis. The remaining coefficients, presented as solid circles, control for the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects, and are plotted against the bottom x-axis. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (b) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.



(A) Balance in discrete variables 1



(B) Balance in discrete variables 2



(C) Balance in continuous variables

FIGURE C.6: Pre-construction trends for the matching strategy.

Notes: The figures display estimates from regressions of pre-construction trends on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes matched individuals in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) and (b) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (c) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.

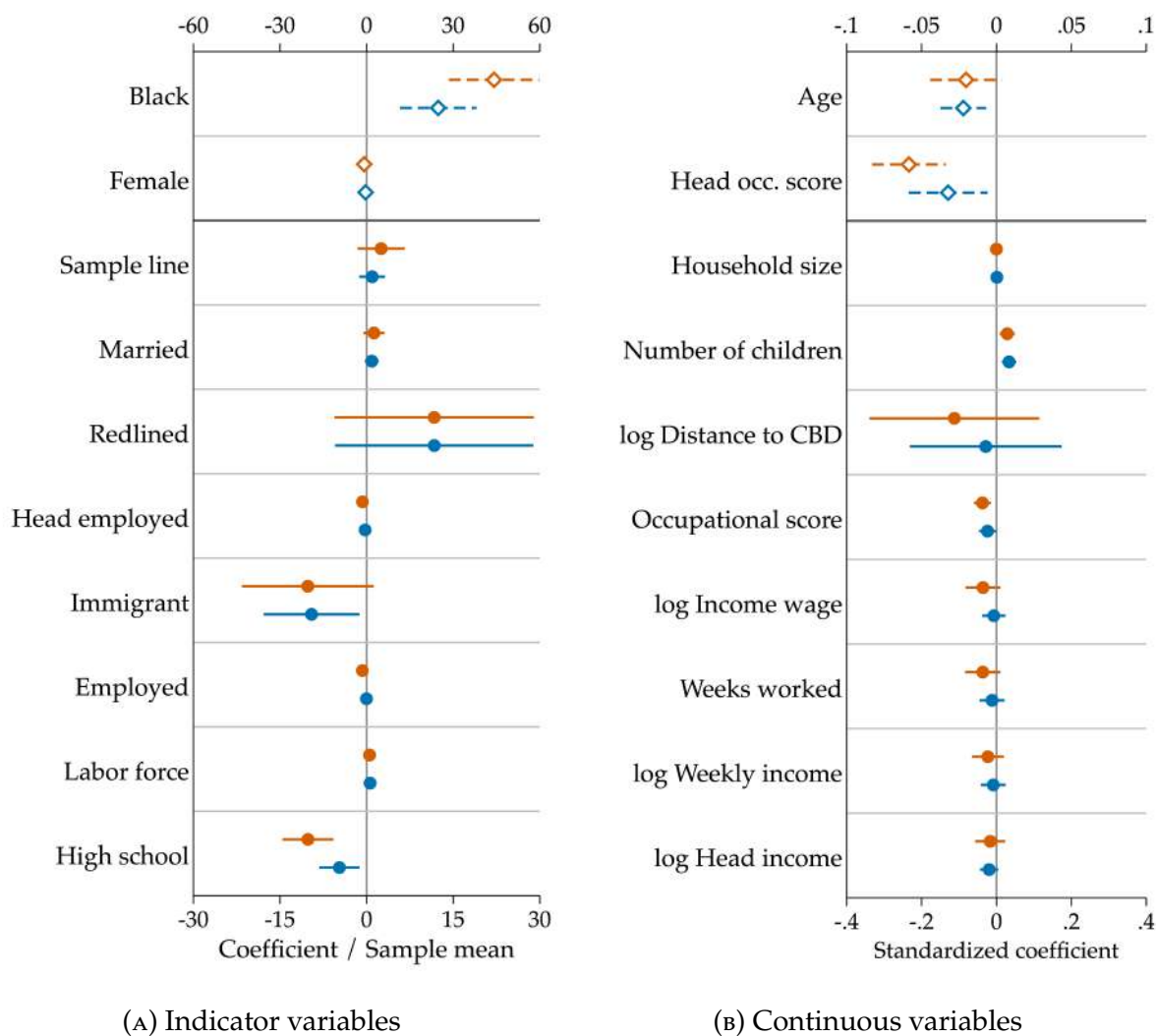
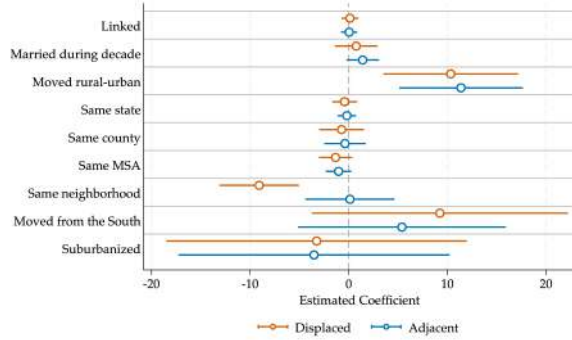
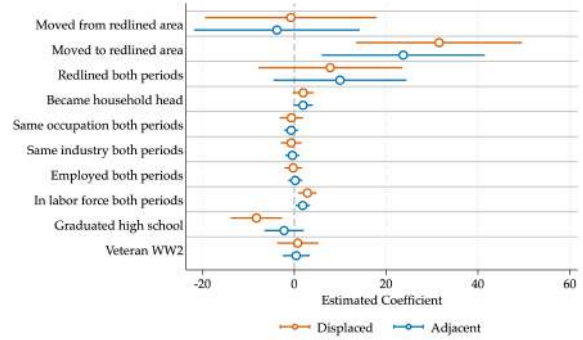


FIGURE C.7: Pre-construction balance for the strategy using the Yellow Book Maps.

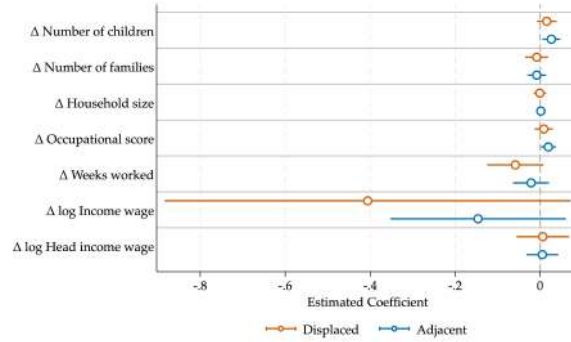
Notes: The figures display estimates from regressions of pre-construction characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals displaced and adjacent to the highway, as well as those that would have been displaced or adjacent to the unbuilt Yellow Book routes. In both panels, the top two coefficients, presented as hollow diamonds, control only for metropolitan area fixed effects and are plotted against the top x-axis. The remaining coefficients, presented as solid circles, control for the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects, and are plotted against the bottom x-axis. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (b) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.



(A) Balance in discrete variables 1



(B) Balance in discrete variables 2



(C) Balance in continuous variables

FIGURE C.8: Pre-construction trends for the strategy using the Yellow Book Maps.

Notes: The figures display estimates from regressions of pre-construction trends on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression. The sample includes individuals in the 1950 census living in homes destroyed by highway construction, those living within 100 meters of constructed highways, and those who would have fallen into either group had the planned highways been built. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. Coefficients and confidence intervals in Panel (a) and (b) are normalized by the sample mean of the outcome, so they can be interpreted as percentage changes. Coefficients in Panel (c) are standardized to have mean zero and standard deviation of one, so they can be interpreted as standard deviation changes.

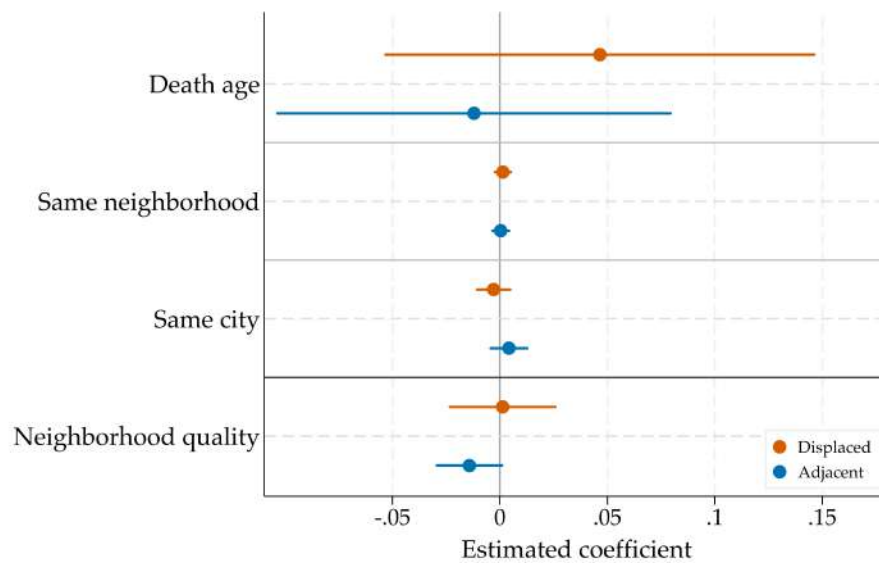
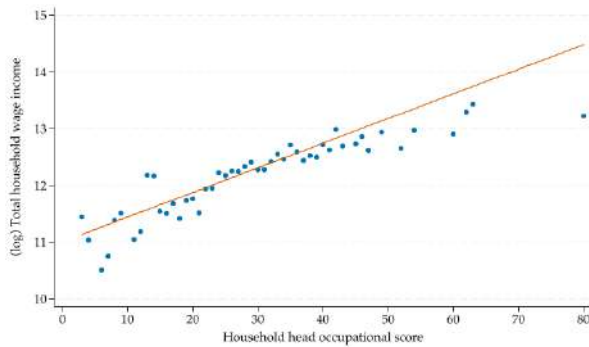
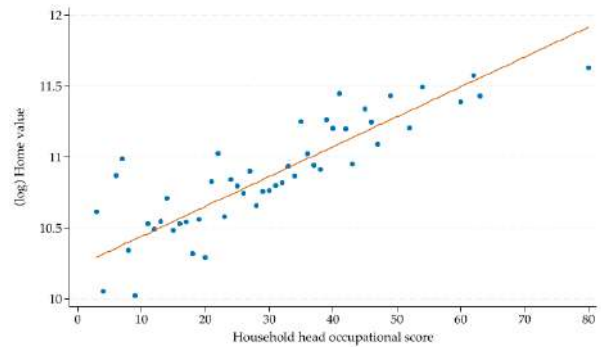


FIGURE C.9: Placebo test using the planned highway strategy.

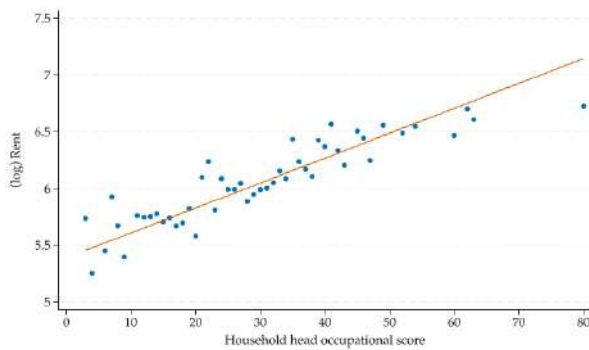
Notes: The figures display estimates from regressions of long-run characteristics on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression, and the sample includes individuals living within 200 meters of the planned highways in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The units for the coefficients and confidence intervals in the first row are years, while those for the two rows below are percentage points. Finally, the units for the coefficients and confidence intervals in the bottom row correspond to standard deviations.



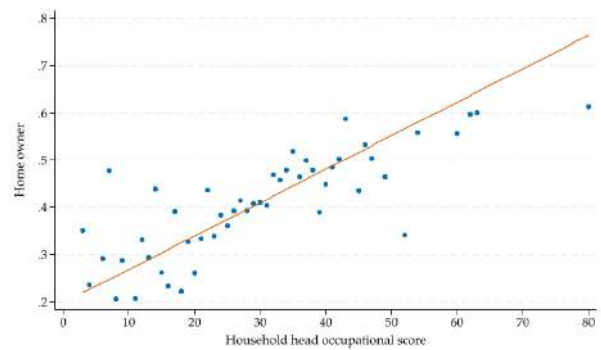
(A) Total household income (log)



(B) Home value (log)



(C) Monthly rent (log)



(D) Home ownership

FIGURE C.10: Correlation between household head's occupational score and wealth proxies.

Notes: Each figure presents a binscatter plot of the relationship between the household head's occupational score and the indicated outcome. The sample consists of household heads in the 1940 Census with non-missing MSA of residence. Monetary outcomes are reported in 2010 dollars. Home ownership is an indicator variable equal to one if the household head owns the home. The fitted line is estimated via OLS.

D. DISPLACEMENT APPENDIX

D.1 Additional tables

TABLE D.1: Effects of Interstate construction on longevity

	Death Age (1)	Survive 75 (2)	Survive 80 (3)
<i>Panel A. Neighbors sample</i>			
Displaced	-0.092 ^a (0.032)	-0.002 (0.003)	-0.009 ^a (0.002)
Adjacent	-0.066 ^a (0.021)	-0.003 (0.002)	-0.005 ^a (0.002)
Observations	154,869	154,869	154,869
Adj. R ²	0.492	0.345	0.328
Mean of dep. var.	76.368	0.540	0.303
<i>Panel B. Yellow Book sample</i>			
Displaced	-0.151 ^a (0.033)	-0.005 ^c (0.003)	-0.008 ^a (0.003)
Adjacent	-0.096 ^a (0.029)	-0.005 ^b (0.002)	-0.003 ^c (0.002)
Observations	155,007	155,007	155,007
Adj. R ²	0.492	0.346	0.328
Mean of dep. var.	76.471	0.546	0.308
<i>Panel C. Matching sample</i>			
Displaced	-0.132 ^a (0.042)	-0.002 (0.004)	-0.010 ^a (0.003)
Adjacent	-0.106 ^a (0.032)	-0.005 ^b (0.003)	-0.006 ^a (0.002)
Observations	102,284	102,284	102,284
Adj. R ²	0.494	0.344	0.329
Mean of dep. var.	76.634	0.557	0.318

Note: OLS estimates are reported. Coefficients are reported with standard errors, clustered at the MSA level, in parentheses. The unit of observation is an individual in the sample of linked census-mortality records. The dependent variables are reported at the top of each column. Panel A reports the results comparing treated individuals to their neighbors living between 100 and 200 meters from future highway construction, Panel B compares them to those who would have been displaced by, or living adjacent to, the Yellow Book routes, and Panel C to matched individuals. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

D.2 Additional figures

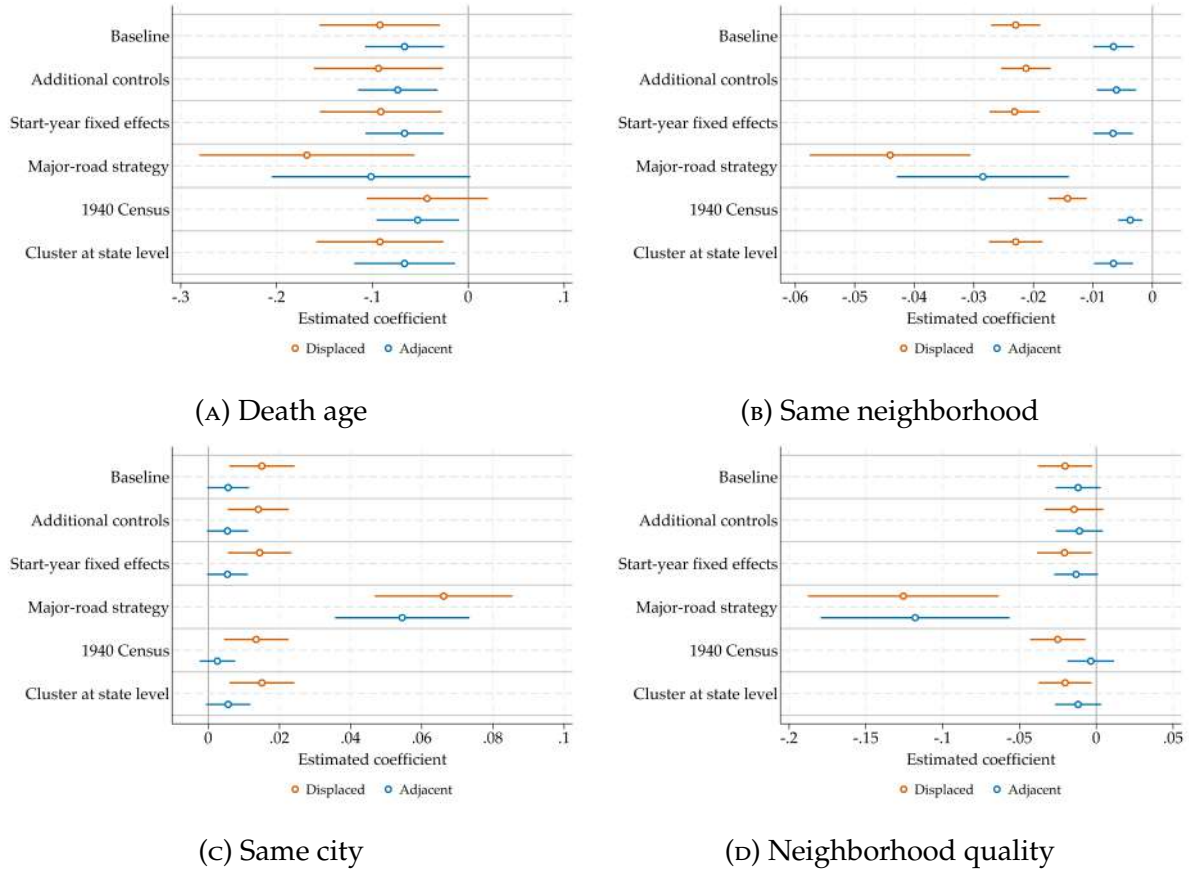


FIGURE D.1: Robustness of the results.

Notes: The figures display estimates from regressions of the main outcomes on the two treatment indicators in equation 2, along with 95% confidence intervals. Each row corresponds to a different regression. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census, unless otherwise noted. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level. The top row corresponds to the baseline results. *State-year fixed effect* controls for the fixed effects for the years in which construction started. *Major road strategy* compares the treated group to other individuals living near non-converted major roads. *Census 1940* repeats the analysis using the 1940 census instead of the 1950 census to identify the treatment and control groups. *Cluster at state level* clusters standard errors at the state level, instead of the metropolitan area level. *Extra controls* includes in the regression the following controls: employment dummies, middle school and high school completion dummies, log distance to the city center in 1950, number of families in the household, household head employment dummy, household head high school degree dummy, and the number of weeks worked in the last year by the household head.

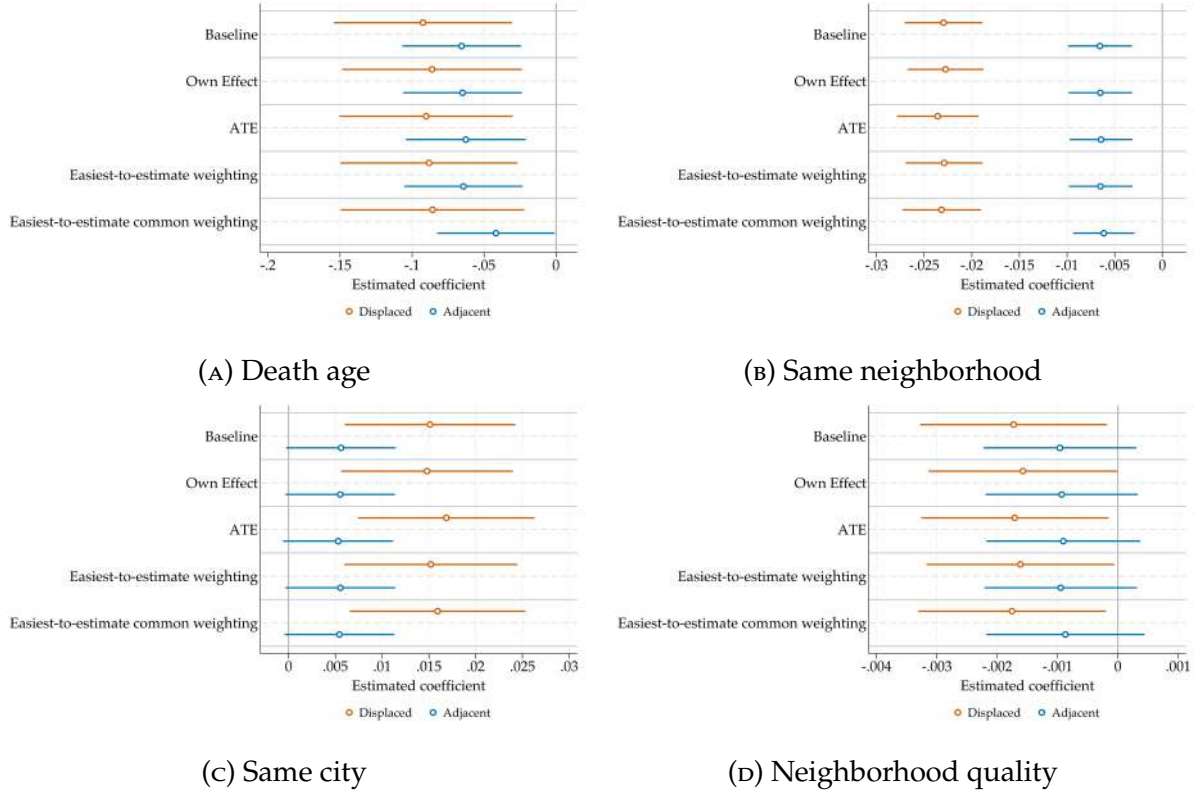


FIGURE D.2: Contamination bias.

Notes: The figures contain the different estimates developed by Goldsmith-Pinkham et al. (2024) to tackle contamination bias across treatments. Each row corresponds to a different regression, with the point estimates plotted along with 95% confidence intervals. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. All regressions include the household head's occupational score, as well as gender, race, birth year, and metropolitan area fixed effects. Standard errors are clustered at the metropolitan area level.

D.3 Duration analysis

Given the nature of the mortality record, I can estimate a duration model to study the effect of displacement on mortality. In this context, I study whether individuals who were displaced by highway construction are more likely to die earlier than their peers who were not displaced. This semi-parametric approach leaves the baseline hazard unrestricted, handles left-truncation through delayed entry at age-specific risk sets, and absorbs high-dimensional fixed effects through stratification without incurring incidental parameters bias.

The hazard function for individual i in stratum s at age t takes the form:

$$h_i(t) = h_{0s}(t) \cdot \exp(\beta_1 \cdot \text{displaced}_i + \beta_2 \cdot \text{adjacent}_i + \beta' \mathbf{X}_i) \quad (\text{D.1})$$

where $h_{0s}(t)$ is a stratum-specific baseline hazard (one per MSA $\times$ birth-year cell) and $\mathbf{X}_i$ includes head-of-household occupational score, race, and sex. I stratify on MSA and birth year rather than including them as covariates, allowing the baseline hazard to vary flexibly

across geographic and cohort dimensions. Standard errors are clustered at the MSA level.

I model age rather than calendar time as the time scale and implement left-truncation to address delayed entry into the observation window. Individuals enter the risk set at the age they attain on January 1, 1988, when BUNMD mortality records begin: an individual born in 1920, for example, enters at age 68. The partial likelihood conditions on survival to the start of observation and appropriately adjusts risk sets for late entry, so that delayed entry does not bias hazard ratio estimates.

An advantage of proportional hazard models is that the identification of β_1 and β_2 does not require the specification of the functional form of $\lambda_0(t, \alpha)$ (Cameron and Trivedi, 2005, ch. 17.8). This functional form permits an easy interpretation of the coefficients β_1 and β_2 . Suppose that two individuals are identical in all aspects except for the displacement indicator. Then, the hazard ratio between the two individuals is:

$$\frac{\lambda(t|\text{Disp} = 1, \mathbf{X}_i)}{\lambda(t|\text{Disp} = 0, \mathbf{X}_i)} = \exp(\beta_1) \quad (\text{D.2})$$

Thus, the new hazard is $\exp(\beta_1)$ times the hazard of the individual who was not displaced. In other words, the change in the hazard for displaced individuals is $\exp(\beta_1) - 1$ times the original hazard. The results are reported in Appendix Table D.4. I report the values of the hazard rates minus one (i.e., $\exp(\beta_1) - 1$), so that the estimates can be interpreted as percentage changes in the hazard of death at a given age, relative to the control group. The estimates suggest that individuals displaced and those living close to future highway construction have between 0.7 and 2.4% higher risk of early death over the study period than their peers.

TABLE D.4: Mortality hazard rates for displaced and adjacent individuals

Variable	Neighbors	Matching	Yellow Book
Displaced	0.013 ^c	0.022 ^b	0.024 ^a
	0.007	0.009	0.006
Adjacent	0.007	0.016 ^a	0.016 ^a
	0.005	0.006	0.006
N	137,410	91,734	137,750
Pseudo R2	0.00060	0.00051	0.00053

Note: The table reports hazard rates from a Cox Proportional Hazard model, where the dependent variable is the hazard of death at a given age. The sample includes individuals in the linked census-mortality records, and the model is stratified by MSA and birth year. The key independent variables are indicators for being displaced by highway construction and for living adjacent to future highway construction in the 1950 census. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

Transforming the increased hazard rate into average expected months of life lost.

To translate the Cox proportional hazard estimates into a measure comparable to the OLS results, I compute the restricted mean survival time (RMST) implied by the model. I use the results from estimating equation D.1 to predict the cumulative hazard function from which the survival function can be derived. I then predict survival curves separately for each of the four demographic cells defined by race (Black/White) and sex (male/female), evaluating the head-of-household occupational income score at its sample mean within each cell. Within each cell, I construct predicted curves for each of the three mutually exclusive treatment groups (displaced, adjacent, and control) and numerically integrate the survival function over the observed age range using the trapezoidal rule to obtain the RMST. The difference in RMST between each treatment group and the control group provides a cell-specific estimate of months of life lost to highway construction. The results are reported in Appendix Table D.5.

TABLE D.5: Expected months of life lost implied by the hazard estimates

Variable	Non-Black Male	Non-Black Female	Black Male	Black Female
Displaced	-1.0928	-1.0401	-1.1174	-1.0686
Adjacent	-0.3808	-0.3624	-0.3895	-0.3723

Note: The table reports the associated reduction in expected months of life implied by the Cox Proportional Hazard model estimates in Appendix Table D.4, for the neighbor strategy. The sample includes individuals in the linked census-mortality records, and the model is stratified by MSA and birth year.

D.4 Predicted mortality

The causal effects of displacement on mortality could be confounded if individuals with higher mortality risk sorted into areas in which highways were later built. I address this concern by predicting the death age of individuals using 1950 baseline characteristics, and then testing whether predicted mortality differs between displaced individuals and their neighbors. To do so, I model the relationship between mortality and baseline characteristics as:

$$\text{Death age}_i = \mathbf{X}_i\Gamma + \epsilon_i$$

I estimate this model using 1950 characteristics of individuals living more than 2,000 meters away from any highway built. I predict mortality by fitting an OLS regression of death age on a set of demographic and socioeconomic variables, along with their interactions. The regression also includes three labor-market variables for the individual (employment status, labor-force participation, and the occupational income score) and the

corresponding employment indicator and occupational income score for the household head. Finally, it also includes metropolitan-area fixed effects. The fitted model is used to generate out-of-sample mortality predictions for individuals in the estimation sample (i.e., those living within 200 meters of the highway). Appendix Figure D.3 compares predicted with actual death age. A regression between the two values has a coefficient of 0.94, with an R^2 of 0.324.

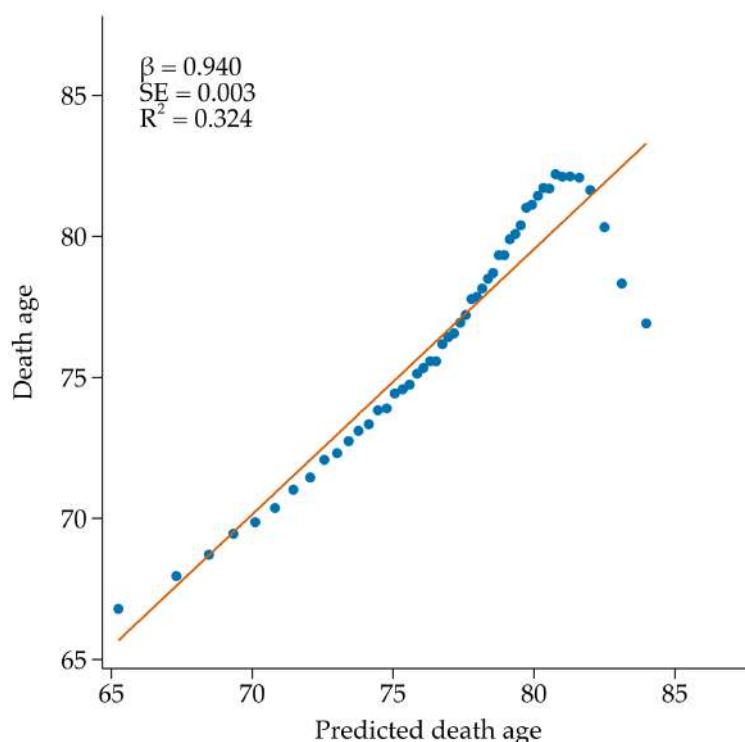


FIGURE D.3: Predicted versus actual death age

Notes: This figure shows the relationship between predicted and actual death age for individuals in the sample of linked census-mortality records. Predicted death age is constructed by fitting an OLS regression of death age on a set of baseline characteristics.

D.5 Results corrected by the probability of staying

In this section, I apply the correction derived in Appendix Section A.7.2 to recover the point estimates of the non-pecuniary costs of displacement.

D.6 Principal component analysis

To measure the quality of the neighborhood at the time of death, I use a principal component analysis (PCA) of the neighborhood characteristics. I create the index separately for Black and White individuals. Each variable is normalized to have a mean of zero and a standard deviation of one. I include the share of adults with a high school degree, the share of adults with a college degree, the employment rate, the median income, the median house

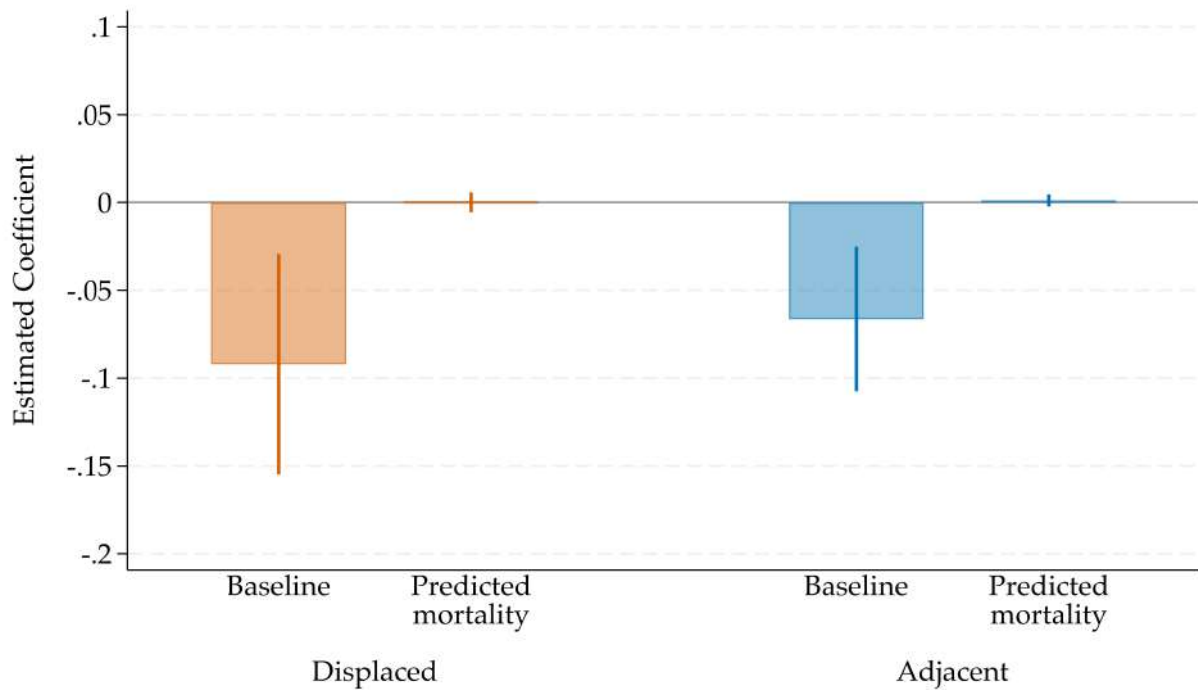


FIGURE D.4: Estimated effects on predicted mortality

Notes: The figure presents the coefficient and 95% confidence intervals for the baseline results and the results on predicted mortality. Predicted mortality is constructed for individuals living more than 2,000 meters away from any highway ever built. The fitted model is then used to predict mortality of those displaced or living within 200 meters of any highway segment built before 1970. More details on the construction of the predicted mortality can be found in Appendix Section D.4. All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan area level.

value, and the share of homeowners. I then normalize the first principal component to have a mean of zero and a standard deviation of one.

I find that the first principal component is positively related to the six variables used for both Black and White individuals. Appendix Table D.6 shows the factor loadings for the first two principal components. The first principal component also has explanatory power for the neighborhood characteristics (i.e., the eigenvalues are 3.01 and 2.37 for White and Black individuals). The estimated weights for all characteristics have the same sign and are very similar in magnitude for both groups.

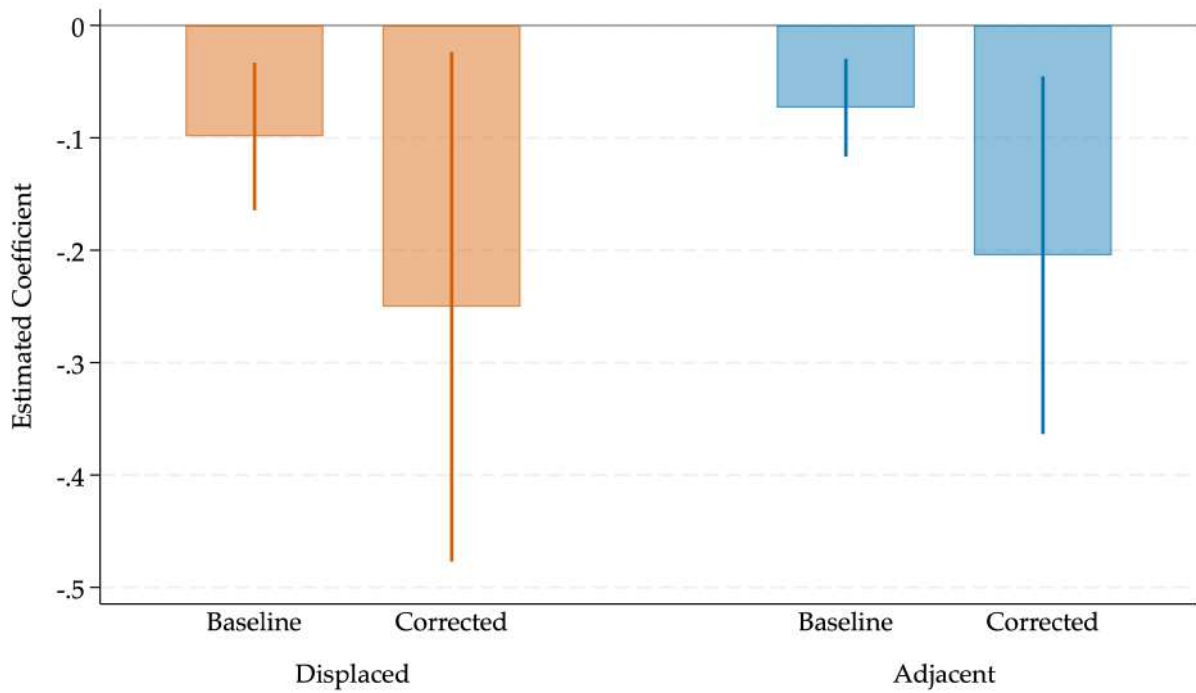


FIGURE D.5: Corrected mortality effects

Notes: The figure presents the coefficient and 95% confidence intervals for the baseline estimates and the estimates correcting for the probability of outmigration. Details about the correction and how it recovers the *true* coefficients are in Appendix Section A.7.2. All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan area level.

TABLE D.6: Factor Loadings for the first two principal components by race

	White		Black	
	(1)	(2)	(3)	(4)
	1st PC	2nd PC	1st PC	2nd PC
Eigenvalue	3.01	1.17	2.37	1.05
Household median income	0.48	0.02	0.51	-0.13
Median home value	0.41	-0.26	0.09	0.68
Employment share	0.41	-0.34	0.42	-0.06
Homeownership share	0.24	0.51	0.28	-0.01
College share	0.19	0.69	0.28	-0.64
High school share	0.44	-0.22	0.47	0.19

Notes: Factor loadings for the first two principal components.

Appendix Figure D.6 shows the eigenvalues of the first six principal components by race. It shows that the first principal component explains most of the variation in the neighborhood characteristics, for both racial groups.

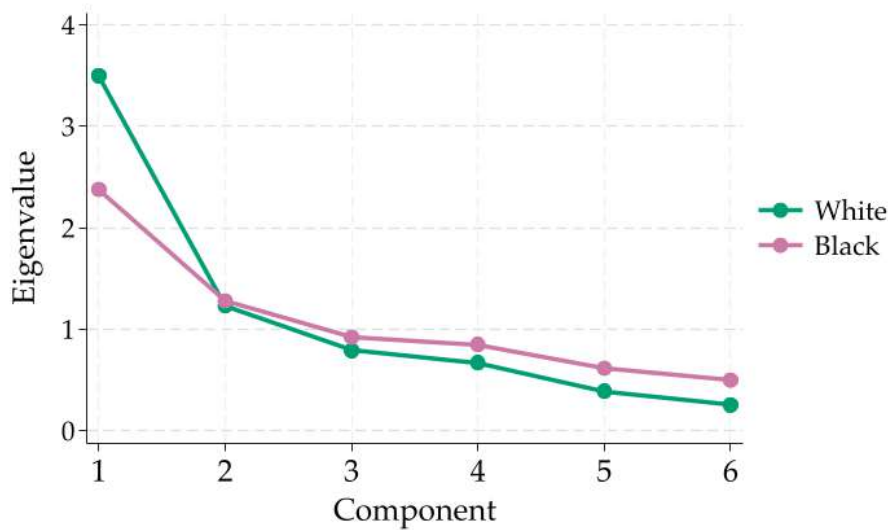


FIGURE D.6: PCA Eigenvalues for the first six principal components by race

Notes: Eigenvalues of the first six principal components.

D.7 Racial differences in the effects

To test whether the racial gap in the estimates comes from pre-construction differences in individual endowments, I construct DiNardo et al.-style (1996) reweighting weights based on cells defined by metropolitan area, sex, quintile of the household head's occupational score, five-year birth cohort, and homeownership status in 1940, where individuals in households that I do not link to the 1940 census form their own tenure category. The weights are constructed so the white population matches the Black population distribution. Together, these weights hold observable composition fixed, so that any remaining difference between the two groups is not driven by the characteristics that define the cells. Appendix Figure D.7 shows that the pre-construction endowment played little to no role in shaping the effects faced by white individuals.

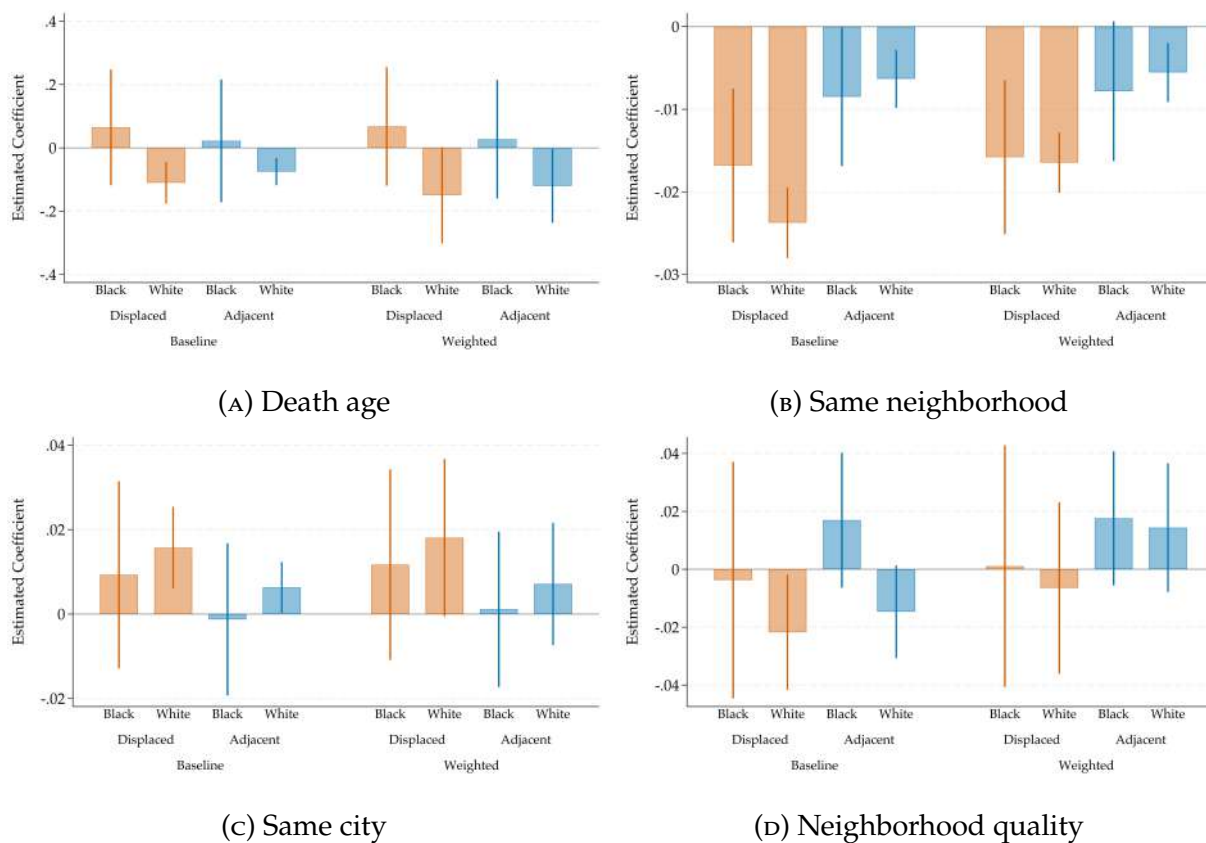
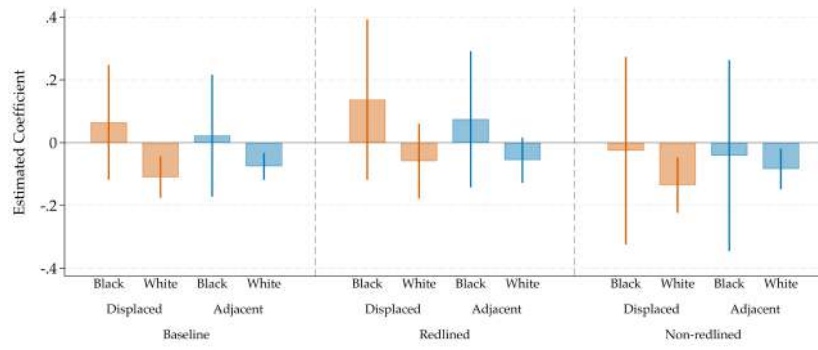
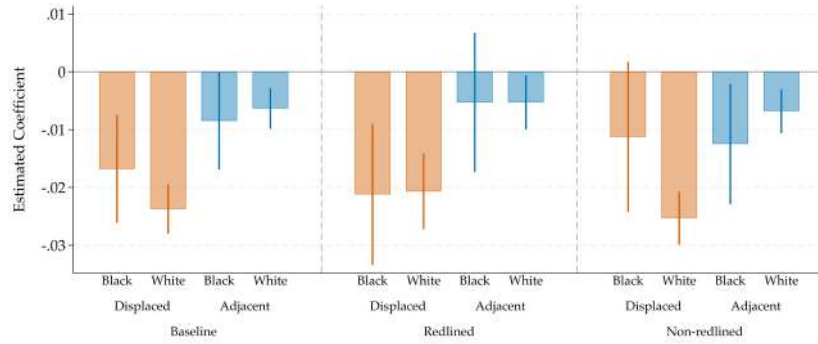


FIGURE D.7: Effects by race — Re-weighted.

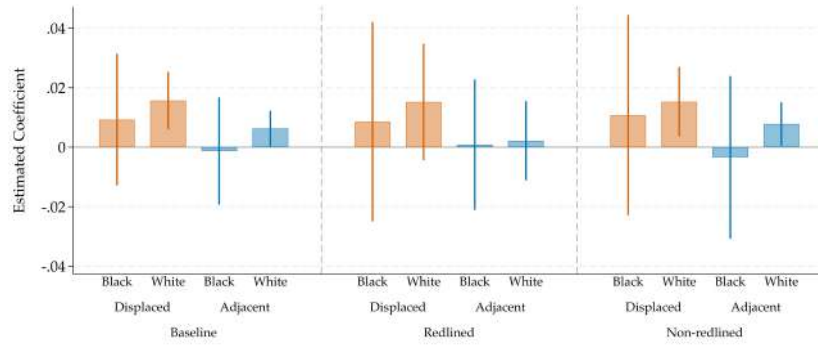
Notes: This figure compares the estimated effects by race to a reweighted sample where white individuals are reweighted to the distribution of metropolitan area, sex, quintile of the household head's occupational score, five-year birth cohort, and homeownership status in 1940, where individuals in households that I do not link to the 1940 census form their own tenure category, following DiNardo et al. (1996). All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan area level.



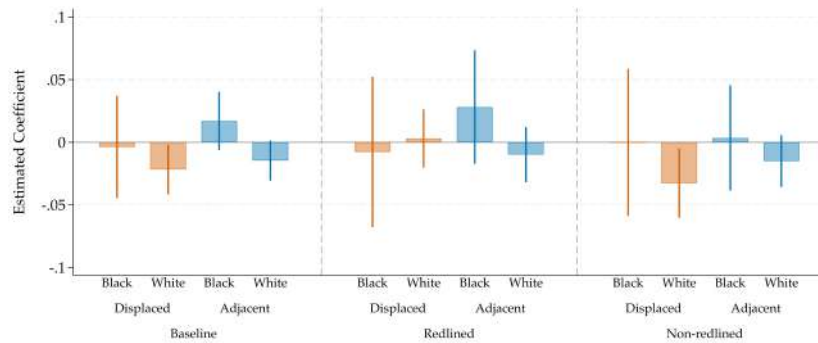
(A) Death age



(B) Same neighborhood



(C) Same city



(D) Neighborhood quality

FIGURE D.8: Effects by race and redlined status.

Notes: This figure compares the estimated effects by race to the effect by race and redlined status. All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan area level.

TABLE D.2: Effects of Interstate construction on mobility

	Same Neighborhood (1)	Same City (2)	Suburban 2000 (3)	log Dist to CBD (4)
<i>Panel A. Neighbors sample</i>				
Displaced	-0.023 ^a (0.002)	0.015 ^a (0.005)	-0.000 (0.004)	-0.007 (0.009)
Adjacent	-0.007 ^a (0.002)	0.006 ^c (0.003)	-0.004 (0.003)	-0.010 ^c (0.006)
Observations	122,682	122,682	121,322	121,322
Adj. R ²	0.025	0.121	0.120	0.151
Mean of dep. var.	0.039	0.620	0.721	9.523
<i>Panel B. Yellow Book sample</i>				
Displaced	-0.022 ^a (0.004)	0.015 ^c (0.008)	0.001 (0.006)	0.005 (0.012)
Adjacent	-0.006 ^c (0.003)	0.006 (0.008)	-0.004 (0.005)	0.001 (0.013)
Observations	123,141	123,141	121,812	121,812
Adj. R ²	0.022	0.109	0.109	0.138
Mean of dep. var.	0.039	0.627	0.746	9.574
<i>Panel C. Matching sample</i>				
Displaced	-0.027 ^a (0.004)	0.020 ^a (0.006)	0.001 (0.007)	0.008 (0.013)
Adjacent	-0.008 ^a (0.003)	0.010 (0.007)	-0.000 (0.005)	0.011 (0.009)
Observations	81,220	81,220	80,295	80,295
Adj. R ²	0.027	0.090	0.093	0.120
Mean of dep. var.	0.044	0.614	0.797	9.708

Note: OLS estimates are reported. Coefficients are reported with standard errors, clustered at the MSA level, in parentheses. The unit of observation is an individual in the sample of linked census-mortality records. The dependent variables are reported at the top of each column. Panel A reports the results comparing treated individuals to their neighbors living between 100 and 200 meters from future highway construction, Panel B compares them to those who would have been displaced by, or living adjacent to, the Yellow Book routes, and Panel C to matched individuals. More details on each of these specifications can be found in the main text and in Appendix Section C. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

TABLE D.3: Effects of Interstate construction on neighborhood characteristics

	Unemploy. Share (1)	College Share (2)	Homeowner. Share (3)	log Median Income (4)	log Median Home Value (5)	log Median Rent (6)	Neighborhood Quality (7)
<i>Panel A. Neighbors sample</i>							
Displaced	0.018 ^b (0.008)	-0.028 ^a (0.009)	-0.004 (0.009)	-0.010 (0.008)	-0.007 (0.010)	-0.002 (0.010)	-0.020 ^b (0.009)
Adjacent	0.013 ^b (0.005)	-0.008 (0.007)	-0.011 ^c (0.006)	-0.009 (0.007)	0.001 (0.007)	-0.004 (0.007)	-0.012 (0.007)
Observations	122,299	122,406	122,546	122,299	120,548	121,665	122,283
Adj. R ²	0.204	0.115	0.123	0.189	0.292	0.215	0.129
Mean of dep. var.	-0.004	0.004	0.003	0.008	0.007	0.009	0.007
<i>Panel B. Yellow Book sample</i>							
Displaced	0.026 ^b (0.011)	-0.078 ^a (0.013)	0.008 (0.013)	-0.026 ^b (0.012)	-0.055 ^a (0.013)	-0.033 ^a (0.011)	-0.062 ^a (0.012)
Adjacent	0.023 ^a (0.007)	-0.063 ^a (0.013)	-0.000 (0.008)	-0.027 ^a (0.010)	-0.055 ^a (0.011)	-0.037 ^a (0.009)	-0.059 ^a (0.011)
Observations	122,687	122,805	122,950	122,688	120,633	121,988	122,670
Adj. R ²	0.201	0.112	0.119	0.177	0.292	0.208	0.121
Mean of dep. var.	-0.002	0.003	0.001	0.007	0.006	0.008	0.006
<i>Panel C. Matching sample</i>							
Displaced	0.027 ^b (0.014)	-0.096 ^a (0.023)	-0.017 (0.022)	-0.038 ^b (0.015)	-0.074 ^a (0.018)	-0.047 ^a (0.014)	-0.083 ^a (0.018)
Adjacent	0.024 ^b (0.011)	-0.070 ^a (0.018)	-0.037 (0.025)	-0.042 ^a (0.009)	-0.063 ^a (0.011)	-0.044 ^a (0.008)	-0.076 ^a (0.013)
Observations	80,979	81,065	81,147	80,996	79,855	80,607	80,973
Adj. R ²	0.142	0.085	0.086	0.116	0.255	0.192	0.106
Mean of dep. var.	0.000	-0.000	-0.000	-0.000	-0.000	0.000	-0.000

Note: OLS estimates are reported. Coefficients are reported with standard errors, clustered at the MSA level, in parentheses. The unit of observation is an individual in the sample of linked census-mortality records. The dependent variables are reported at the top of each column. Panel A reports the results comparing treated individuals to their neighbors living between 100 and 200 meters from future highway construction, Panel B compares them to those who would have been displaced by, or living adjacent to, the Yellow Book routes, and Panel C to matched individuals. More details on each of these specifications can be found in the main text and in Appendix Section C. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

E. MECHANISMS APPENDIX

This appendix provides additional details on the mechanisms driving the mortality results.

E.1 Additional supportive evidence

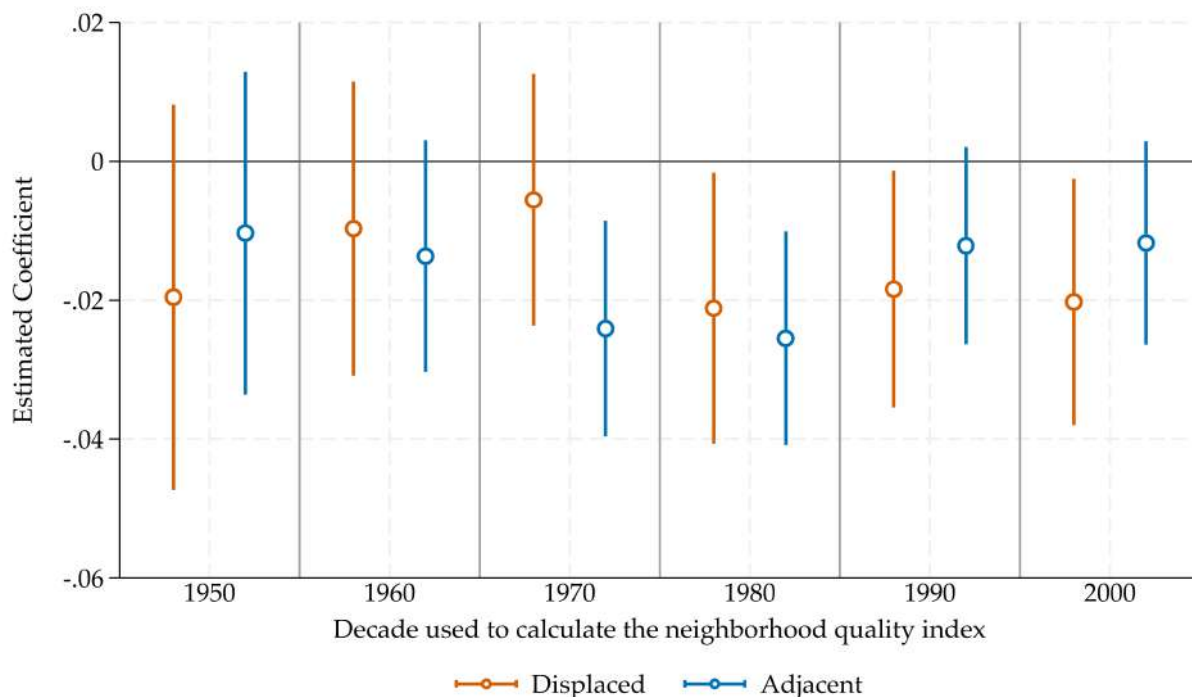


FIGURE E.1: Evolution of neighborhood quality over time

Notes: The figure displays estimates from regressions of neighborhood quality on the two treatment indicators in equation 2, along with 95% confidence intervals. Each column corresponds to a different regression. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variable corresponds to the neighborhood quality index described in Appendix Section D.6, calculated for the decade denoted in the x-axis and normalized to have mean zero and standard deviation one. The index is imputed to each individual based on the neighborhood in which they lived at the time of death. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

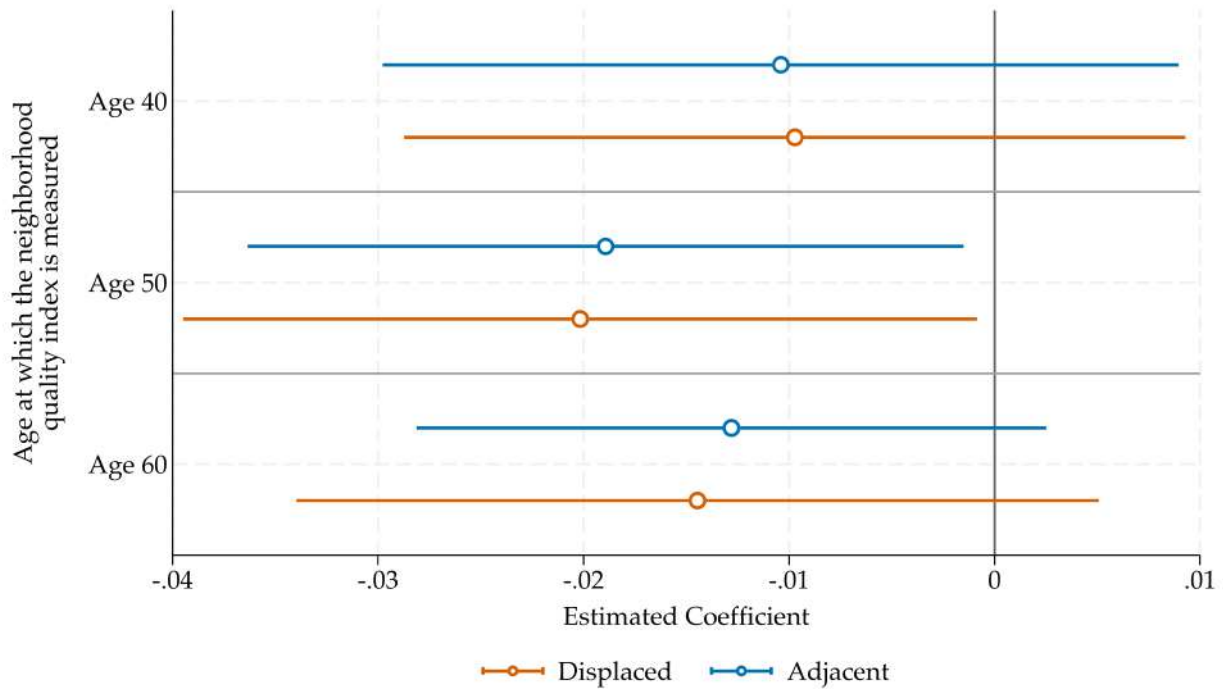


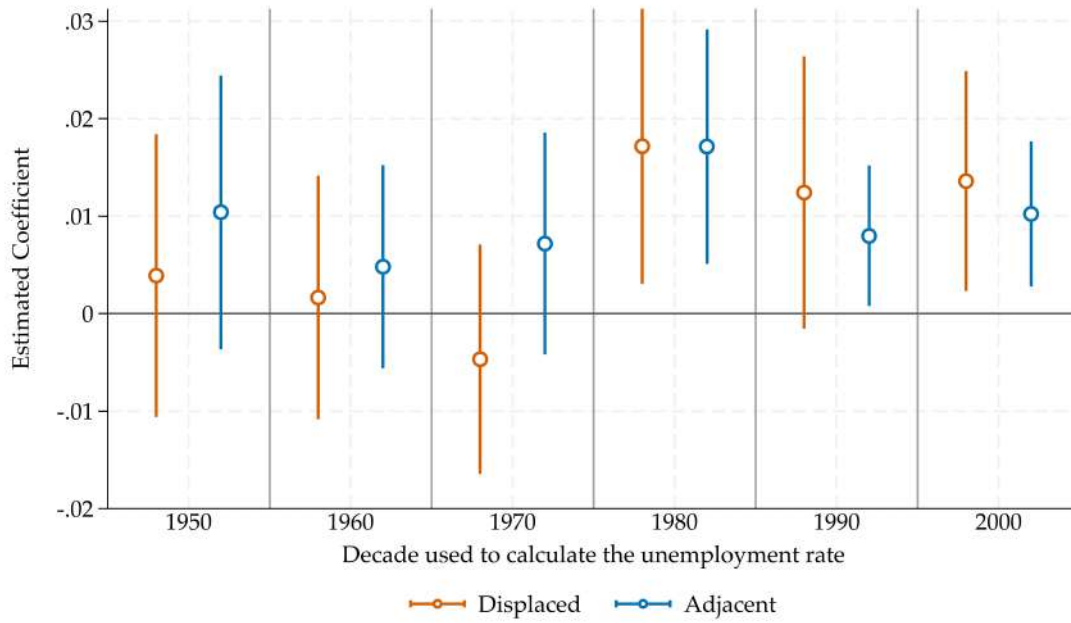
FIGURE E.2: Neighborhood quality at different ages

Notes: The figure displays estimates from regressions of neighborhood quality on the two treatment indicators in equation 2, along with 95% confidence intervals. Each column corresponds to a different regression. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variable is the neighborhood quality index, measured in the neighborhood where the individual lived when they turned the age denoted in the x-axis, and normalized to have mean zero and standard deviation one. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

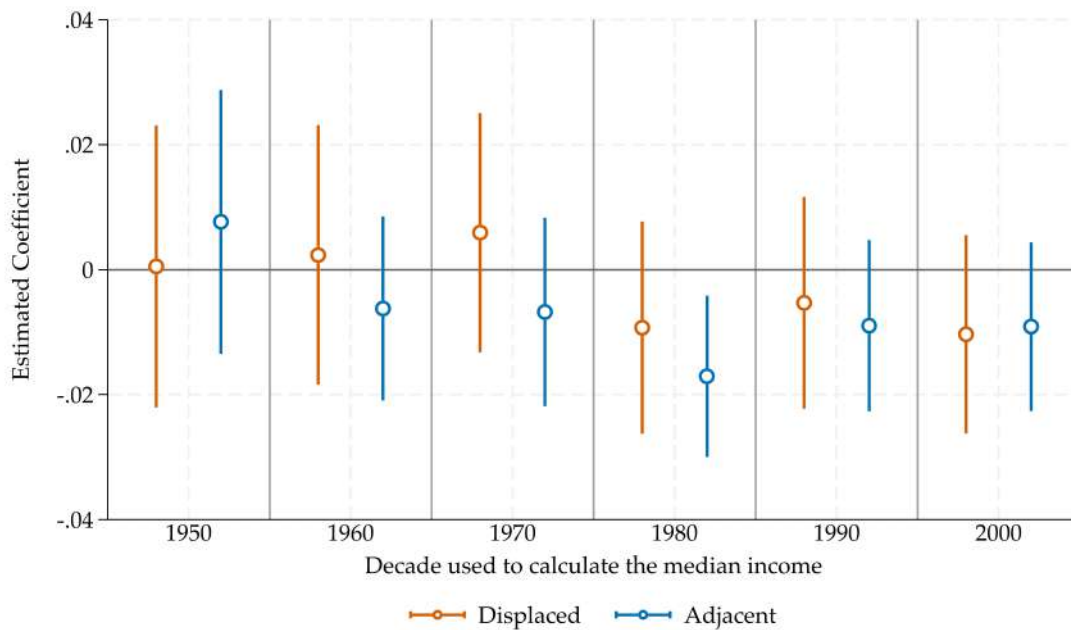
TABLE E.1: Other neighborhood characteristics

	PM _{2.5}	log Distance to CBD 2000
	(1)	(2)
Displaced	0.014 ^c (0.008)	-0.007 (0.009)
Adjacent	0.005 (0.005)	-0.010 ^c (0.006)
Observations	121,124	121,322
R-squared (adjusted)	0.016	0.151

Note: OLS estimates are reported. Coefficients of estimating equation 2 are reported with standard errors in parentheses, clustered at the MSA level. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. The dependent variables are reported at the top of each column and are standardized to have mean zero and standard deviation one. PM_{2.5} (column 1) is the five-year average (2000-2005) of the season-demeaned PM_{2.5} concentration in the neighborhood of residence, which is then standardized. Distance to the CBD (column 2) is the distance from the individual's last recorded residence to their city's central business district. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.



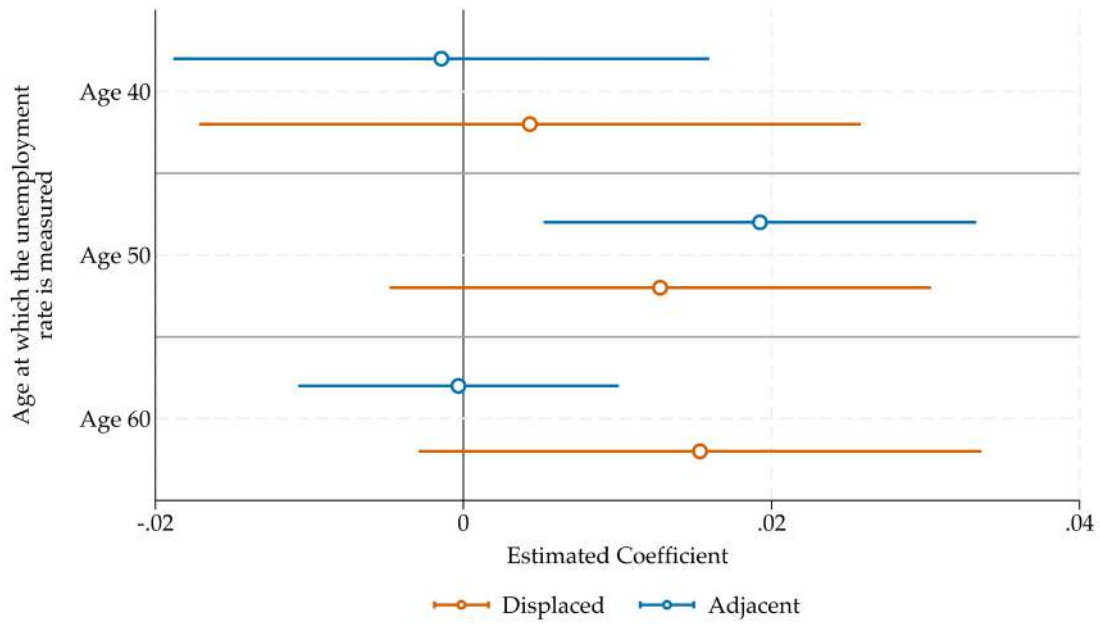
(A) Unemployment rate



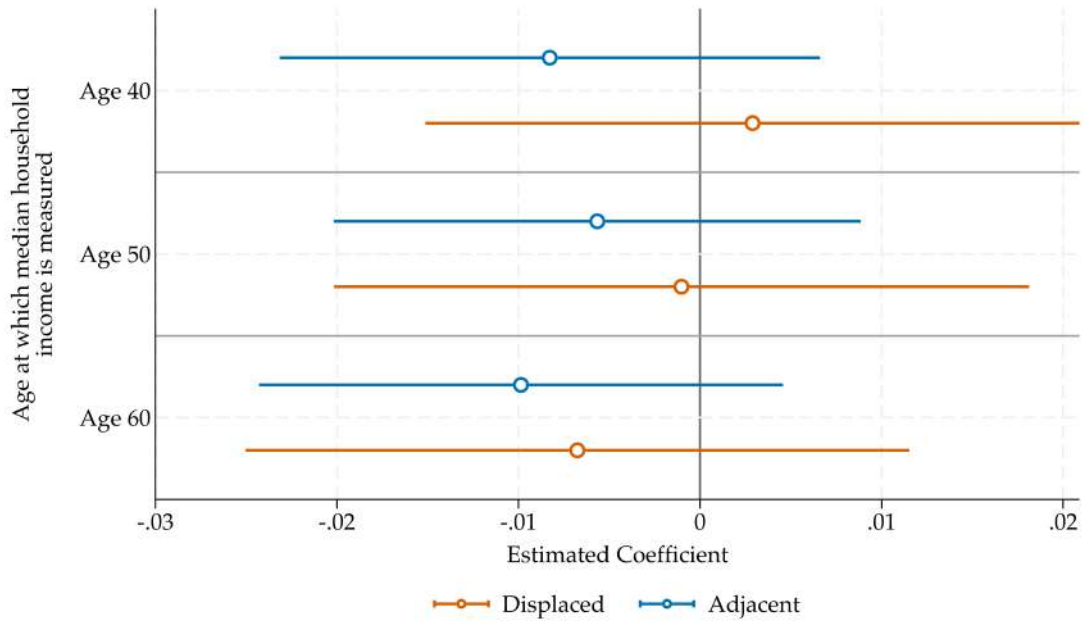
(B) Median income (log)

FIGURE E.3: Evolution of neighborhood labor characteristics over time

Notes: The figures display estimates from regressions of neighborhood unemployment rate or log median income on the two treatment indicators in equation 2, along with 95% confidence intervals. Each column corresponds to a different regression. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variable corresponds to the unemployment rate (panel (a)) or the log median income (panel (b)), calculated for the decade denoted in the x-axis and normalized to have mean zero and standard deviation one. The values are imputed to each individual based on the neighborhood in which they lived at the time of death. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.



(A) Unemployment rate



(B) Median income (log)

FIGURE E.4: Neighborhood labor characteristics at different ages

Notes: The figures display estimates from regressions of neighborhood unemployment rate or log median income on the two treatment indicators in equation 2, along with 95% confidence intervals. Each column corresponds to a different regression. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. The dependent variable is the unemployment rate (panel (a)) or the log median income (panel (b)), measured in the neighborhood where the individual lived when they turned the age denoted in the x-axis, and normalized to have mean zero and standard deviation one. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

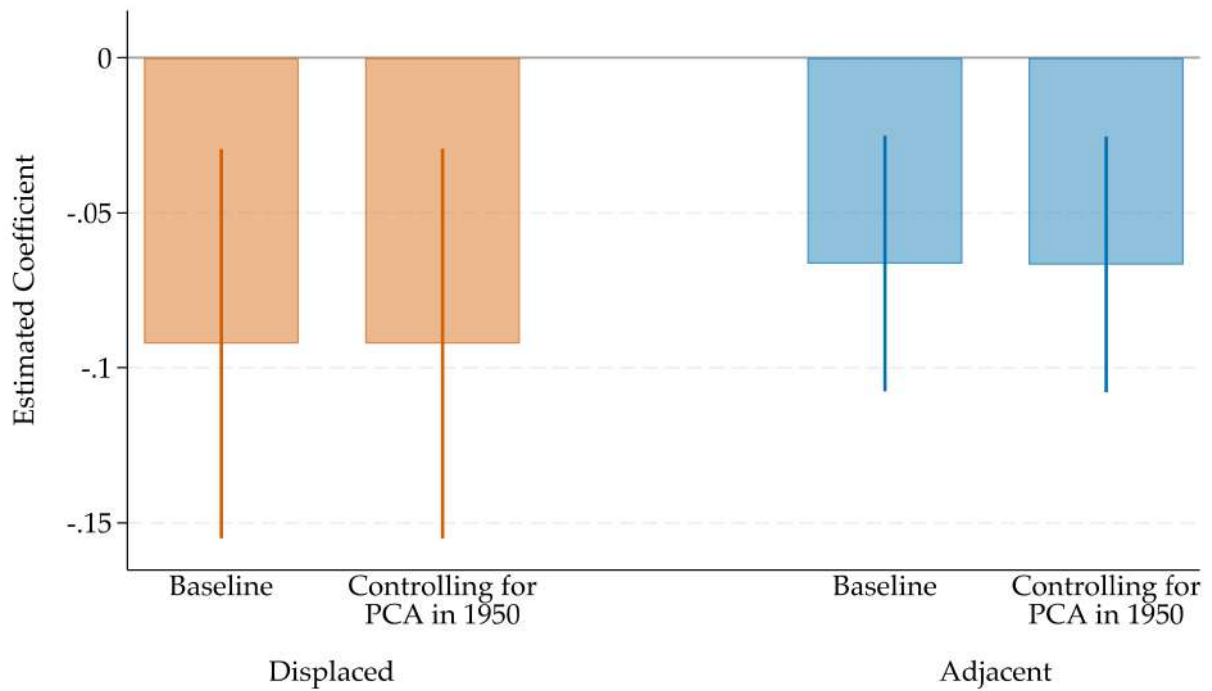
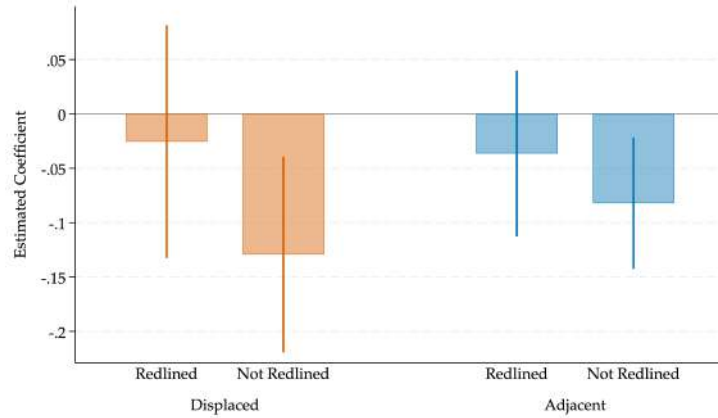
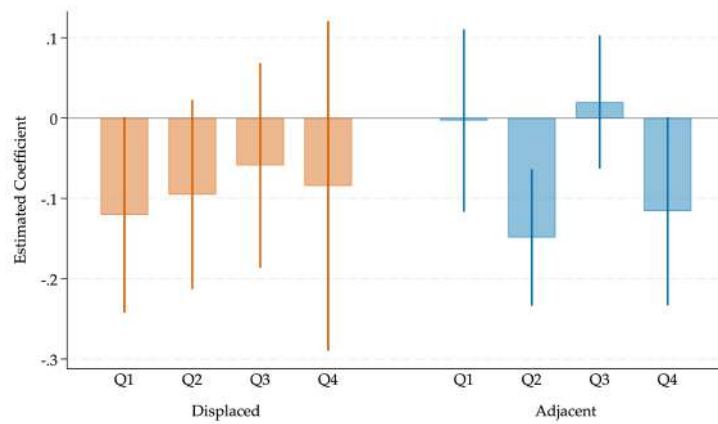


FIGURE E.5: Robustness controlling for 1950 neighborhood quality

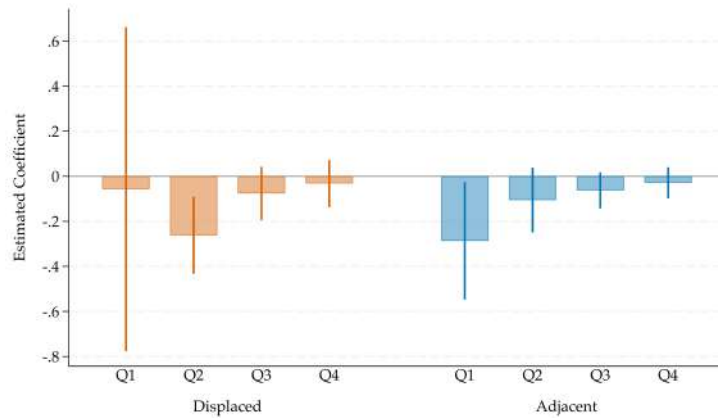
Notes: The figure displays estimates from equation 2 for longevity controlling for 1950 neighborhood quality, along with 95% confidence intervals. Neighborhood quality in 1950 is the first principal component of 1950 neighborhood characteristics, excluding the housing variables, which are not available in the 1950 census. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each row corresponds to a different regression, with standard errors clustered at the metropolitan area level. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender.



(A) Redlining status



(B) Income quartiles



(C) Unemployment quartiles

FIGURE E.6: Heterogeneity by neighborhood characteristics in 1950

Notes: The figures display estimates of the effect on longevity from regressions in which the sample is split by neighborhood characteristics in 1950: redlining status, quartiles of male income, and quartiles of the male unemployment rate. A neighborhood is classified as redlined if it was assigned a grade of D by the Home Owners' Loan Corporation. Each panel corresponds to a different regression using a different proxy for neighborhood quality. Within each panel, each bar is an estimate along with 95% confidence intervals. The sample corresponds to individuals living within 200 meters of the newly constructed highway in the 1950 census. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

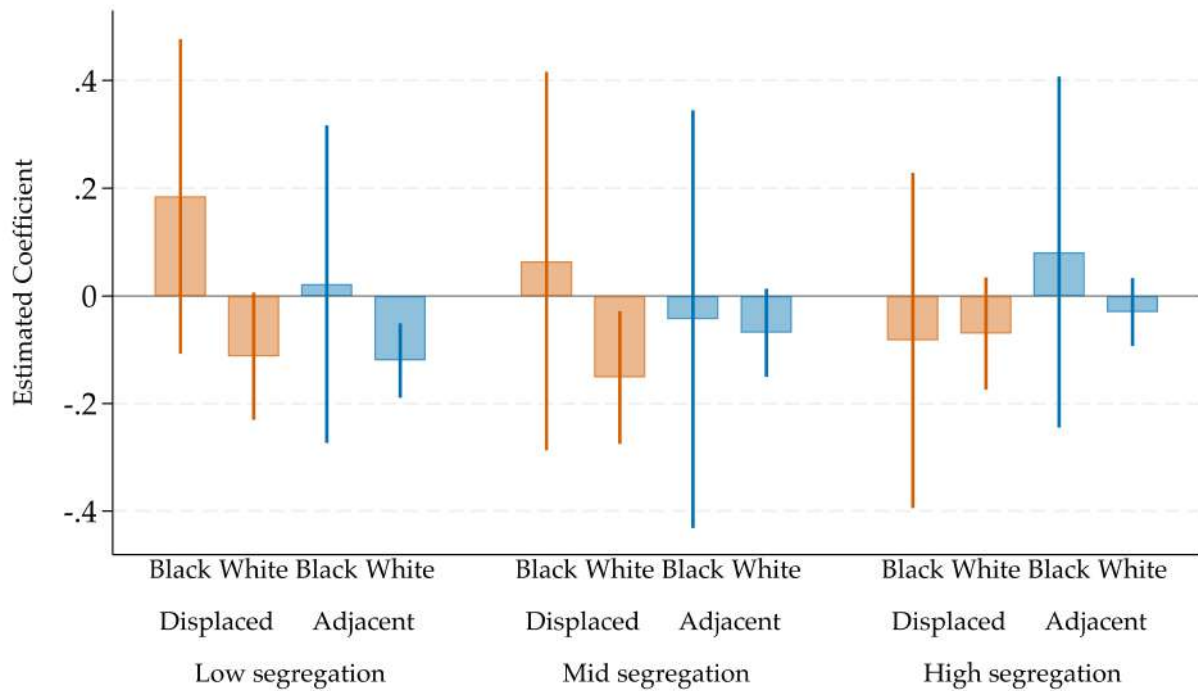


FIGURE E.7: Heterogeneity by race and racial segregation

Notes: The figure displays estimates of the effect on longevity from regressions in which the sample is split by race and by the level of racial segregation in the metropolitan area. Racial segregation is measured using the dissimilarity index, with metropolitan areas divided into terciles of that index. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each bar represents an estimate along with 95% confidence intervals. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

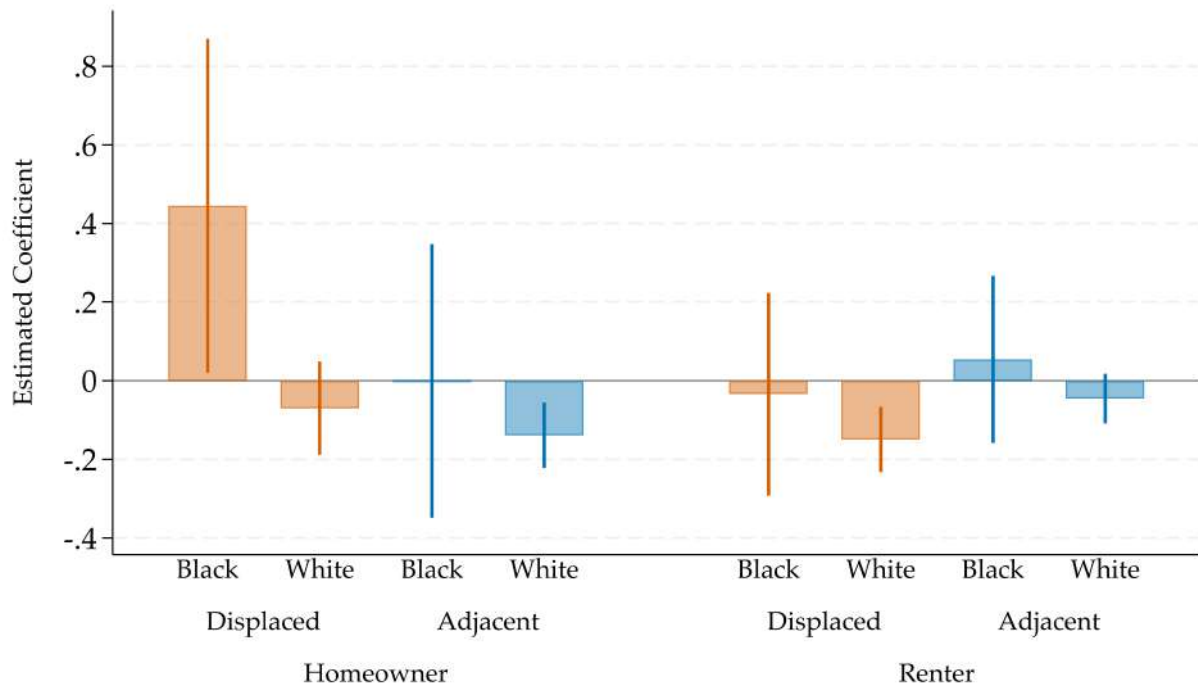


FIGURE E.8: Heterogeneity by race and homeownership

Notes: The figure displays estimates of the effect on longevity from regressions in which the sample is split by race and homeownership status. Homeownership is measured using linked 1940 and 1950 census data: an individual is classified as a homeowner if anyone in their 1950 household was a homeowner in 1940. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each bar represents an estimate along with 95% confidence intervals. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

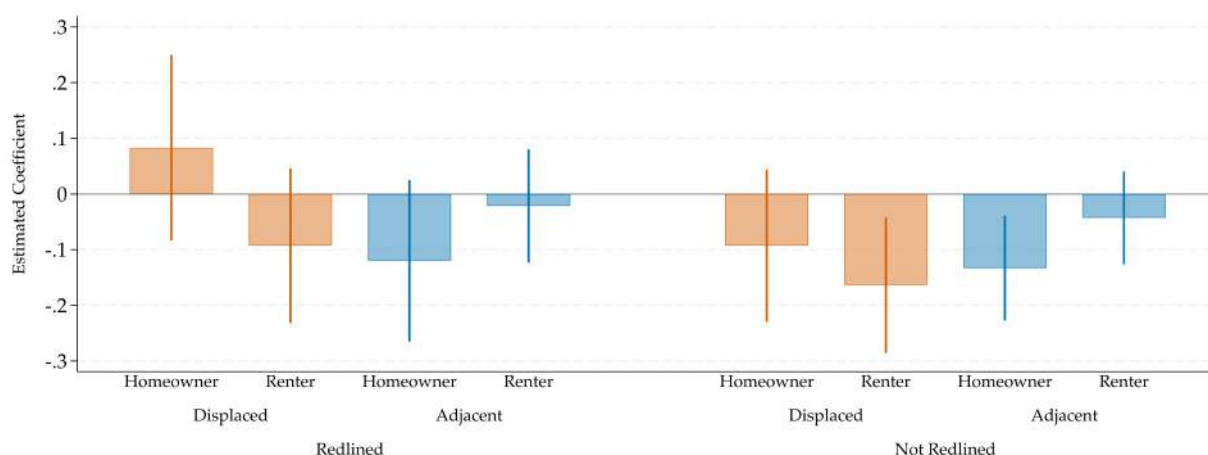


FIGURE E.9: Heterogeneity by homeownership and redlining status

Notes: The figure displays estimates of the effect on longevity from regressions in which the sample is split by homeownership and redlining status. Homeownership is measured using linked 1940 and 1950 census data: an individual is classified as a homeowner if anyone in their 1950 household was a homeowner in 1940. The sample includes those individuals displaced by highway construction and those living within 200 meters of these projects in the 1950 census, linked to administrative mortality records from 1988 to 2007. Each bar represents an estimate along with 95% confidence intervals. All specifications include MSA fixed effects and controls for head-of-household occupational score, race, birth year, and gender. Standard errors are clustered at the metropolitan area level.

TABLE E.2: Displacement and public housing

	Public Housing	log Home Value	log Monthly Rent	log Annual Income
	(1)	(2)	(3)	(4)
Displaced by highway	0.097 ^b (0.046)	-0.201 ^c (0.107)	-0.176 (0.118)	-0.330 ^a (0.115)
Previous homeowner	-0.001 (0.007)	0.213 ^a (0.019)	0.057 ^b (0.024)	0.291 ^a (0.019)
Observations	8,652	4,204	4,098	12,396
R-squared (adjusted)	0.026	0.079	0.229	0.055
Mean dependent variable	0.051	11.853	6.267	10.458

Note: OLS estimates are reported. The data come from the 1973 and 1974 American Housing Survey (AHS), which records whether a respondent moved because of highway displacement. The sample includes individuals who moved in the previous 12 months and who were living in a metropolitan area at the time of the 1973 or 1974 AHS. The dependent variables are reported at the top of each column. The key regressor is an indicator for having been displaced by highway construction in the previous 12 months. Every regression controls for survey dummies, race, a second-order polynomial in age, and an indicator for whether the individual owned their home before moving. Standard errors are robust. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.

F. RELOCATION ASSISTANCE APPENDIX

F.1 Building the list of states providing relocation assistance

To study whether relocation payments improved long-run outcomes, I construct a novel dataset documenting when each state began providing relocation payments to displaced individuals. To the best of my knowledge, this is the first systematic compilation of state-level policy adoption timing. I begin by reviewing federal and state legislative records to identify when each state enacted laws authorizing such payments. I then cross-reference these records with four official sources documenting the state of relocation assistance programs as of January 1963, January 1965, and January 1967, as well as reported start years from the 1968 Federal-Aid Highway Act (Highway Research Board, 1963; US Advisory Commission on Intergovernmental Relations, 1965; U.S. Department of Transportation, 1967; United States. Department of Transportation, 1970). Together, these sources provide comprehensive coverage of state-level policy adoption, allowing me to construct a complete timeline of when each state began providing relocation assistance to displaced individuals. Appendix Table F.1 summarizes the resulting dataset.

One limitation is that legislative enactment does not guarantee implementation. South Dakota passed authorizing legislation in 1963 but declined to operate the program (US Advisory Commission on Intergovernmental Relations, 1965, pg. 70). However, this is the only documented case of non-implementation, and South Dakota is excluded from my sample because it had no metropolitan areas in 1950.

Appendix Figure F.4 shows the number of states enacting relocation assistance programs by year. Most states adopted these programs between 1963 and 1966, although a quarter had not done so by 1968. As seen in Appendix Figure F.5, most late adopters are in the South and non-coastal Western states. Finally, Appendix Table F.2 shows that states with earlier adoption dates had higher GDP per capita and larger populations in 1960, and smaller Black populations.

F.2 Event study design

I can exploit the staggered timing of state relocation assistance to estimate the causal effect of relocation on long-run outcomes. To do so, I use an event-study design that compares households displaced by highway construction before and after their state enacted relocation assistance relative to the year before enactment, with an analogous comparison for adjacent households. The estimating equation follows,

$$\begin{aligned}
y_i = & \sum_{\substack{k=-6 \\ k \neq -1}}^3 \beta_k^D (1\{\tau_i = k\} \times D_i) + \sum_{\substack{k=-6 \\ k \neq -1}}^3 \beta_k^A (1\{\tau_i = k\} \times A_i) \\
& + D_i + A_i + \sum_{k=-6}^3 \gamma_k 1\{\tau_i = k\} + \lambda_{t(i)} + \mathbf{X}_i' \Gamma + \varepsilon_i,
\end{aligned} \tag{F.1}$$

where i indexes individuals in the 1950 linked sample. The outcome y_i includes age at death, mobility outcomes, and the neighborhood quality index. D_i indicates whether the individual was displaced by highway construction and A_i indicates an individual living adjacent to the newly constructed highway. The comparison group corresponds to individuals living between 100 and 200 meters from the newly constructed highway. The event time τ_i indicates the year in which the individual was displaced (or adjacent) relative to the year of enactment of relocation assistance in their state, with negative values indicating displacement before enactment and positive values indicating displacement after enactment. Events occurring more than six years before or more than three years after enactment are binned into these endpoint categories. The indicators $1\{\tau_i = k\}$ correspond to dummies relative to the event. The fixed effect $\lambda_{t(i)}$ corresponds to the year of construction fixed effect. As relocation assistance occurred later in time, the year fixed effect implies that identification comes from the change in relocation laws per se rather than the date of construction. The vector $\mathbf{X}_i$ includes controls for the household head's occupational score, as well as fixed effects for gender, race, birth year, and metropolitan area of residence. Standard errors are clustered at the metropolitan-area level.

The coefficients of interest are β_k^D and β_k^A , the effect of relocation assistance on those displaced and adjacent relative to the year prior to enactment. The identifying assumption is that, absent the relocation law, the outcomes of households displaced just after enactment would have continued to track those of households displaced just before. Although the assumption is not directly testable, I can examine the pre-enactment coefficients to see whether they are statistically different from zero.

The results of estimating equation F.1 are presented in Appendix Figure F.1. Across the four outcomes, the pre-enactment coefficients for displaced individuals are not statistically different from zero, lending credibility to the parallel trends assumption. For adjacent individuals, some pre-enactment coefficients differ from zero individually, but for all four outcomes I fail to reject the joint hypothesis that they are all zero. Panel A shows that, in the two years following enactment, the effect of relocation assistance on age at death is positive and statistically different from zero. The increase in longevity after enactment suggests that relocation assistance was an effective policy tool for mitigating the mortality consequences of displacement. The effect fades in the third year after enactment, although the confidence interval is wide and includes the point estimates from the first two years.

Panels B, C, and D show that relocation assistance had no statistically significant effect on the probability of living in the same neighborhood or city, or on the neighborhood quality index. As discussed in Section 7.3, this is consistent with the fact that relocation assistance actually provided a cash payment to displaced households, rather than providing them with housing in the same neighborhood or city, or helping them find a better neighborhood.

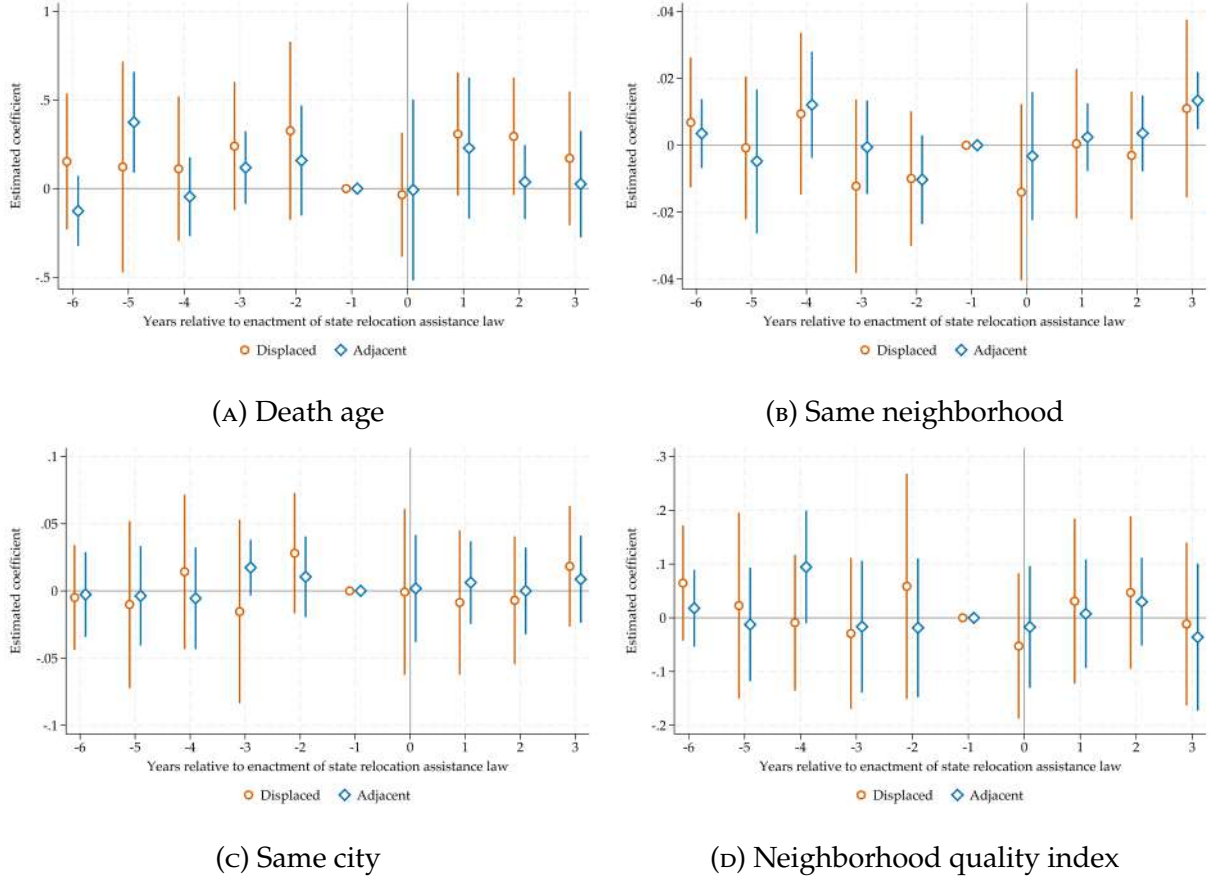


FIGURE F.1: Event study results for relocation assistance.

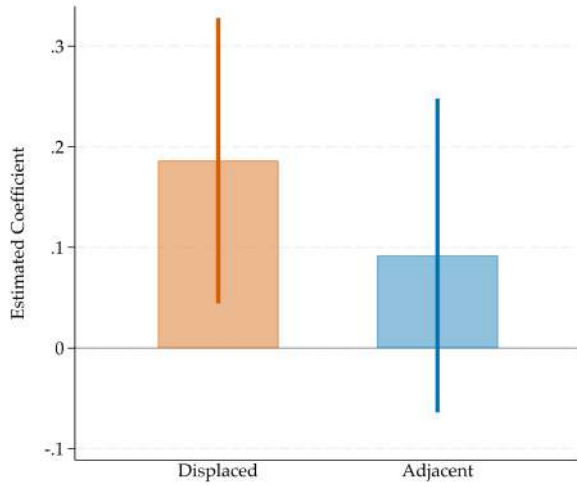
Notes: Each panel plots the event-study coefficients from estimating equation (F.1) on the 1950 linked sample. The horizontal axis is time relative to the event τ , the number of years between the start of nearby highway construction and the enactment of the individual's state relocation-assistance law. The vertical line at 0 marks the enactment period and the coefficients are relative to period $\tau = -1$, so every coefficient is read relative to the year just before the law applies. Red circles denote the displaced coefficients β_k^D and blue diamonds the adjacent coefficients β_k^A , with vertical spikes corresponding to 95% confidence intervals. Panel A presents the results for age at death in years, Panel B for an indicator for residing in the same neighborhood, Panel C for an indicator for residing in the same city, and Panel D for the neighborhood quality index. All specifications control for the household head's occupational score and absorb metropolitan-area, birth-year, race, sex, and year of construction fixed effects. Standard errors are clustered by metropolitan area.

F.3 Additional tables and figures

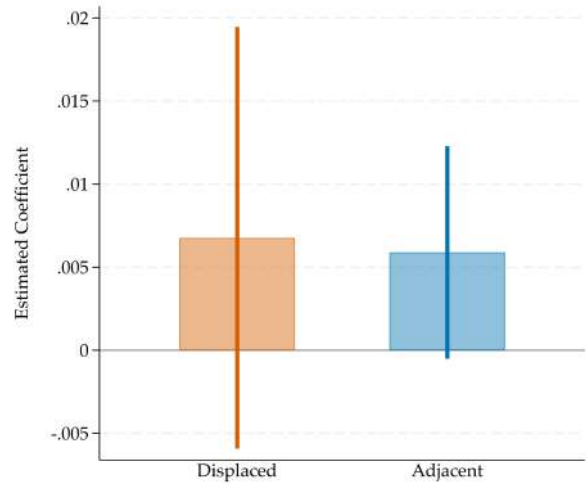
TABLE F.2: Correlations between state characteristics and relocation assistance enactment.

	log GDP per capita (1963)	log Total Population (1960)	log White Population (1960)	log Non-White Population (1960)
Year of enactment	-2.258 ^c (1.312)	-0.308 (0.381)	-0.430 (0.376)	0.240 (0.199)
Observations	48	48	48	48
R-squared	0.036	0.012	0.024	0.025

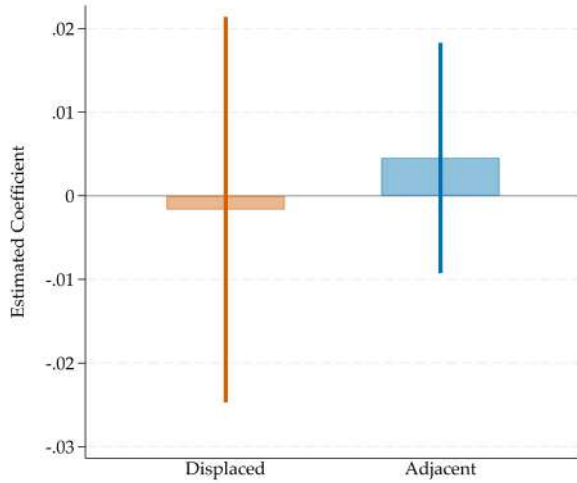
Note: OLS estimates are reported. Each coefficient corresponds to a different regression and is reported with robust standard errors in parentheses. The unit of observation is a state. The dependent variable is the year of enactment of relocation assistance. Each independent variable is denoted in the column header. ^a indicates the coefficient is significant at the 1%, ^b at the 5%, and ^c at the 10% level.



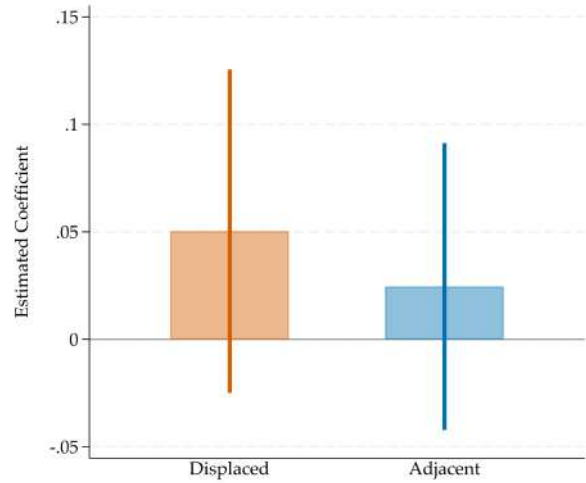
(A) Death age



(B) Same neighborhood



(C) Same city



(D) Neighborhood quality index

FIGURE F.2: Effect of relocation assistance — State-year fixed effects.

Notes: The figure compares the effect of Interstate highway construction for displaced individuals and those residing within 100 meters of the highway, with and without relocation assistance laws in place. In particular, it plots the estimates of γ_2 and δ_2 from equation 3 for each outcome. The dependent variable in Panel (a) is the age at which the individual died. The dependent variable in Panel (b) is an indicator for whether the individual lived in the same neighborhood (census tract) at the time of the census and at the time of death. The dependent variable in Panel (c) is an indicator for whether the individual lived in the same city (metropolitan area) at the time of the census and at the time of death. The dependent variable in Panel (d) is an index of neighborhood quality, constructed as the first principal component of the neighborhood characteristics described in Section 6.3. The comparison group consists of unaffected neighbors living between 100 and 200 meters from the highway. All regressions control for the household head's occupational score and include fixed effects for state-year of construction, gender, race, and birth year. Standard errors are clustered at the metropolitan area level.

TABLE F.1: Years of enactment of relocation assistance laws by state.

State	Year	Source
Alabama	1965	Acts of Alabama 1965, Number 228
Arizona	1969	United States. Department of Transportation (1970)
Arkansas	1969	Arkansas Acts 1969, Number 62
California	1965	The California Relocation Assistance Act of 1965 (Chapter 1650)
Colorado	1969	Colorado Session Laws 1970, Chapter 95
Connecticut	1962	U.S. Department of Transportation (1967)
Delaware	1970	Delaware Laws 1970, Chapter 634, codified in 29 Delaware Code § 8004
District Of Columbia	1964	Public Law 88-629, 78 Statute 1015
Florida	1969	United States. Department of Transportation (1970)
Georgia	1966	Georgia Laws 1966, p. 60, also known as House Bill 105
Idaho	1969	Idaho Session Laws 1969, Chapter 295
Illinois	1965	Illinois Laws 1965, p. 3377, House Bill 2210
Indiana	1967	Indiana Acts 1967, Chapter 273, codified as IC 8-13-18.5
Iowa	1965	Iowa Code Ann. § 316.6, as established by 61 G.A. ch. 319 (1965)
Kansas	1969	United States. Department of Transportation (1970)
Kentucky	1964	Kentucky Acts 1964, Chapter 15
Louisiana	1972	Louisiana Act 121 of 1972, codified as Louisiana Revised Statute 19:201
Maine	1965	U.S. Department of Transportation (1967)
Maryland	1962	Maryland Laws 1963, Chapter 36, Article 33A, §12
Massachusetts	1963	Massachusetts Acts of 1963, Chapter 718
Michigan	1966	U.S. Department of Transportation (1967)
Minnesota	1962	Minnesota Statutes § 117.50 et seq.
Mississippi	1970	Mississippi Acts 1971, Chapter 520
Missouri	1968	United States. Department of Transportation (1970)
Montana	1969	United States. Department of Transportation (1970)
Nebraska	1963	Nebraska Revised Statute § 39-1329
Nevada	1963	Nevada Statutes 1963, Chapter 408
New Hampshire	1965	New Hampshire Laws 1965, Ch. 433, codified as NH RSA 233-A
New Jersey	1963	New Jersey P.L. 1963, c. 79, codified as N.J.S.A. 27:7-44.3
New Mexico	1969	United States. Department of Transportation (1970)
New York	1962	Gov. Rockefeller's NY Relocation Department (1962)
North Carolina	1965	U.S. Department of Transportation (1967)
North Dakota	1963	North Dakota Century Code 24-01-41.1
Ohio	1963	Ohio Administrative Code 5501:2-5-03
Oklahoma	1963	Highway Research Board (1963)
Oregon	1963	Oregon Revised Statute § 366.515, modified by Ore. Laws 1963, ch. 472
Pennsylvania	1964	Pennsylvania Eminent Domain Code of 1964
Rhode Island	1962	Highway Research Board (1963), R.I. General Laws § 37-6.1-1 et seq.
South Carolina	1968	South Carolina Act N. 1233, codified as S.C. Code § 57-5-1610 et seq.
South Dakota	1963	US Advisory Commission on Intergovernmental Relations (1965)
Tennessee	1951	Public Acts of Tennessee 1951, Chapter 176
Texas	1969	Texas Acts 1969, Chapter 363, codified as Article 6674n-4
Utah	1963	Highway Research Board (1963)
Vermont	1963	Vermont Acts 1963, Number 110, codified at 19 V.S.A. § 2001 et seq.
Virginia	1964	Virginia Acts 1964, Chapter 46
Washington	1965	U.S. Department of Transportation (1967)
West Virginia	1963	West Virginia Acts 1963, Chapter 60
Wisconsin	1961	Highway Research Board (1963)
Wyoming	1969	United States. Department of Transportation (1970)

Note: Years of enactment of relocation assistance laws for each state, as documented in the sources listed in the third column.

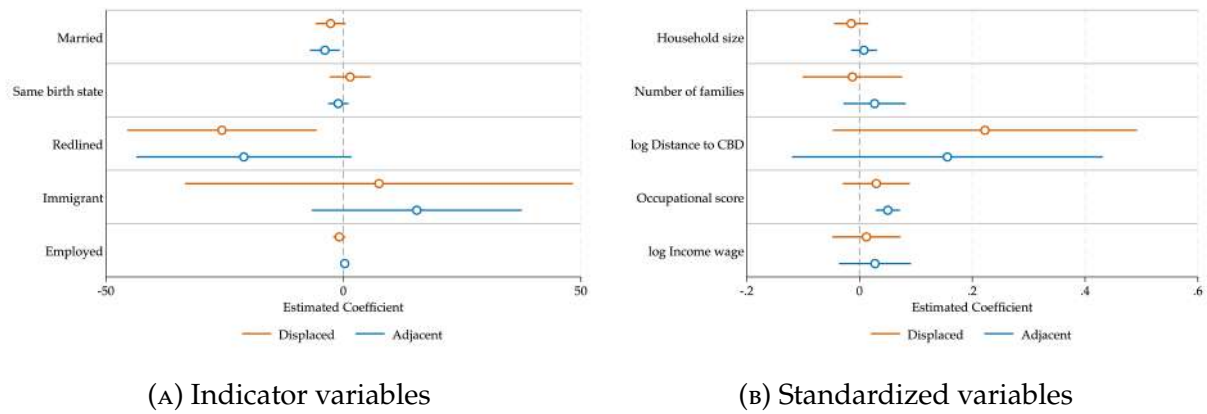


FIGURE F.3: Balance tests for relocation assistance.

Notes: The figure presents the coefficient and confidence intervals for being treated while eligible for relocation assistance, compared to being treated while ineligible, for a set of pre-treatment characteristics. Each coefficient corresponds to a different regression and is reported with standard errors clustered at the metropolitan area level. The coefficients and confidence intervals in panel (a) are normalized by the sample mean of the dependent variable, while the dependent variables in panel (b) are standardized to have mean zero and standard deviation one. All regressions control for the household head's occupational score and include fixed effects for gender, race, birth year, and metropolitan area of residence.

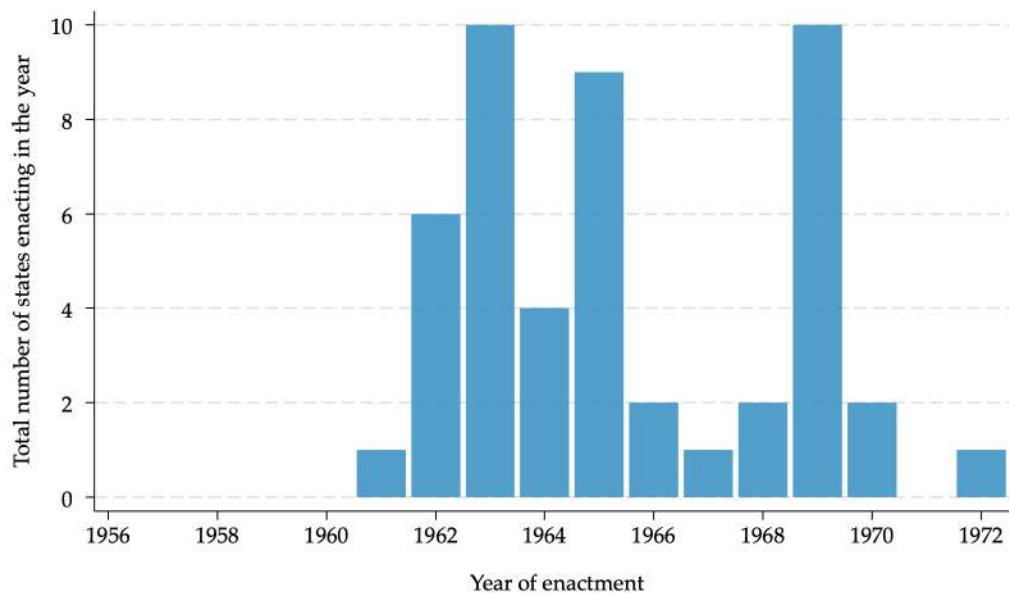


FIGURE F.4: Number of states enacting relocation assistance by year.

Notes: Each bar corresponds to the number of states enacting relocation assistance in a given year, as shown in Appendix Table F.1. Tennessee is not included in the figure because it enacted relocation assistance in 1951, before the start of my sample period.

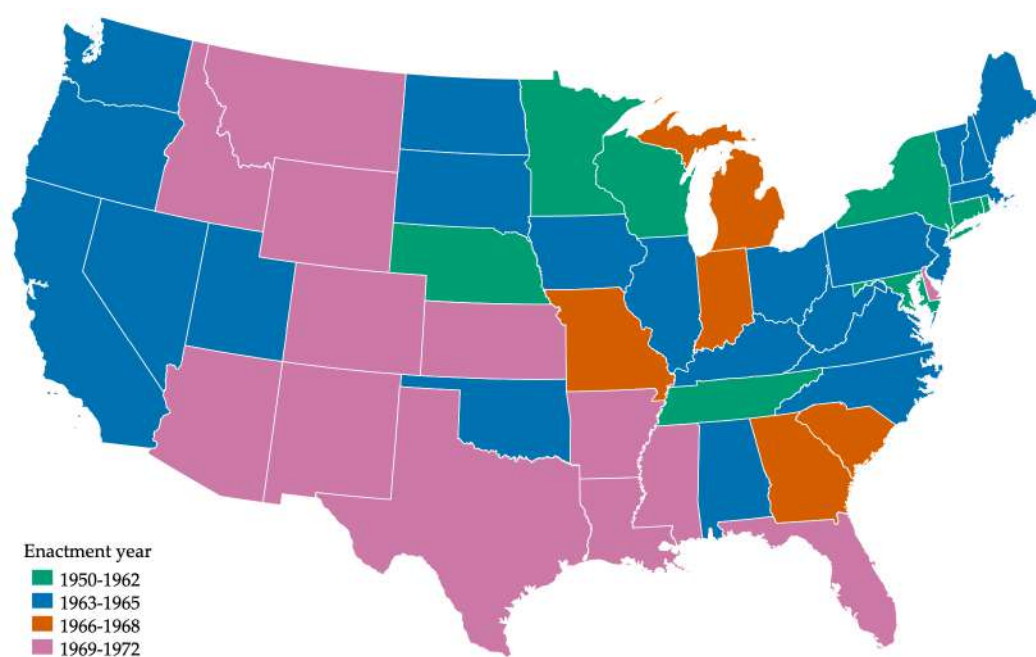


FIGURE F.5: Spatial distribution of years of enactment of relocation assistance.

Notes: Map showing the years of enactment of relocation assistance for each state, as shown in Appendix Table F.1.

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